ST7305

Ultra-Low Power Active Matrix 408H x 408V TFT Display Driver with Controller

Datasheet

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Version 1.2

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1 INTRODUCTION

The ST7305 is a single-chip controller/driver for small and medium TFT LCD display that operate at very low frame frequency to conserve power. This chip is capable of supporting up to 408H x 408V pixels and provides a standard 8-bit 8080 parallel interface, 3-wire SPI serial interface and single/dual 4-wire SPI serial interface to configure the system and update the graphics content. The 408-channel source driver supports 1-bit per pixel (or sub-pixel) control to simplify the driver construction and lower power dissipation.

ST7305 is controlled using a conventional mobile driver SoC interface which supports an interface I/O voltage (VDDI) of 1.8V ~ 3.3V. ST7305 is equipped with all required charge pumps, buffer amplifiers and regulators to run directly from an unregulated coin cell battery source (Normal mode VDDA=2.8V ~ 3.3V; 1.8V mode VDDA=1.8V). All voltages used by the TFT channel drivers are close loop regulated by ST7305 and can be adjusted in fine increments by user controlled digital register settings to optimize power and display characteristics. The built-in timing controller (TCON) supports a wide range of output timing for various pixel organizations and driver waveform generation. The combination of timing and voltage controls allows the ST7305 driver SoC to have wide compatibility with various LCD and TFT types.

ST7305 is designed for consumer applications; this product is not designed for use in cars, motorcycles, marine equipment, aircraft equipment, military equipment and other applications in extreme environment.

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2 FEATURES

Design for Low Power Active Matrix a-Si TFT Display

Single-chip TFT-LCD Controller/Driver with On-chip Frame Memory

Support for 408H x 408V Mono pixels

Outputs bi-level 1-bit per pixel source drive

Support multiple frame rate (0.25Hz ~ 53Hz)

Supports multiple LCD modes

- ECB or MTN
- Normally black or white

Built-in 408 x 408 x 1b internal SRAM

Dot/Column inversion DC drive of TFTs at low refresh rates

Built-in Sequential/Interlace Gate Scan Function

Microprocessor Interface

- 80 parallel 8bits
- 3/ 4-line serial interface support write-operation and read- status

Wide Power Supply Range

- Normal Mode
  VDDI(Digital) : 1.8V ~ 3.3V
  VDDA(Analog): 2.8V ~ 3.3V
- 1.8V Mode
  VDDI(Digital)=VDDA(Analog)=1.8V

On-chip Build-In Circuits

- DC/DC Converter
- Non-Volatile (NV) Memory to Store Initial Register Setting and Factory Default Value (Module ID, Module Version, etc)
- Internal Oscillator for Display Clock Generation
- Timing Controller

On-chip power management system

Built in charge pump circuits

- 2x boost generates: 6.2V (Normal Mode)
- High voltage +16.5V/-15V charge pump

Built in low power analog circuit

- On-chip oscillator circuit.
- Built-in voltage regulator with programmable contrast
- On-chip power system
  - VCOM level : GND
  - Source Voltage (VSHP/VSHN/VSLP/VSLN): +6.0V ~ -5.0V

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- Gate Driver HIGH Level (VGH to AGND): +8V ~ +16.5V
- Gate Driver LOW Level (VGL to AGND): -6V ~ -15V
- Max. VGH – VGL: 31.5V

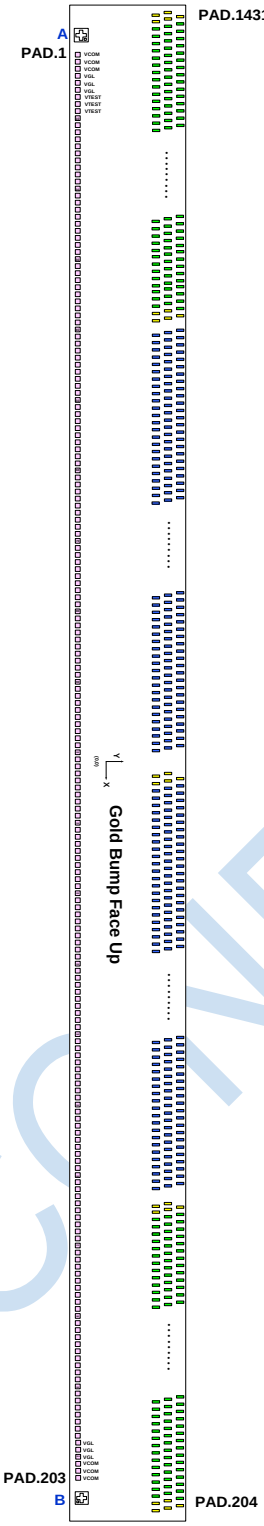
- **Built-in NV memory for VSHP/VSLP/VSHN/VSLN amplitude adjustment and ID setting.**
- **COG packaging**
- **Wide operating temperature range: -30°C ~ +85°C**

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3 PAD ARRANGEMENT

3.1 Configuration of Signal Pads



ST7305	
Chip Thinness	200
Bump Height	12
Output Pad	13(X) x 95(Y)
I/O Pad	40(X) x 50(Y)

Unit: um

Figure 1 Chip

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3.2 Bump

3.2.1 Output Pads

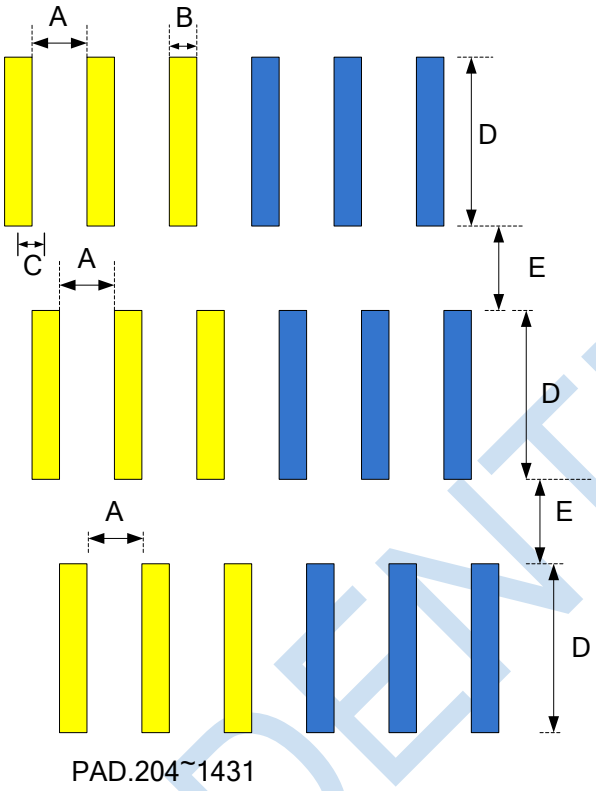


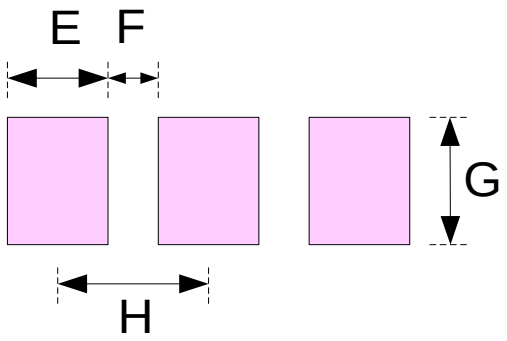
Figure 2 Output Pads Outline

Symbol	Item	Size (um)
A	Bump Gap (H)	17
B	Bump Width	13
C	Bump Pitch	10
D	Bump Height	95
E	Bump Gap (V)	20

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3.2.2 Input Pads



Symbol	Item	Size (um)
E	Bump Width	40
F	Bump Gap	20
G	Bump Height	50
H	Bump Pitch	60

Figure 3 Input Pads Outline

3.2.3 Alignment Marks

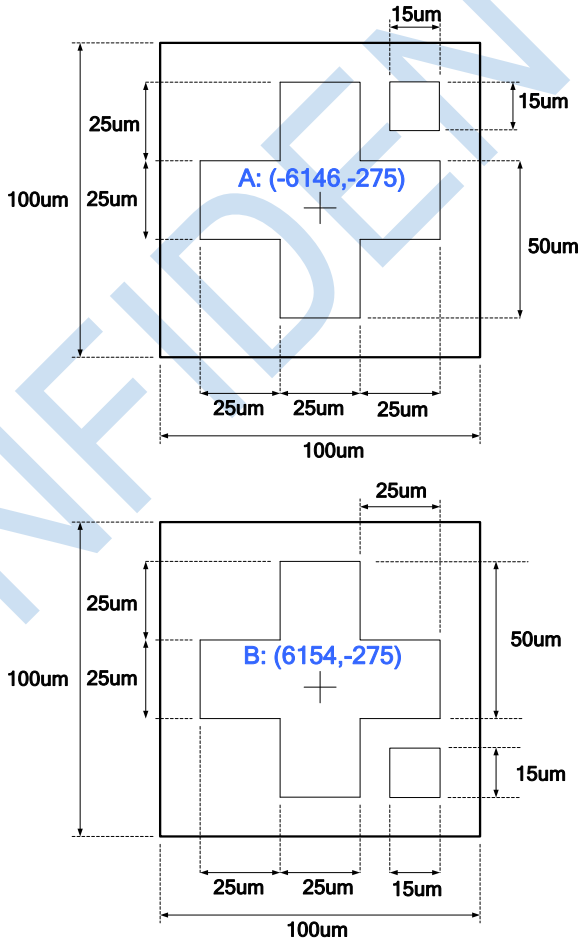


Figure 4 Alignment Marks Outline

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4 PAD CENTER COORDINATES

PAD.	Name	X	Y
1	VCOM	-6056	-292.62
2	VCOM	-5996	-292.62
3	VCOM	-5936	-292.62
4	VGL	-5876	-292.62
5	VGL	-5816	-292.62
6	VGL	-5756	-292.62
7	VTEST	-5696	-292.62
8	VTEST	-5636	-292.62
9	VTEST	-5576	-292.62
10	VTEST	-5516	-292.62
11	VTEST	-5456	-292.62
12	VTEST	-5396	-292.62
13	VTEST	-5336	-292.62
14	VTEST	-5276	-292.62
15	VCOM	-5216	-292.62
16	VCOM	-5156	-292.62
17	VCOM	-5096	-292.62
18	VCOM	-5036	-292.62
19	VCOM	-4976	-292.62
20	VCOM	-4916	-292.62
21	VDDA	-4856	-292.62
22	VDDA	-4796	-292.62
23	VDDA	-4736	-292.62
24	VAGND	-4676	-292.62
25	VAGND	-4616	-292.62
26	VAGND	-4556	-292.62
27	CA1N	-4496	-292.62
28	CA1N	-4436	-292.62
29	CA1N	-4376	-292.62
30	CA1N	-4316	-292.62
31	CA1P	-4256	-292.62
32	CA1P	-4196	-292.62
33	CA1P	-4136	-292.62
34	CA1P	-4076	-292.62
35	CA1P	-4016	-292.62
36	CA1P	-3956	-292.62
37	VPV	-3896	-292.62
38	VPV	-3836	-292.62
39	VPV	-3776	-292.62
40	VPV	-3716	-292.62
41	VPV	-3656	-292.62
42	VPV	-3596	-292.62
43	CA2P	-3536	-292.62
44	CA2P	-3476	-292.62
45	CA2P	-3416	-292.62
46	CA2P	-3356	-292.62
47	CA2P	-3296	-292.62
48	CA2P	-3236	-292.62
49	CA2N	-3176	-292.62
50	CA2N	-3116	-292.62
51	CA2N	-3056	-292.62
52	CA2N	-2996	-292.62
53	CA2N	-2936	-292.62
54	CA2N	-2876	-292.62
55	VNV	-2816	-292.62
56	VNV	-2756	-292.62
57	VNV	-2696	-292.62
58	VNV	-2636	-292.62
59	VNV	-2576	-292.62
60	VNV	-2516	-292.62
61	CF1N	-2456	-292.62
62	CF1N	-2396	-292.62
63	CF1N	-2336	-292.62
64	CF1N	-2276	-292.62
65	CF1P	-2216	-292.62
66	CF1P	-2156	-292.62
67	CF1P	-2096	-292.62
68	CF1P	-2036	-292.62
69	AVDD	-1976	-292.62
70	AVDD	-1916	-292.62
71	AVDD	-1856	-292.62
72	AVDD	-1796	-292.62
73	AVDD	-1736	-292.62
74	AVDD	-1676	-292.62
75	CB2P	-1616	-292.62
76	CB2P	-1556	-292.62
77	CB2P	-1496	-292.62
78	CB2P	-1436	-292.62
79	CB2N	-1376	-292.62
80	CB2N	-1316	-292.62
81	CB2N	-1256	-292.62
82	CB2N	-1196	-292.62
83	CB1P	-1136	-292.62
84	CB1P	-1076	-292.62
85	CB1N	-1016	-292.62
86	CB1N	-956	-292.62
87	CB1N	-896	-292.62
88	CB1N	-836	-292.62
89	V1V	-776	-292.62
90	V1V	-716	-292.62
91	CE1P	-656	-292.62
92	CE1P	-596	-292.62
93	CE1N	-536	-292.62
94	CE1N	-476	-292.62
95	NAVDD	-416	-292.62
96	NAVDD	-356	-292.62
97	CE2N	-296	-292.62
98	CE2N	-236	-292.62
99	CE2P	-176	-292.62

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PAD.	Name	X	Y
100	CE2P	-116	-292.62
101	CC1N	-56	-292.62
102	CC1N	4	-292.62
103	CC1P	64	-292.62
104	CC1P	124	-292.62
105	VGH	184	-292.62
106	VGH	244	-292.62
107	CD1P	304	-292.62
108	CD1P	364	-292.62
109	CD1N	424	-292.62
110	CD1N	484	-292.62
111	VGL	544	-292.62
112	VGL	604	-292.62
113	VCCO	664	-292.62
114	VCCO	724	-292.62
115	VCCI	784	-292.62
116	VCCI	844	-292.62
117	VDDA	904	-292.62
118	VDDA	964	-292.62
119	VDDA	1024	-292.62
120	VDDA	1084	-292.62
121	VDDA	1144	-292.62
122	VDDA	1204	-292.62
123	VDDR	1264	-292.62
124	VDDR	1324	-292.62
125	VDDI	1384	-292.62
126	VDDI	1444	-292.62
127	VDDI	1504	-292.62
128	VDDI	1564	-292.62
129	VDGND	1624	-292.62
130	VDGND	1684	-292.62
131	VDGND	1744	-292.62
132	VDGND	1804	-292.62
133	VAGND	1864	-292.62
134	VAGND	1924	-292.62
135	VAGND	1984	-292.62
136	VAGND	2044	-292.62
137	VAGND	2104	-292.62
138	VAGND	2164	-292.62
139	VRGND	2224	-292.62
140	VRGND	2284	-292.62
141	VDDI	2344	-292.62
142	CSB	2404	-292.62
143	A0	2464	-292.62
144	ERD	2524	-292.62
145	RWR	2584	-292.62
146	TESTENB	2644	-292.62
147	MTest5	2704	-292.62
148	MTest4	2764	-292.62
149	MTest3	2824	-292.62
150	RSTB	2884	-292.62
151	VDGND	2944	-292.62
152	D0	3004	-292.62
153	D1	3064	-292.62
154	D2	3124	-292.62
155	D3	3184	-292.62
156	D4	3244	-292.62
157	D5	3304	-292.62
158	D6	3364	-292.62
159	D7	3424	-292.62
160	VDDI	3484	-292.62
161	ModeSel	3544	-292.62
162	IF1	3604	-292.62
163	IF0	3664	-292.62
164	CLS	3724	-292.62
165	CL	3784	-292.62
166	TE	3844	-292.62
167	EXTB	3904	-292.62
168	VPP	3964	-292.62
169	VPP	4024	-292.62
170	VPP	4084	-292.62
171	VPP	4144	-292.62
172	VPP	4204	-292.62
173	VPP	4264	-292.62
174	VDGND	4324	-292.62
175	TESTB	4384	-292.62
176	TESTA	4444	-292.62
177	MTest2	4504	-292.62
178	MTest1	4564	-292.62
179	MTest0	4624	-292.62
180	VDDI	4684	-292.62
181	VCOM	4744	-292.62
182	VCOM	4804	-292.62
183	VCOM	4864	-292.62
184	VCOM	4924	-292.62
185	VCOM	4984	-292.62
186	VCOM	5044	-292.62
187	DUMMY	5104	-292.62
188	DUMMY	5164	-292.62
189	DUMMY	5224	-292.62
190	DUMMY	5284	-292.62
191	DUMMY	5344	-292.62
192	DUMMY	5404	-292.62
193	DUMMY	5464	-292.62
194	DUMMY	5524	-292.62
195	DUMMY	5584	-292.62
196	DUMMY	5644	-292.62
197	DUMMY	5704	-292.62
198	VGL	5764	-292.62
199	VGL	5824	-292.62
200	VGL	5884	-292.62
201	VCOM	5944	-292.62

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PAD.	Name	X	Y
202	VCOM	6004	-292.62
203	VCOM	6064	-292.62
204	DUMMY	6239	35.58
205	DUMMY	6229	150.58
206	Gtest	6219	265.58
207	DUMMY	6209	35.58
208	Gtest	6199	150.58
209	G2	6189	265.58
210	G4	6179	35.58
211	G6	6169	150.58
212	G8	6159	265.58
213	G10	6149	35.58
214	G12	6139	150.58
215	G14	6129	265.58
216	G16	6119	35.58
217	G18	6109	150.58
218	G20	6099	265.58
219	G22	6089	35.58
220	G24	6079	150.58
221	G26	6069	265.58
222	G28	6059	35.58
223	G30	6049	150.58
224	G32	6039	265.58
225	G34	6029	35.58
226	G36	6019	150.58
227	G38	6009	265.58
228	G40	5999	35.58
229	G42	5989	150.58
230	G44	5979	265.58
231	G46	5969	35.58
232	G48	5959	150.58
233	G50	5949	265.58
234	G52	5939	35.58
235	G54	5929	150.58
236	G56	5919	265.58
237	G58	5909	35.58
238	G60	5899	150.58
239	G62	5889	265.58
240	G64	5879	35.58
241	G66	5869	150.58
242	G68	5859	265.58
243	G70	5849	35.58
244	G72	5839	150.58
245	G74	5829	265.58
246	G76	5819	35.58
247	G78	5809	150.58
248	G80	5799	265.58
249	G82	5789	35.58
250	G84	5779	150.58
251	G86	5769	265.58
252	G88	5759	35.58
253	G90	5749	150.58
254	G92	5739	265.58
255	G94	5729	35.58
256	G96	5719	150.58
257	G98	5709	265.58
258	G100	5699	35.58
259	G102	5689	150.58
260	G104	5679	265.58
261	G106	5669	35.58
262	G108	5659	150.58
263	G110	5649	265.58
264	G112	5639	35.58
265	G114	5629	150.58
266	G116	5619	265.58
267	G118	5609	35.58
268	G120	5599	150.58
269	G122	5589	265.58
270	G124	5579	35.58
271	G126	5569	150.58
272	G128	5559	265.58
273	G130	5549	35.58
274	G132	5539	150.58
275	G134	5529	265.58
276	G136	5519	35.58
277	G138	5509	150.58
278	G140	5499	265.58
279	G142	5489	35.58
280	G144	5479	150.58
281	G146	5469	265.58
282	G148	5459	35.58
283	G150	5449	150.58
284	G152	5439	265.58
285	G154	5429	35.58
286	G156	5419	150.58
287	G158	5409	265.58
288	G160	5399	35.58
289	G162	5389	150.58
290	G164	5379	265.58
291	G166	5369	35.58
292	G168	5359	150.58
293	G170	5349	265.58
294	G172	5339	35.58
295	G174	5329	150.58
296	G176	5319	265.58
297	G178	5309	35.58
298	G180	5299	150.58
299	G182	5289	265.58
300	G184	5279	35.58
301	G186	5269	150.58
302	G188	5259	265.58
303	G190	5249	35.58

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PAD.	Name	X	Y
304	G192	5239	150.58
305	G194	5229	265.58
306	G196	5219	35.58
307	G198	5209	150.58
308	G200	5199	265.58
309	G202	5189	35.58
310	G204	5179	150.58
311	G206	5169	265.58
312	G208	5159	35.58
313	G210	5149	150.58
314	G212	5139	265.58
315	G214	5129	35.58
316	G216	5119	150.58
317	G218	5109	265.58
318	G220	5099	35.58
319	G222	5089	150.58
320	G224	5079	265.58
321	G226	5069	35.58
322	G228	5059	150.58
323	G230	5049	265.58
324	G232	5039	35.58
325	G234	5029	150.58
326	G236	5019	265.58
327	G238	5009	35.58
328	G240	4999	150.58
329	G242	4989	265.58
330	G244	4979	35.58
331	G246	4969	150.58
332	G248	4959	265.58
333	G250	4949	35.58
334	G252	4939	150.58
335	G254	4929	265.58
336	G256	4919	35.58
337	G258	4909	150.58
338	G260	4899	265.58
339	G262	4889	35.58
340	G264	4879	150.58
341	G266	4869	265.58
342	G268	4859	35.58
343	G270	4849	150.58
344	G272	4839	265.58
345	G274	4829	35.58
346	G276	4819	150.58
347	G278	4809	265.58
348	G280	4799	35.58
349	G282	4789	150.58
350	G284	4779	265.58
351	G286	4769	35.58
352	G288	4759	150.58
353	G290	4749	265.58
354	G292	4739	35.58
355	G294	4729	150.58
356	G296	4719	265.58
357	G298	4709	35.58
358	G300	4699	150.58
359	G302	4689	265.58
360	G304	4679	35.58
361	G306	4669	150.58
362	G308	4659	265.58
363	G310	4649	35.58
364	G312	4639	150.58
365	G314	4629	265.58
366	G316	4619	35.58
367	G318	4609	150.58
368	G320	4599	265.58
369	G322	4589	35.58
370	G324	4579	150.58
371	G326	4569	265.58
372	G328	4559	35.58
373	G330	4549	150.58
374	G332	4539	265.58
375	G334	4529	35.58
376	G336	4519	150.58
377	G338	4509	265.58
378	G340	4499	35.58
379	G342	4489	150.58
380	G344	4479	265.58
381	G346	4469	35.58
382	G348	4459	150.58
383	G350	4449	265.58
384	G352	4439	35.58
385	G354	4429	150.58
386	G356	4419	265.58
387	G358	4409	35.58
388	G360	4399	150.58
389	G362	4389	265.58
390	G364	4379	35.58
391	G366	4369	150.58
392	G368	4359	265.58
393	G370	4349	35.58
394	G372	4339	150.58
395	G374	4329	265.58
396	G376	4319	35.58
397	G378	4309	150.58
398	G380	4299	265.58
399	G382	4289	35.58
400	G384	4279	150.58
401	G386	4269	265.58
402	G388	4259	35.58
403	G390	4249	150.58
404	G392	4239	265.58
405	G394	4229	35.58

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PAD.	Name	X	Y
406	G396	4219	150.58
407	G398	4209	265.58
408	G400	4199	35.58
409	G402	4189	150.58
410	G404	4179	265.58
411	G406	4169	35.58
412	G408	4159	150.58
413	G410	4149	265.58
414	RESERVED	4139	35.58
415	RESERVED	4129	150.58
416	RESERVED	4119	265.58
417	RESERVED	4109	35.58
418	RESERVED	4099	150.58
419	RESERVED	4089	265.58
420	RESERVED	4079	35.58
421	RESERVED	4069	150.58
422	RESERVED	4059	265.58
423	RESERVED	4049	35.58
424	RESERVED	4039	150.58
425	RESERVED	4029	265.58
426	RESERVED	4019	35.58
427	RESERVED	4009	150.58
428	RESERVED	3999	265.58
429	RESERVED	3989	35.58
430	RESERVED	3979	150.58
431	RESERVED	3969	265.58
432	RESERVED	3959	35.58
433	RESERVED	3949	150.58
434	RESERVED	3939	265.58
435	RESERVED	3929	35.58
436	RESERVED	3919	150.58
437	RESERVED	3909	265.58
438	RESERVED	3899	35.58
439	RESERVED	3889	150.58
440	RESERVED	3879	265.58
441	RESERVED	3869	35.58
442	RESERVED	3859	150.58
443	RESERVED	3849	265.58
444	RESERVED	3839	35.58
445	RESERVED	3829	150.58
446	RESERVED	3819	265.58
447	RESERVED	3809	35.58
448	RESERVED	3799	150.58
449	RESERVED	3789	265.58
450	DUMMY	3779	35.58
451	DUMMY	3769	150.58
452	DUMMY	3759	265.58
453	VGL	3749	35.58
454	DUMMY	3739	150.58
455	DUMMY	3699	265.58
456	RESERVED	3689	35.58
457	RESERVED	3679	150.58
458	RESERVED	3669	265.58
459	RESERVED	3659	35.58
460	RESERVED	3649	150.58
461	RESERVED	3639	265.58
462	RESERVED	3629	35.58
463	RESERVED	3619	150.58
464	RESERVED	3609	265.58
465	RESERVED	3599	35.58
466	RESERVED	3589	150.58
467	RESERVED	3579	265.58
468	RESERVED	3569	35.58
469	RESERVED	3559	150.58
470	RESERVED	3549	265.58
471	RESERVED	3539	35.58
472	RESERVED	3529	150.58
473	RESERVED	3519	265.58
474	RESERVED	3509	35.58
475	RESERVED	3499	150.58
476	RESERVED	3489	265.58
477	RESERVED	3479	35.58
478	RESERVED	3469	150.58
479	RESERVED	3459	265.58
480	RESERVED	3449	35.58
481	RESERVED	3439	150.58
482	RESERVED	3429	265.58
483	RESERVED	3419	35.58
484	RESERVED	3409	150.58
485	RESERVED	3399	265.58
486	RESERVED	3389	35.58
487	RESERVED	3379	150.58
488	RESERVED	3369	265.58
489	RESERVED	3359	35.58
490	RESERVED	3349	150.58
491	RESERVED	3339	265.58
492	RESERVED	3329	35.58
493	RESERVED	3319	150.58
494	RESERVED	3309	265.58
495	RESERVED	3299	35.58
496	RESERVED	3289	150.58
497	RESERVED	3279	265.58
498	RESERVED	3269	35.58
499	RESERVED	3259	150.58
500	RESERVED	3249	265.58
501	RESERVED	3239	35.58
502	RESERVED	3229	150.58
503	RESERVED	3219	265.58
504	RESERVED	3209	35.58
505	RESERVED	3199	150.58
506	RESERVED	3189	265.58
507	RESERVED	3179	35.58

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PAD.	Name	X	Y
508	RESERVED	3169	150.58
509	RESERVED	3159	265.58
510	RESERVED	3149	35.58
511	RESERVED	3139	150.58
512	RESERVED	3129	265.58
513	RESERVED	3119	35.58
514	RESERVED	3109	150.58
515	RESERVED	3099	265.58
516	RESERVED	3089	35.58
517	RESERVED	3079	150.58
518	RESERVED	3069	265.58
519	RESERVED	3059	35.58
520	RESERVED	3049	150.58
521	RESERVED	3039	265.58
522	RESERVED	3029	35.58
523	RESERVED	3019	150.58
524	RESERVED	3009	265.58
525	RESERVED	2999	35.58
526	RESERVED	2989	150.58
527	RESERVED	2979	265.58
528	RESERVED	2969	35.58
529	RESERVED	2959	150.58
530	RESERVED	2949	265.58
531	RESERVED	2939	35.58
532	RESERVED	2929	150.58
533	RESERVED	2919	265.58
534	RESERVED	2909	35.58
535	RESERVED	2899	150.58
536	RESERVED	2889	265.58
537	RESERVED	2879	35.58
538	RESERVED	2869	150.58
539	RESERVED	2859	265.58
540	RESERVED	2849	35.58
541	RESERVED	2839	150.58
542	RESERVED	2829	265.58
543	RESERVED	2819	35.58
544	RESERVED	2809	150.58
545	RESERVED	2799	265.58
546	RESERVED	2789	35.58
547	RESERVED	2779	150.58
548	RESERVED	2769	265.58
549	RESERVED	2759	35.58
550	RESERVED	2749	150.58
551	RESERVED	2739	265.58
552	RESERVED	2729	35.58
553	RESERVED	2719	150.58
554	RESERVED	2709	265.58
555	RESERVED	2699	35.58
556	RESERVED	2689	150.58
557	RESERVED	2679	265.58
558	RESERVED	2669	35.58
559	RESERVED	2659	150.58
560	RESERVED	2649	265.58
561	RESERVED	2639	35.58
562	RESERVED	2629	150.58
563	RESERVED	2619	265.58
564	RESERVED	2609	35.58
565	RESERVED	2599	150.58
566	RESERVED	2589	265.58
567	RESERVED	2579	35.58
568	RESERVED	2569	150.58
569	RESERVED	2559	265.58
570	RESERVED	2549	35.58
571	RESERVED	2539	150.58
572	RESERVED	2529	265.58
573	RESERVED	2519	35.58
574	RESERVED	2509	150.58
575	RESERVED	2499	265.58
576	RESERVED	2489	35.58
577	RESERVED	2479	150.58
578	RESERVED	2469	265.58
579	RESERVED	2459	35.58
580	RESERVED	2449	150.58
581	RESERVED	2439	265.58
582	RESERVED	2429	35.58
583	RESERVED	2419	150.58
584	RESERVED	2409	265.58
585	RESERVED	2399	35.58
586	RESERVED	2389	150.58
587	RESERVED	2379	265.58
588	RESERVED	2369	35.58
589	RESERVED	2359	150.58
590	RESERVED	2349	265.58
591	RESERVED	2339	35.58
592	RESERVED	2329	150.58
593	RESERVED	2319	265.58
594	RESERVED	2309	35.58
595	RESERVED	2299	150.58
596	RESERVED	2289	265.58
597	RESERVED	2279	35.58
598	RESERVED	2269	150.58
599	RESERVED	2259	265.58
600	RESERVED	2249	35.58
601	RESERVED	2239	150.58
602	RESERVED	2229	265.58
603	RESERVED	2219	35.58
604	RESERVED	2209	150.58
605	RESERVED	2199	265.58
606	RESERVED	2189	35.58
607	RESERVED	2179	150.58
608	RESERVED	2169	265.58
609	RESERVED	2159	35.58

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PAD.	Name	X	Y
610	RESERVED	2149	150.58
611	RESERVED	2139	265.58
612	S408	2129	35.58
613	S407	2119	150.58
614	S406	2109	265.58
615	S405	2099	35.58
616	S404	2089	150.58
617	S403	2079	265.58
618	S402	2069	35.58
619	S401	2059	150.58
620	S400	2049	265.58
621	S399	2039	35.58
622	S398	2029	150.58
623	S397	2019	265.58
624	S396	2009	35.58
625	S395	1999	150.58
626	S394	1989	265.58
627	S393	1979	35.58
628	S392	1969	150.58
629	S391	1959	265.58
630	S390	1949	35.58
631	S389	1939	150.58
632	S388	1929	265.58
633	S387	1919	35.58
634	S386	1909	150.58
635	S385	1899	265.58
636	S384	1889	35.58
637	S383	1879	150.58
638	S382	1869	265.58
639	S381	1859	35.58
640	S380	1849	150.58
641	S379	1839	265.58
642	S378	1829	35.58
643	S377	1819	150.58
644	S376	1809	265.58
645	S375	1799	35.58
646	S374	1789	150.58
647	S373	1779	265.58
648	S372	1769	35.58
649	S371	1759	150.58
650	S370	1749	265.58
651	S369	1739	35.58
652	S368	1729	150.58
653	S367	1719	265.58
654	S366	1709	35.58
655	S365	1699	150.58
656	S364	1689	265.58
657	S363	1679	35.58
658	S362	1669	150.58
659	S361	1659	265.58
660	S360	1649	35.58
661	S359	1639	150.58
662	S358	1629	265.58
663	S357	1619	35.58
664	S356	1609	150.58
665	S355	1599	265.58
666	S354	1589	35.58
667	S353	1579	150.58
668	S352	1569	265.58
669	S351	1559	35.58
670	S350	1549	150.58
671	S349	1539	265.58
672	S348	1529	35.58
673	S347	1519	150.58
674	S346	1509	265.58
675	S345	1499	35.58
676	S344	1489	150.58
677	S343	1479	265.58
678	S342	1469	35.58
679	S341	1459	150.58
680	S340	1449	265.58
681	S339	1439	35.58
682	S338	1429	150.58
683	S337	1419	265.58
684	S336	1409	35.58
685	S335	1399	150.58
686	S334	1389	265.58
687	S333	1379	35.58
688	S332	1369	150.58
689	S331	1359	265.58
690	S330	1349	35.58
691	S329	1339	150.58
692	S328	1329	265.58
693	S327	1319	35.58
694	S326	1309	150.58
695	S325	1299	265.58
696	S324	1289	35.58
697	S323	1279	150.58
698	S322	1269	265.58
699	S321	1259	35.58
700	S320	1249	150.58
701	S319	1239	265.58
702	S318	1229	35.58
703	S317	1219	150.58
704	S316	1209	265.58
705	S315	1199	35.58
706	S314	1189	150.58
707	S313	1179	265.58
708	S312	1169	35.58
709	S311	1159	150.58
710	S310	1149	265.58
711	S309	1139	35.58

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PAD.	Name	X	Y
712	S308	1129	150.58
713	S307	1119	265.58
714	S306	1109	35.58
715	S305	1099	150.58
716	S304	1089	265.58
717	S303	1079	35.58
718	S302	1069	150.58
719	S301	1059	265.58
720	S300	1049	35.58
721	S299	1039	150.58
722	S298	1029	265.58
723	S297	1019	35.58
724	S296	1009	150.58
725	S295	999	265.58
726	S294	989	35.58
727	S293	979	150.58
728	S292	969	265.58
729	S291	959	35.58
730	S290	949	150.58
731	S289	939	265.58
732	S288	929	35.58
733	S287	919	150.58
734	S286	909	265.58
735	S285	899	35.58
736	S284	889	150.58
737	S283	879	265.58
738	S282	869	35.58
739	S281	859	150.58
740	S280	849	265.58
741	S279	839	35.58
742	S278	829	150.58
743	S277	819	265.58
744	S276	809	35.58
745	S275	799	150.58
746	S274	789	265.58
747	S273	779	35.58
748	S272	769	150.58
749	S271	759	265.58
750	S270	749	35.58
751	S269	739	150.58
752	S268	729	265.58
753	S267	719	35.58
754	S266	709	150.58
755	S265	699	265.58
756	S264	689	35.58
757	S263	679	150.58
758	S262	669	265.58
759	S261	659	35.58
760	S260	649	150.58
761	S259	639	265.58
762	S258	629	35.58
763	S257	619	150.58
764	S256	609	265.58
765	S255	599	35.58
766	S254	589	150.58
767	S253	579	265.58
768	S252	569	35.58
769	S251	559	150.58
770	S250	549	265.58
771	S249	539	35.58
772	S248	529	150.58
773	S247	519	265.58
774	S246	509	35.58
775	S245	499	150.58
776	S244	489	265.58
777	S243	479	35.58
778	S242	469	150.58
779	S241	459	265.58
780	S240	449	35.58
781	S239	439	150.58
782	S238	429	265.58
783	S237	419	35.58
784	S236	409	150.58
785	S235	399	265.58
786	S234	389	35.58
787	S233	379	150.58
788	S232	369	265.58
789	S231	359	35.58
790	S230	349	150.58
791	S229	339	265.58
792	S228	329	35.58
793	S227	319	150.58
794	S226	309	265.58
795	S225	299	35.58
796	S224	289	150.58
797	S223	279	265.58
798	S222	269	35.58
799	S221	259	150.58
800	S220	249	265.58
801	S219	239	35.58
802	S218	229	150.58
803	S217	219	265.58
804	S216	209	35.58
805	S215	199	150.58
806	S214	189	265.58
807	S213	179	35.58
808	S212	169	150.58
809	S211	159	265.58
810	S210	149	35.58
811	S209	139	150.58
812	S208	129	265.58
813	S207	119	35.58

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PAD.	Name	X	Y
814	S206	109	150.58
815	S205	99	265.58
816	DUMMY	89	35.58
817	DUMMY	79	150.58
818	GTest	69	265.58
819	DUMMY	59	35.58
820	GTest	49	150.58
821	S204	-99	35.58
822	S203	-109	150.58
823	S202	-119	265.58
824	S201	-129	35.58
825	S200	-139	150.58
826	S199	-149	265.58
827	S198	-159	35.58
828	S197	-169	150.58
829	S196	-179	265.58
830	S195	-189	35.58
831	S194	-199	150.58
832	S193	-209	265.58
833	S192	-219	35.58
834	S191	-229	150.58
835	S190	-239	265.58
836	S189	-249	35.58
837	S188	-259	150.58
838	S187	-269	265.58
839	S186	-279	35.58
840	S185	-289	150.58
841	S184	-299	265.58
842	S183	-309	35.58
843	S182	-319	150.58
844	S181	-329	265.58
845	S180	-339	35.58
846	S179	-349	150.58
847	S178	-359	265.58
848	S177	-369	35.58
849	S176	-379	150.58
850	S175	-389	265.58
851	S174	-399	35.58
852	S173	-409	150.58
853	S172	-419	265.58
854	S171	-429	35.58
855	S170	-439	150.58
856	S169	-449	265.58
857	S168	-459	35.58
858	S167	-469	150.58
859	S166	-479	265.58
860	S165	-489	35.58
861	S164	-499	150.58
862	S163	-509	265.58
863	S162	-519	35.58
864	S161	-529	150.58
865	S160	-539	265.58
866	S159	-549	35.58
867	S158	-559	150.58
868	S157	-569	265.58
869	S156	-579	35.58
870	S155	-589	150.58
871	S154	-599	265.58
872	S153	-609	35.58
873	S152	-619	150.58
874	S151	-629	265.58
875	S150	-639	35.58
876	S149	-649	150.58
877	S148	-659	265.58
878	S147	-669	35.58
879	S146	-679	150.58
880	S145	-689	265.58
881	S144	-699	35.58
882	S143	-709	150.58
883	S142	-719	265.58
884	S141	-729	35.58
885	S140	-739	150.58
886	S139	-749	265.58
887	S138	-759	35.58
888	S137	-769	150.58
889	S136	-779	265.58
890	S135	-789	35.58
891	S134	-799	150.58
892	S133	-809	265.58
893	S132	-819	35.58
894	S131	-829	150.58
895	S130	-839	265.58
896	S129	-849	35.58
897	S128	-859	150.58
898	S127	-869	265.58
899	S126	-879	35.58
900	S125	-889	150.58
901	S124	-899	265.58
902	S123	-909	35.58
903	S122	-919	150.58
904	S121	-929	265.58
905	S120	-939	35.58
906	S119	-949	150.58
907	S118	-959	265.58
908	S117	-969	35.58
909	S116	-979	150.58
910	S115	-989	265.58
911	S114	-999	35.58
912	S113	-1009	150.58
913	S112	-1019	265.58
914	S111	-1029	35.58
915	S110	-1039	150.58

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PAD.	Name	X	Y
916	S109	-1049	265.58
917	S108	-1059	35.58
918	S107	-1069	150.58
919	S106	-1079	265.58
920	S105	-1089	35.58
921	S104	-1099	150.58
922	S103	-1109	265.58
923	S102	-1119	35.58
924	S101	-1129	150.58
925	S100	-1139	265.58
926	S99	-1149	35.58
927	S98	-1159	150.58
928	S97	-1169	265.58
929	S96	-1179	35.58
930	S95	-1189	150.58
931	S94	-1199	265.58
932	S93	-1209	35.58
933	S92	-1219	150.58
934	S91	-1229	265.58
935	S90	-1239	35.58
936	S89	-1249	150.58
937	S88	-1259	265.58
938	S87	-1269	35.58
939	S86	-1279	150.58
940	S85	-1289	265.58
941	S84	-1299	35.58
942	S83	-1309	150.58
943	S82	-1319	265.58
944	S81	-1329	35.58
945	S80	-1339	150.58
946	S79	-1349	265.58
947	S78	-1359	35.58
948	S77	-1369	150.58
949	S76	-1379	265.58
950	S75	-1389	35.58
951	S74	-1399	150.58
952	S73	-1409	265.58
953	S72	-1419	35.58
954	S71	-1429	150.58
955	S70	-1439	265.58
956	S69	-1449	35.58
957	S68	-1459	150.58
958	S67	-1469	265.58
959	S66	-1479	35.58
960	S65	-1489	150.58
961	S64	-1499	265.58
962	S63	-1509	35.58
963	S62	-1519	150.58
964	S61	-1529	265.58
965	S60	-1539	35.58
966	S59	-1549	150.58
967	S58	-1559	265.58
968	S57	-1569	35.58
969	S56	-1579	150.58
970	S55	-1589	265.58
971	S54	-1599	35.58
972	S53	-1609	150.58
973	S52	-1619	265.58
974	S51	-1629	35.58
975	S50	-1639	150.58
976	S49	-1649	265.58
977	S48	-1659	35.58
978	S47	-1669	150.58
979	S46	-1679	265.58
980	S45	-1689	35.58
981	S44	-1699	150.58
982	S43	-1709	265.58
983	S42	-1719	35.58
984	S41	-1729	150.58
985	S40	-1739	265.58
986	S39	-1749	35.58
987	S38	-1759	150.58
988	S37	-1769	265.58
989	S36	-1779	35.58
990	S35	-1789	150.58
991	S34	-1799	265.58
992	S33	-1809	35.58
993	S32	-1819	150.58
994	S31	-1829	265.58
995	S30	-1839	35.58
996	S29	-1849	150.58
997	S28	-1859	265.58
998	S27	-1869	35.58
999	S26	-1879	150.58
1000	S25	-1889	265.58
1001	S24	-1899	35.58
1002	S23	-1909	150.58
1003	S22	-1919	265.58
1004	S21	-1929	35.58
1005	S20	-1939	150.58
1006	S19	-1949	265.58
1007	S18	-1959	35.58
1008	S17	-1969	150.58
1009	S16	-1979	265.58
1010	S15	-1989	35.58
1011	S14	-1999	150.58
1012	S13	-2009	265.58
1013	S12	-2019	35.58
1014	S11	-2029	150.58
1015	S10	-2039	265.58
1016	S9	-2049	35.58
1017	S8	-2059	150.58

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PAD.	Name	X	Y
1018	S7	-2069	265.58
1019	S6	-2079	35.58
1020	S5	-2089	150.58
1021	S4	-2099	265.58
1022	S3	-2109	35.58
1023	S2	-2119	150.58
1024	S1	-2129	265.58
1025	RESERVED	-2139	35.58
1026	RESERVED	-2149	150.58
1027	RESERVED	-2159	265.58
1028	RESERVED	-2169	35.58
1029	RESERVED	-2179	150.58
1030	RESERVED	-2189	265.58
1031	RESERVED	-2199	35.58
1032	RESERVED	-2209	150.58
1033	RESERVED	-2219	265.58
1034	RESERVED	-2229	35.58
1035	RESERVED	-2239	150.58
1036	RESERVED	-2249	265.58
1037	RESERVED	-2259	35.58
1038	RESERVED	-2269	150.58
1039	RESERVED	-2279	265.58
1040	RESERVED	-2289	35.58
1041	RESERVED	-2299	150.58
1042	RESERVED	-2309	265.58
1043	RESERVED	-2319	35.58
1044	RESERVED	-2329	150.58
1045	RESERVED	-2339	265.58
1046	RESERVED	-2349	35.58
1047	RESERVED	-2359	150.58
1048	RESERVED	-2369	265.58
1049	RESERVED	-2379	35.58
1050	RESERVED	-2389	150.58
1051	RESERVED	-2399	265.58
1052	RESERVED	-2409	35.58
1053	RESERVED	-2419	150.58
1054	RESERVED	-2429	265.58
1055	RESERVED	-2439	35.58
1056	RESERVED	-2449	150.58
1057	RESERVED	-2459	265.58
1058	RESERVED	-2469	35.58
1059	RESERVED	-2479	150.58
1060	RESERVED	-2489	265.58
1061	RESERVED	-2499	35.58
1062	RESERVED	-2509	150.58
1063	RESERVED	-2519	265.58
1064	RESERVED	-2529	35.58
1065	RESERVED	-2539	150.58
1066	RESERVED	-2549	265.58
1067	RESERVED	-2559	35.58
1068	RESERVED	-2569	150.58
1069	RESERVED	-2579	265.58
1070	RESERVED	-2589	35.58
1071	RESERVED	-2599	150.58
1072	RESERVED	-2609	265.58
1073	RESERVED	-2619	35.58
1074	RESERVED	-2629	150.58
1075	RESERVED	-2639	265.58
1076	RESERVED	-2649	35.58
1077	RESERVED	-2659	150.58
1078	RESERVED	-2669	265.58
1079	RESERVED	-2679	35.58
1080	RESERVED	-2689	150.58
1081	RESERVED	-2699	265.58
1082	RESERVED	-2709	35.58
1083	RESERVED	-2719	150.58
1084	RESERVED	-2729	265.58
1085	RESERVED	-2739	35.58
1086	RESERVED	-2749	150.58
1087	RESERVED	-2759	265.58
1088	RESERVED	-2769	35.58
1089	RESERVED	-2779	150.58
1090	RESERVED	-2789	265.58
1091	RESERVED	-2799	35.58
1092	RESERVED	-2809	150.58
1093	RESERVED	-2819	265.58
1094	RESERVED	-2829	35.58
1095	RESERVED	-2839	150.58
1096	RESERVED	-2849	265.58
1097	RESERVED	-2859	35.58
1098	RESERVED	-2869	150.58
1099	RESERVED	-2879	265.58
1100	RESERVED	-2889	35.58
1101	RESERVED	-2899	150.58
1102	RESERVED	-2909	265.58
1103	RESERVED	-2919	35.58
1104	RESERVED	-2929	150.58
1105	RESERVED	-2939	265.58
1106	RESERVED	-2949	35.58
1107	RESERVED	-2959	150.58
1108	RESERVED	-2969	265.58
1109	RESERVED	-2979	35.58
1110	RESERVED	-2989	150.58
1111	RESERVED	-2999	265.58
1112	RESERVED	-3009	35.58
1113	RESERVED	-3019	150.58
1114	RESERVED	-3029	265.58
1115	RESERVED	-3039	35.58
1116	RESERVED	-3049	150.58
1117	RESERVED	-3059	265.58
1118	RESERVED	-3069	35.58
1119	RESERVED	-3079	150.58

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PAD.	Name	X	Y
1120	RESERVED	-3089	265.58
1121	RESERVED	-3099	35.58
1122	RESERVED	-3109	150.58
1123	RESERVED	-3119	265.58
1124	RESERVED	-3129	35.58
1125	RESERVED	-3139	150.58
1126	RESERVED	-3149	265.58
1127	RESERVED	-3159	35.58
1128	RESERVED	-3169	150.58
1129	RESERVED	-3179	265.58
1130	RESERVED	-3189	35.58
1131	RESERVED	-3199	150.58
1132	RESERVED	-3209	265.58
1133	RESERVED	-3219	35.58
1134	RESERVED	-3229	150.58
1135	RESERVED	-3239	265.58
1136	RESERVED	-3249	35.58
1137	RESERVED	-3259	150.58
1138	RESERVED	-3269	265.58
1139	RESERVED	-3279	35.58
1140	RESERVED	-3289	150.58
1141	RESERVED	-3299	265.58
1142	RESERVED	-3309	35.58
1143	RESERVED	-3319	150.58
1144	RESERVED	-3329	265.58
1145	RESERVED	-3339	35.58
1146	RESERVED	-3349	150.58
1147	RESERVED	-3359	265.58
1148	RESERVED	-3369	35.58
1149	RESERVED	-3379	150.58
1150	RESERVED	-3389	265.58
1151	RESERVED	-3399	35.58
1152	RESERVED	-3409	150.58
1153	RESERVED	-3419	265.58
1154	RESERVED	-3429	35.58
1155	RESERVED	-3439	150.58
1156	RESERVED	-3449	265.58
1157	RESERVED	-3459	35.58
1158	RESERVED	-3469	150.58
1159	RESERVED	-3479	265.58
1160	RESERVED	-3489	35.58
1161	RESERVED	-3499	150.58
1162	RESERVED	-3509	265.58
1163	RESERVED	-3519	35.58
1164	RESERVED	-3529	150.58
1165	RESERVED	-3539	265.58
1166	RESERVED	-3549	35.58
1167	RESERVED	-3559	150.58
1168	RESERVED	-3569	265.58
1169	RESERVED	-3579	35.58
1170	RESERVED	-3589	150.58
1171	RESERVED	-3599	265.58
1172	RESERVED	-3609	35.58
1173	RESERVED	-3619	150.58
1174	RESERVED	-3629	265.58
1175	RESERVED	-3639	35.58
1176	RESERVED	-3649	150.58
1177	RESERVED	-3659	265.58
1178	RESERVED	-3669	35.58
1179	RESERVED	-3679	150.58
1180	RESERVED	-3689	265.58
1181	VGL	-3739	35.58
1182	DUMMY	-3749	150.58
1183	DUMMY	-3759	265.58
1184	DUMMY	-3769	35.58
1185	DUMMY	-3779	150.58
1186	RESERVED	-3789	265.58
1187	RESERVED	-3799	35.58
1188	RESERVED	-3809	150.58
1189	RESERVED	-3819	265.58
1190	RESERVED	-3829	35.58
1191	RESERVED	-3839	150.58
1192	RESERVED	-3849	265.58
1193	RESERVED	-3859	35.58
1194	RESERVED	-3869	150.58
1195	RESERVED	-3879	265.58
1196	RESERVED	-3889	35.58
1197	RESERVED	-3899	150.58
1198	RESERVED	-3909	265.58
1199	RESERVED	-3919	35.58
1200	RESERVED	-3929	150.58
1201	RESERVED	-3939	265.58
1202	RESERVED	-3949	35.58
1203	RESERVED	-3959	150.58
1204	RESERVED	-3969	265.58
1205	RESERVED	-3979	35.58
1206	RESERVED	-3989	150.58
1207	RESERVED	-3999	265.58
1208	RESERVED	-4009	35.58
1209	RESERVED	-4019	150.58
1210	RESERVED	-4029	265.58
1211	RESERVED	-4039	35.58
1212	RESERVED	-4049	150.58
1213	RESERVED	-4059	265.58
1214	RESERVED	-4069	35.58
1215	RESERVED	-4079	150.58
1216	RESERVED	-4089	265.58
1217	RESERVED	-4099	35.58
1218	RESERVED	-4109	150.58
1219	RESERVED	-4119	265.58
1220	RESERVED	-4129	35.58
1221	RESERVED	-4139	150.58

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PAD.	Name	X	Y
1222	G409	-4149	265.58
1223	G407	-4159	35.58
1224	G405	-4169	150.58
1225	G403	-4179	265.58
1226	G401	-4189	35.58
1227	G399	-4199	150.58
1228	G397	-4209	265.58
1229	G395	-4219	35.58
1230	G393	-4229	150.58
1231	G391	-4239	265.58
1232	G389	-4249	35.58
1233	G387	-4259	150.58
1234	G385	-4269	265.58
1235	G383	-4279	35.58
1236	G381	-4289	150.58
1237	G379	-4299	265.58
1238	G377	-4309	35.58
1239	G375	-4319	150.58
1240	G373	-4329	265.58
1241	G371	-4339	35.58
1242	G369	-4349	150.58
1243	G367	-4359	265.58
1244	G365	-4369	35.58
1245	G363	-4379	150.58
1246	G361	-4389	265.58
1247	G359	-4399	35.58
1248	G357	-4409	150.58
1249	G355	-4419	265.58
1250	G353	-4429	35.58
1251	G351	-4439	150.58
1252	G349	-4449	265.58
1253	G347	-4459	35.58
1254	G345	-4469	150.58
1255	G343	-4479	265.58
1256	G341	-4489	35.58
1257	G339	-4499	150.58
1258	G337	-4509	265.58
1259	G335	-4519	35.58
1260	G333	-4529	150.58
1261	G331	-4539	265.58
1262	G329	-4549	35.58
1263	G327	-4559	150.58
1264	G325	-4569	265.58
1265	G323	-4579	35.58
1266	G321	-4589	150.58
1267	G319	-4599	265.58
1268	G317	-4609	35.58
1269	G315	-4619	150.58
1270	G313	-4629	265.58
1271	G311	-4639	35.58
1272	G309	-4649	150.58
1273	G307	-4659	265.58
1274	G305	-4669	35.58
1275	G303	-4679	150.58
1276	G301	-4689	265.58
1277	G299	-4699	35.58
1278	G297	-4709	150.58
1279	G295	-4719	265.58
1280	G293	-4729	35.58
1281	G291	-4739	150.58
1282	G289	-4749	265.58
1283	G287	-4759	35.58
1284	G285	-4769	150.58
1285	G283	-4779	265.58
1286	G281	-4789	35.58
1287	G279	-4799	150.58
1288	G277	-4809	265.58
1289	G275	-4819	35.58
1290	G273	-4829	150.58
1291	G271	-4839	265.58
1292	G269	-4849	35.58
1293	G267	-4859	150.58
1294	G265	-4869	265.58
1295	G263	-4879	35.58
1296	G261	-4889	150.58
1297	G259	-4899	265.58
1298	G257	-4909	35.58
1299	G255	-4919	150.58
1300	G253	-4929	265.58
1301	G251	-4939	35.58
1302	G249	-4949	150.58
1303	G247	-4959	265.58
1304	G245	-4969	35.58
1305	G243	-4979	150.58
1306	G241	-4989	265.58
1307	G239	-4999	35.58
1308	G237	-5009	150.58
1309	G235	-5019	265.58
1310	G233	-5029	35.58
1311	G231	-5039	150.58
1312	G229	-5049	265.58
1313	G227	-5059	35.58
1314	G225	-5069	150.58
1315	G223	-5079	265.58
1316	G221	-5089	35.58
1317	G219	-5099	150.58
1318	G217	-5109	265.58
1319	G215	-5119	35.58
1320	G213	-5129	150.58
1321	G211	-5139	265.58
1322	G209	-5149	35.58
1323	G207	-5159	150.58

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PAD.	Name	X	Y
1324	G205	-5169	265.58
1325	G203	-5179	35.58
1326	G201	-5189	150.58
1327	G199	-5199	265.58
1328	G197	-5209	35.58
1329	G195	-5219	150.58
1330	G193	-5229	265.58
1331	G191	-5239	35.58
1332	G189	-5249	150.58
1333	G187	-5259	265.58
1334	G185	-5269	35.58
1335	G183	-5279	150.58
1336	G181	-5289	265.58
1337	G179	-5299	35.58
1338	G177	-5309	150.58
1339	G175	-5319	265.58
1340	G173	-5329	35.58
1341	G171	-5339	150.58
1342	G169	-5349	265.58
1343	G167	-5359	35.58
1344	G165	-5369	150.58
1345	G163	-5379	265.58
1346	G161	-5389	35.58
1347	G159	-5399	150.58
1348	G157	-5409	265.58
1349	G155	-5419	35.58
1350	G153	-5429	150.58
1351	G151	-5439	265.58
1352	G149	-5449	35.58
1353	G147	-5459	150.58
1354	G145	-5469	265.58
1355	G143	-5479	35.58
1356	G141	-5489	150.58
1357	G139	-5499	265.58
1358	G137	-5509	35.58
1359	G135	-5519	150.58
1360	G133	-5529	265.58
1361	G131	-5539	35.58
1362	G129	-5549	150.58
1363	G127	-5559	265.58
1364	G125	-5569	35.58
1365	G123	-5579	150.58
1366	G121	-5589	265.58
1367	G119	-5599	35.58
1368	G117	-5609	150.58
1369	G115	-5619	265.58
1370	G113	-5629	35.58
1371	G111	-5639	150.58
1372	G109	-5649	265.58
1373	G107	-5659	35.58
1374	G105	-5669	150.58
1375	G103	-5679	265.58
1376	G101	-5689	35.58
1377	G99	-5699	150.58
1378	G97	-5709	265.58
1379	G95	-5719	35.58
1380	G93	-5729	150.58
1381	G91	-5739	265.58
1382	G89	-5749	35.58
1383	G87	-5759	150.58
1384	G85	-5769	265.58
1385	G83	-5779	35.58
1386	G81	-5789	150.58
1387	G79	-5799	265.58
1388	G77	-5809	35.58
1389	G75	-5819	150.58
1390	G73	-5829	265.58
1391	G71	-5839	35.58
1392	G69	-5849	150.58
1393	G67	-5859	265.58
1394	G65	-5869	35.58
1395	G63	-5879	150.58
1396	G61	-5889	265.58
1397	G59	-5899	35.58
1398	G57	-5909	150.58
1399	G55	-5919	265.58
1400	G53	-5929	35.58
1401	G51	-5939	150.58
1402	G49	-5949	265.58
1403	G47	-5959	35.58
1404	G45	-5969	150.58
1405	G43	-5979	265.58
1406	G41	-5989	35.58
1407	G39	-5999	150.58
1408	G37	-6009	265.58
1409	G35	-6019	35.58
1410	G33	-6029	150.58
1411	G31	-6039	265.58
1412	G29	-6049	35.58
1413	G27	-6059	150.58
1414	G25	-6069	265.58
1415	G23	-6079	35.58
1416	G21	-6089	150.58
1417	G19	-6099	265.58
1418	G17	-6109	35.58
1419	G15	-6119	150.58
1420	G13	-6129	265.58
1421	G11	-6139	35.58
1422	G9	-6149	150.58
1423	G7	-6159	265.58
1424	G5	-6169	35.58
1425	G3	-6179	150.58

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PAD.	Name	X	Y
1426	G1	-6189	265.58
1427	DUMMY	-6199	35.58
1428	GTest	-6209	150.58

PAD.	Name	X	Y
1429	GTest	-6219	265.58
1430	DUMMY	-6229	35.58
1431	DUMMY	-6239	150.58

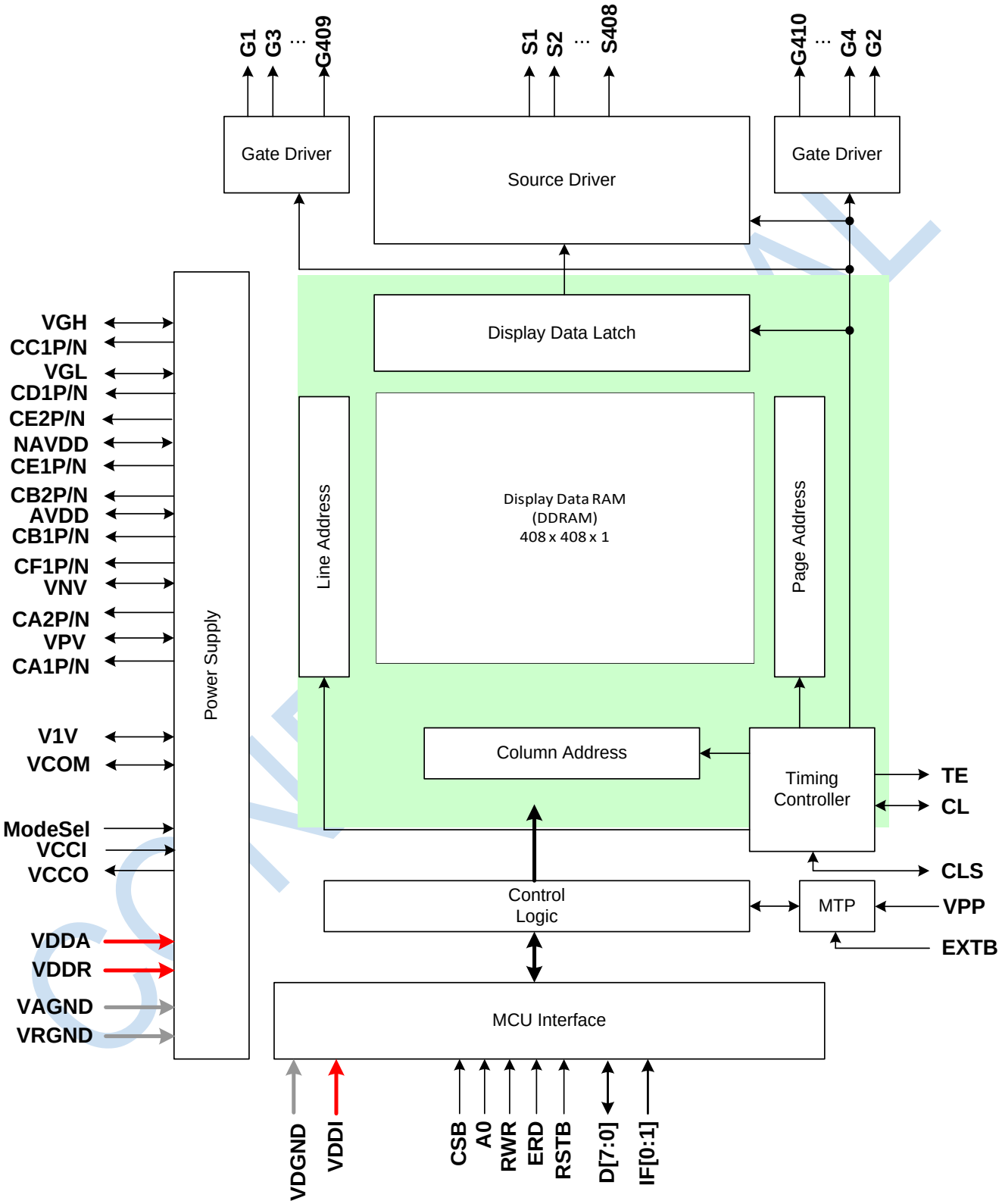
PAD.	Name	X	Y
Mark	ALIGN_L	-6146	-275
Mark	ALIGN_R	6154	-275

Unit: um

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5 BLOCK DIAGRAM



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6 PIN DESCRIPTION

6.1 Power supply Pin

Name	Type	Description
VDDI	Power	Power Supply (Digital)
VDDA	Power	Power Supply (Analog)
VDDR	Power	Power Supply (Reference)
VCCI/VCCO	Power	VCCI is the power source of digital circuits. VCCO is the VCC output. VCCI and VCCO should be connected together.
ModeSel	I	Power Mode Selection. ModeSel = “VAGND”: 1.8V mode (VDDI=VDDA=1.8V) ModeSel = “VDDA”: Normal mode (VDDI=1.8V~3.6V ; VDDA=2.55~3.6V)
VDGND	Power	Ground (Digital)
VAGND	Power	Ground (Analog)
VRGND	Power	Ground (Reference)

6.2 Digital I/O

<table>
<tr><th>Name</th><th>Type</th><th>Description</th></tr>
<tr><td>RSTB</td><td>I</td><td>Reset input pin. When RSTB is “L”, internal initialization procedure is executed.</td></tr>
<tr><td>IF[1:0]</td><td>I</td><td>These pins select interface operation mode.
<table>
<tr><th>IF1</th><th>IF0</th><th>MPU interface type</th></tr>
<tr><td>L</td><td>L</td><td>8-bit 8080 parallel interface</td></tr>
<tr><td>L</td><td>H</td><td>Single 4-line serial interface</td></tr>
<tr><td>H</td><td>L</td><td>Dual 4-line serial interface</td></tr>
<tr><td>H</td><td>H</td><td>3-line serial interface</td></tr>
</table></td></tr>
<tr><td>CSB</td><td>I</td><td>Chip select input pin.
CSB=“L”: This chip is selected and the MPU interface is active.
CSB=“H”: This chip is not selected and the MPU interface is disabled (D[7:0] are high impedance).</td></tr>
<tr><td>A0</td><td>I</td><td>When using 8080, Single 4SPI and Dual 4SPI mode.
It determines whether the access is related to data or command.
A0 = "H": Indicates that D[7:0] are display data;
A0 = "L": Indicates that D[7:0] are control data.
When using 3SPI.
There is no A0 pin in 3-Line. A0 should be fixed to “H” by VDDI.</td></tr>
</table>

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Name	Type	Description
RWR	I	**When using 8080** Write enable in 8080 parallel interface. **When using Dual 4SPI** Second data lane in Dual 4SPI interface. This pin is not used in 3-Line and Single 4-Line SPI. RWR should be connected to VDDI
ERD	I	Read enable in 8080 interface. This pin is not used in serial interfaces and should be connected to VDDI.
D[7:0]	I/O	**When using 8-bit parallel interface: 8080 mode** 8 bit bi-directional data bus. Connect to the data bus of 8-bit microprocessor. When CSB is “H”, D[7:0] are high impedance.
	I/O	**When using serial interface: Single/Dual 4-line SPI, 3-line SPI** D[7:4] : fix to “H” by VDDI. D[3:2] : serial output data (SDA_OUT). D[1] : serial input data (SDA_IN). D[0] : serial input clock (SCLK). **D1 to D3 must be connected together (SDA)** When CSB is “H”, D[7:0] are high impedance.
TE	O	Tearing effect signal is used to synchronize MCU to frame memory writing. If not used, please let this pin open.

Note:

1. After VDDI is turned ON, all MPU interface pins should not be left OPEN.

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6.3 Clock System Input

Name	Type	Description
CLS	I	Clock source selection pin. CLS=”H”: enable internal clock. CLS=”L”: disable internal clock and use external clock.
CL	I/O	For external clock. If CLS=”H”: this pin is open. If CLS=”L”: this pin is the input of oscillator.

6.4 Driving Output Pin

Name	Type	Description
S1 to S408	O	Source driver output pins.
G1 to G410	O	Gate driver output pins.
VPV VNV V1V AVDD NAVDD	O	Power output pin for analog circuit.
VGH	O	Power output (positive) pin for gate driver.
VGL	O	Power output (negative) pin for gate driver.
VCOM	O	COM outputs. A power supply for the TFT-LCD common electrode. **Must be connected to GND of FPC.**
CA1P/N CA2P/N	O	Capacitor connecting pins for step-up circuit. (for VPV)
CB1P/N CB2P/N	O	Capacitor connecting pins for step-up circuit. (for AVDD)
CC1P/N	O	Capacitor connecting pins for step-up circuit. (for VGH)
CD1P/N	O	Capacitor connecting pins for step-up circuit. (for VGL)
CE1P/N CE2P/N	O	Capacitor connecting pins for step-up circuit. (for NAVDD)
CF1P/N	O	Capacitor connecting pins for step-up circuit. (for VNV)

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6.5 MTP Pin

Name	Type	Description
VPP	Power	The programming power supply of the built-in NVM. Apply external power 7.5V here when programming (> 8mA for successful programming). If not used, left this pin open.
EXTB	I	EXTB="L": Enable the extension operation mode. When programming MTP, connect EXTB to VDGND externally. This pin has an internal pull-high resistor. Please leave this pin OPEN after special operation.

6.6 Others

Name	Type	Description
VTEST TESTENB TESTA TESTB GTest DUMMY RESERVED MTest[5:0]	O	Reserved for test only. Please leave these pins open.

6.7 Recommend Resistance

Item	Pin name	Resistance
Powers	VDDA, VAGND	< 3 ohm
	VCCO, VCCI	< 20 ohm
	VDDI, VDGND, VDDR, VRGND	< 50 ohm
Driving Output	CAxN/P, VPV, VNV, AVDD	< 3 ohm
	CBxN/P, CF1N/P	< 5 ohm
	V1V, CExP/N, NAVDD	< 10 ohm
	CC1N/P, VGH, CD1N/P, VGL	< 20 ohm
	VCOM	< 50 ohm
Digital	D[7:0], CSB, RSTB, A0, RWR, ERD	< 100 ohm
	IF[1:0], CLS, CL, TE, EXTB, TESTB, TESTA, ModeSel	< 200 ohm
MTP	VPP	< 10 ohm

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7 FUNCTION DESCRIPTION

ST7305 supports 8-bit parallel data bus for 8080 series CPU, 3-line and single/dual 4-line interface.

7.1 Microprocessor Interface

7.1.1 Chip Select Input

CSB pin is used for chip selection. This driver can interface with an MPU when CSB is "L". If CSB is “H”, the inputs of A0, ERD and RWR with any combination will be ignored and D[7:0] are high impedance. In 4-Line and 3-Line serial interface, the internal shift register and serial counter are reset when CSB is “H”.

7.1.2 Interface Selection

The interface selection is controlled by IF[1:0] pins. Please refer to the table below:

Table 1 Parallel/Serial Interface Mode

Setting		MCU Type	Interface Pin Function				
IF1	IF0		CSB	A0	RWR	ERD	D[7:0]
L	L	8-bit 8080 Parallel interface	CSB	A0	/WR	/RD	D[7:0]
L	H	Single 4-line serial interface			-	-	D[0]=SCL, D[1:3]=SDA, D[4:7]= -
H	L	Dual 4-line serial interface			SDA2		
H	H	3-line serial interface		-	-	-	

Note: The un-used pins are marked as “-” and should be connected to “H” by VDDI.

7.1.3 8080 Parallel Interface

ST7305 uses bus holder and internal data bus for data transfer with MPU. When writing data from the MPU to on-chip RAM, data is automatically transferred from the bus holder to the RAM as shown in Fig1.

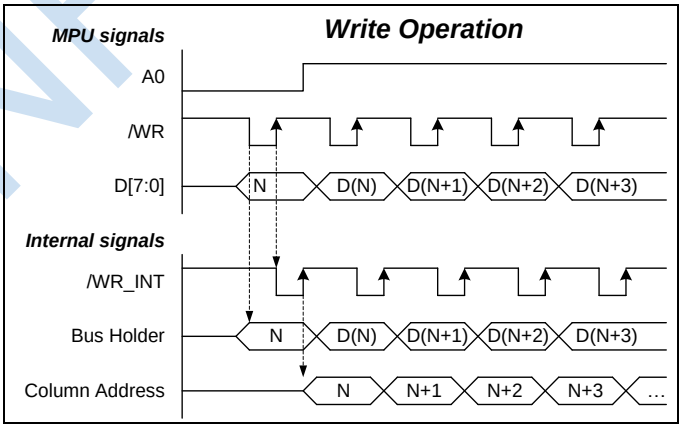


Fig1. Data Transfer: Write

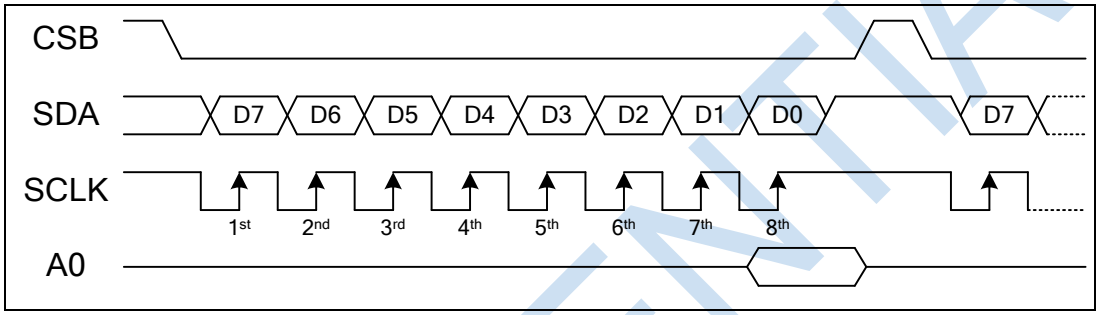
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7.1.4 Single 4-Line Serial Interface

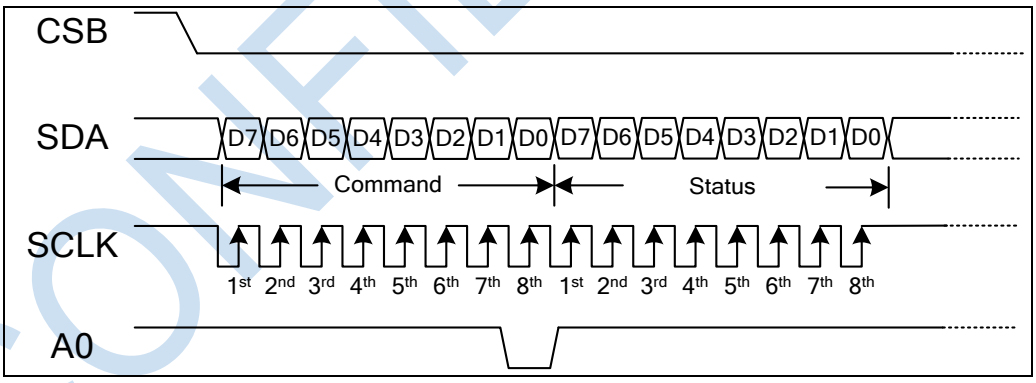
ST7305 is active when CSB is “L”, serial data (SDA) and serial clock (SCLK) inputs are enabled. When CSB is “H”, ST7305 is not active, the internal 8-bit shift register and 3-bit counter are reset. The DDRAM column address pointer will be increased by one automatically after writing each byte of DDRAM.

The display data/command indication is controlled by the register selection pin (A0). The signals transferred on data bus will be display data when A0 is high and will be instruction when A0 is low. Serial data (SDA) is latched at the rising edge of serial clock (SCLK). After the 8th serial clock, the serial data will be processed as 8-bit parallel data. The DDRAM column address pointer will be increased by one automatically after each byte of DDRAM access.



Write Operation of 4-Line SPI

After entering the “Read Status” instruction to read IC status, the information is shifted out as shown below. CSB signal must be kept at “L” during this period. All read out data will be 8 bits.



Read Status Operation of 4-Line SPI

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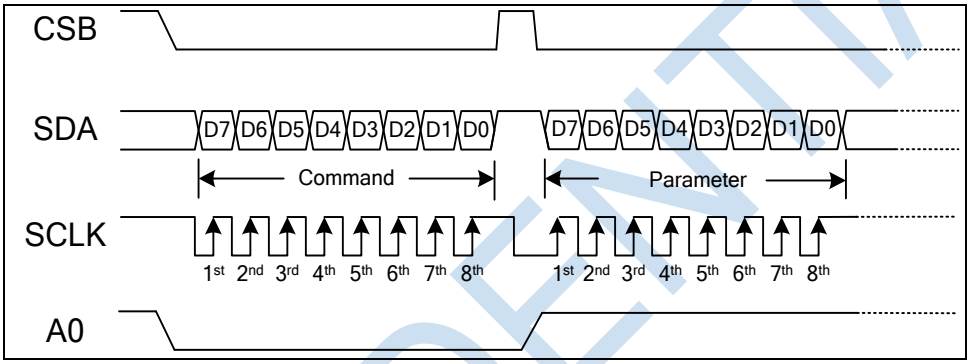
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7.1.5 Dual 4-Line Serial Interface

Dual 4-line serial interface use: CSB (chip enable), SCLK (serial clock), SDA (serial data input/output 1), RWR (serial data input 2) and A0.

Command write mode:

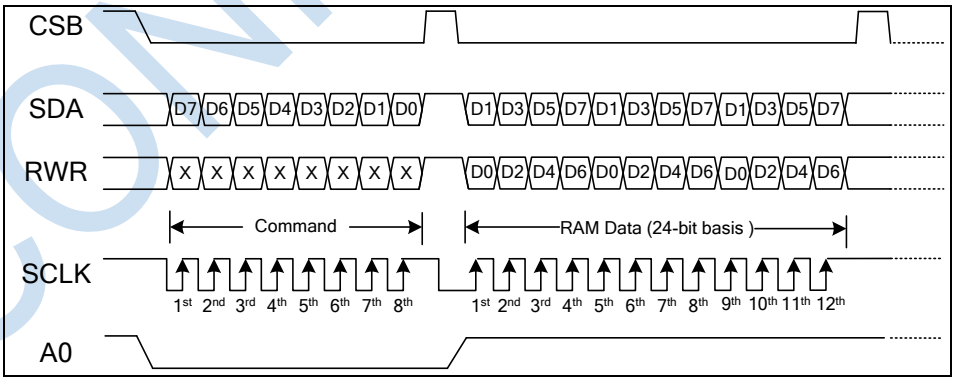
The command write protocol of dual 4-line serial interface is the same with the single 4-line serial interface. The display data/command indication is controlled by the register selection pin (A0). Any instruction can be sent in any order to the driver. The MSB is transmitted first. The serial interface is initialized when CSB is high. In this state, SCLK clock pulse or SDA data have no effect. A falling edge on CSB enables the serial interface and indicates the start of data transmission.



Command Write Operation of Dual 4-Line SPI

RAM write mode:

The RAM write mode of dual 4-line serial interface need use SDA pin and RWR pin to be data input pins. The LSB is transmitted first and the data bit will be exchanged between SDA and RWR.



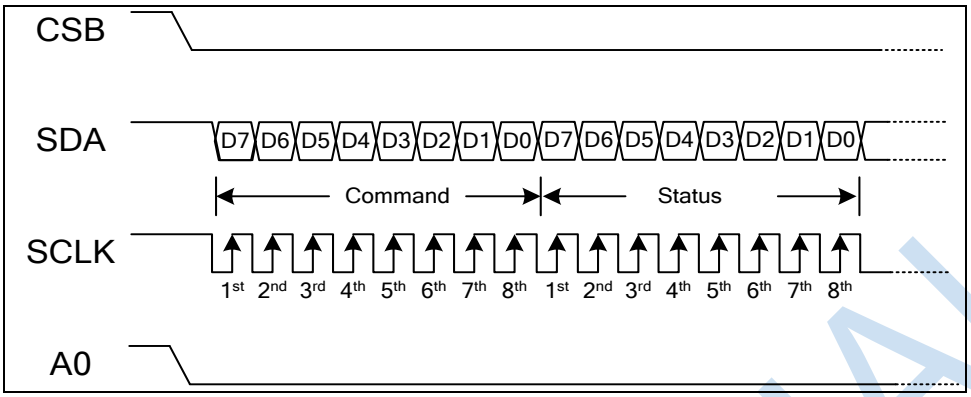
RAM Write Operation of Dual 4-Line SPI

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Read protocol for dual 4-line serial interface:

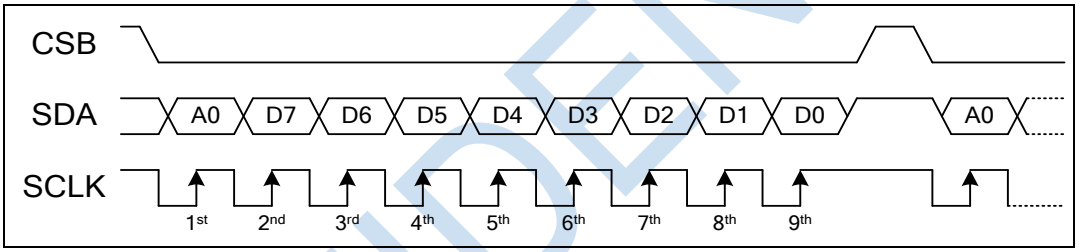
Read protocol (for RDID1/RDID2/RDID3 command: 8-bit read):



Read Status Operation of Dual 4-Line SPI Interface

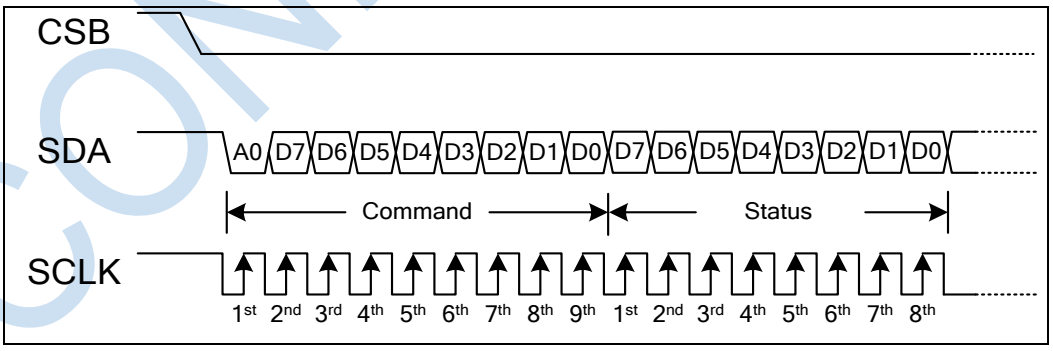
7.1.6 3-Line Serial Interface

In 3-Line interface, A0 pin is not available. The 1st output bit defines command byte or parameter byte.



Write Operation of 3-Line SPI

After entering the “Read Status” instruction to read IC status, the information is shifted out as shown below. CSB signal must be kept at “L” during this period. All read out data will be 8 bits.



Read Status Operation of 3-Line SPI

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7.2 Data Color Coding

Use specific command for switching between data input modes.

7.2.1 Data Input Mode

(1) 8080/Single 4SPI/3SPI Interface

TYPE1: There are 4 write operations for 24-bit data. (Set by “BPS=0” of command 0x3Ah)

Command	A0	D7	D6	D5	D4	D3	D2	D1	D0
DDRAM write	0	0	0	1	0	1	1	0	0
1st write	1	P1	P2	P3	P4	P5	P6	-	-
2nd write	1	P7	P8	P9	P10	P11	P12	-	-
3rd write	1	P13	P14	P15	P16	P17	P18	-	-
4th write	1	P19	P20	P21	P22	P23	P24	-	-

Note: - don’t care

TYPE2: There are 3 write operations for 24-bit data. (Set by “BPS=1” of command 0x3Ah)

Command	A0	D7	D6	D5	D4	D3	D2	D1	D0
DDRAM write	0	0	0	1	0	1	1	0	0
1st write	1	P1	P2	P3	P4	P5	P6	P7	P8
2nd write	1	P9	P10	P11	P12	P13	P14	P15	P16
3rd write	1	P17	P18	P19	P20	P21	P22	P23	P24

(2) Dual 4SP Interface

There are 3 write operations for 24-bit data at the same time.

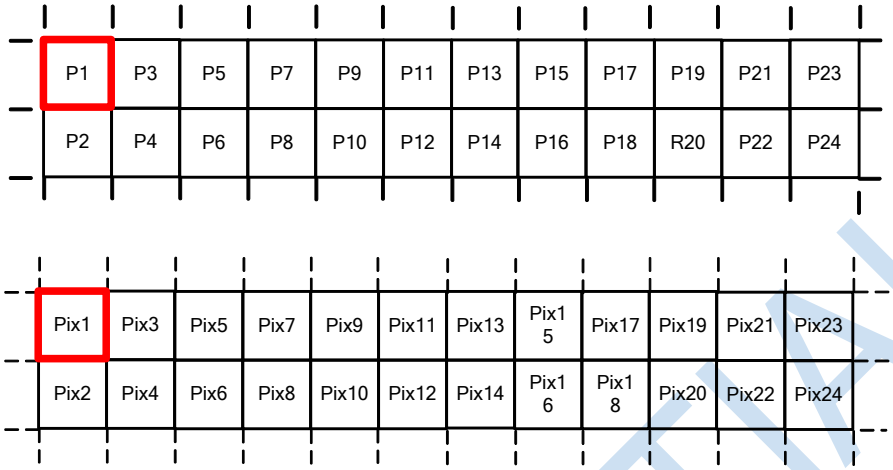
Command	A0	D7	D6	D5	D4	D3	D2	D1	D0
DDRAM write	0	0	0	1	0	1	1	0	0
1st write	1	P1	P2	P3	P4	P5	P6	P7	P8
2nd write	1	P9	P10	P11	P12	P13	P14	P15	P16
3rd write	1	P17	P18	P19	P20	P21	P22	P23	P24

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7.2.2 Data to Display Mapping

The data mapping of mono display is as below.



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7.2.3 Memory to Display Address Mapping

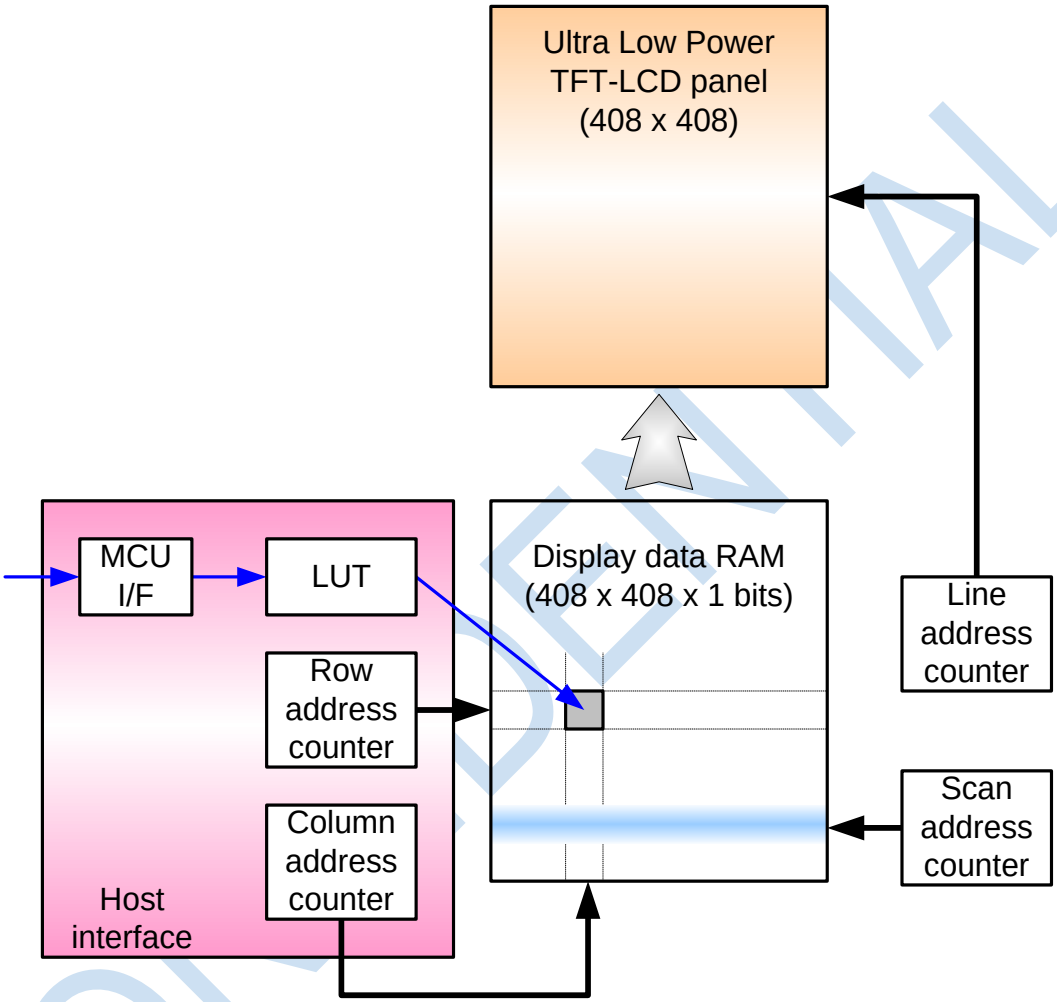
ROW		START CLOUMN ADDRESS												..	END CLOUMN ADDRESS												MX=0	COL
MY=0	MY=1	END CLOUMN ADDRESS												..	START CLOUMN ADDRESS												MX=1	
START ROW ADDRESS	END ROW ADDRESS	P1	P3	P5	P7	P9	P11	P13	P15	P17	P19	P21	P23	..													L1	
		P2	P4	P6	P8	P10	P12	P14	P16	P18	P20	P22	P24	..													L2	
START ROW ADDRESS +1	END ROW ADDRESS -1																										L3	
																											L4	
:	:	:	:	:															:	:	:						:	
END ROW ADDRESS -1	START ROW ADDRESS +1																										L405	
																											L406	
END ROW ADDRESS	START ROW ADDRESS																										L407	
																											L408	
Source		S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	:	S397	S398	S399	S400	S401	S402	S403	S404	S405	S406	S407	S408	Line	

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7.3 Display Data RAM

7.3.1 Configuration

The display module has an integrated 408 x 408 x 1b graphic type static RAM. There will be no abnormal visible effect on the display when there is a simultaneous Panel Read and Write to the same location of the Frame Memory.



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7.4 Address Control

The address counter sets the addresses of the display data RAM for writing and reading. Data is written pixel-wise into the RAM matrix of DRIVER. The locations of RAM are addressed by the address pointers. The address ranges are X=13(0Dh) to X=46 (2Eh) and Y=0 to Y=203 (CBh). Addresses outside these ranges are not allowed. Before writing to the RAM, a window must be defined that will be written. The window is programmable via the command registers XS, YS designating the start address and XE, YE designating the end address.

For example the whole display contents will be written, the window is defined by the following values: XS=13 (0Dh) YS=0 (0h) and XE=46 (2Eh), YE=203 (CBh).

In vertical addressing mode (MV=1), the Y-address increments after each byte, after the last Y-address (Y=YE), Y wraps around to YS and X increments to address the next column. In horizontal addressing mode (V=0), the X-address increments after each byte, after the last X-address (X=XE), X wraps around to XS and Y increments to address the next row. After every last address (X=XE and Y=YE) the address pointers wrap around to address (X=XS and Y=YS).

For flexibility in handling a wide variety of display architectures, the commands “CASET, RASET and MADCTL”, define flags MX and MY, which allows mirroring of the X-address and Y-address. All combinations of flags are allowed. Section 7.4 show the available combinations of writing to the display RAM. When MX, MY and MV will be changed the data bust be rewritten to the display RAM.

For each image condition, the controls for the column and row counters apply as below

Condition	Column Counter	Row Counter
When RAMWR/RAMRD command is accepted	Return to “Start Column (XS)”	Return to “Start Row (YS)”
Complete Pixel Read / Write action	Increment by 1	No change
The Column counter value is larger than “End Column (XE)”	Return to “Start Column (XS)”	Increment by 1
The Column counter value is larger than “End Column (XE)” and the Row counter value is larger than “End Row (YE)”	Return to “Start Column (XS)”	Return to “Start Row (YS)”

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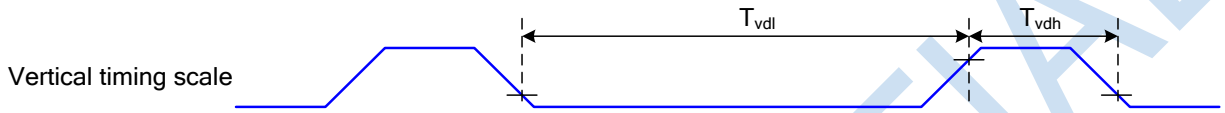
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7.5 Tearing Effect

The Tearing Effect output line supplies to the MPU a Panel synchronization signal. This signal can be enabled or disabled by the Tearing Effect Line Off & On commands. The mode of the Tearing Effect signal is defined by the parameter of the Tearing Effect Line On command. The signal can be used by the MPU to synchronize Frame Memory Writing when displaying video images.

7.5.1 Tearing effect line modes

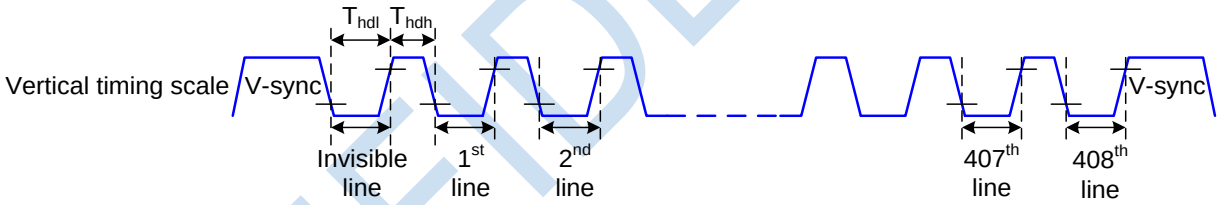
Mode 1, the Tearing Effect Output signal consists of V-Blanking Information only:



tvdh= The LCD display is not updated from the Frame Memory

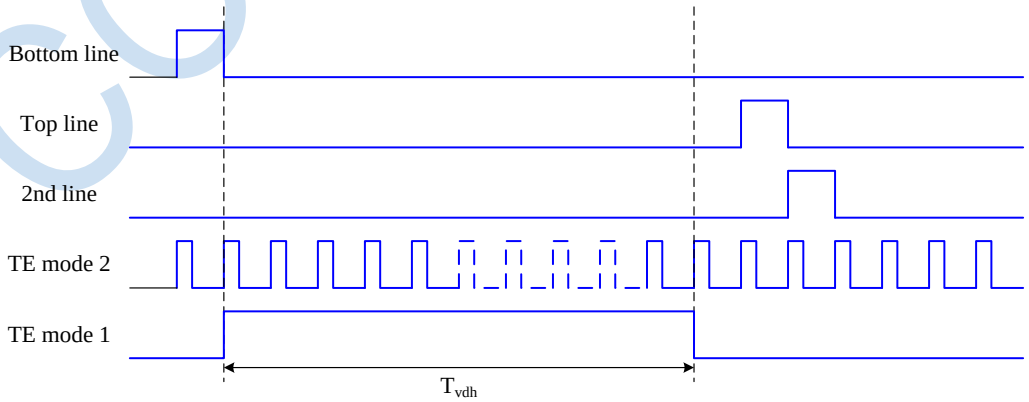
tvdl= The LCD display is updated from the Frame Memory (except Invisible Line – see above)

Mode 2, the Tearing Effect Output signal consists of V-Blanking and H-Blanking Information, there is one V-sync and 408 H-sync pulses per field.



thdh= The LCD display is not updated from the Frame Memory

thdl= The LCD display is updated from the Frame Memory (except Invisible Line – see above)



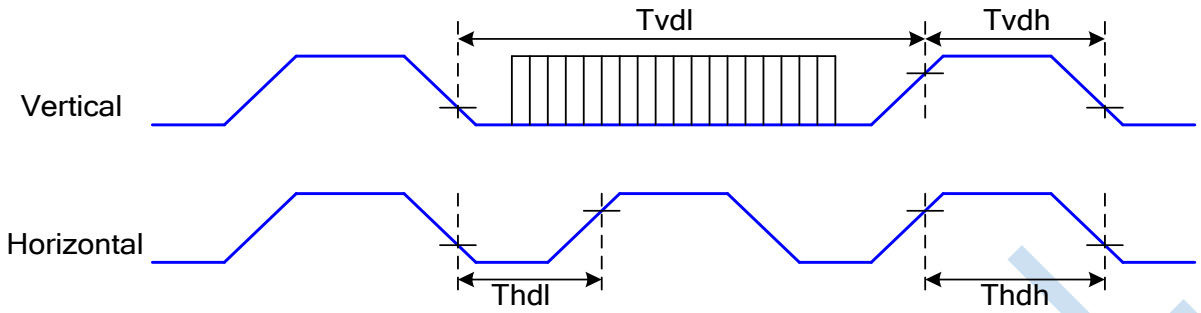
Note: During Sleep In Mode, the Tearing Output Pin is active Low.

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7.5.2 Tearing effect line timings

The Tearing Effect signal is described below:



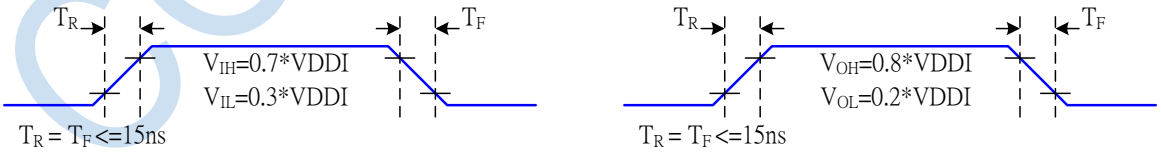
Symbol	Parameter	min	max	unit	description
tvdl	Vertical Timing Low Duration	30	-	ms	
tvdh	Vertical Timing High Duration	1.2	-	ms	
thdl	Horizontal Timing Low Duration	31.2	-	µs	
thdh	Horizontal Timing Low Duration	31.2	-	µs	

AC characteristics of Tearing Effect Signal HPM Mode (Frame Rate = 32 Hz, Ta=25°C)

Symbol	Parameter	min	max	unit	description
tvdl	Vertical Timing Low Duration	990.5	-	ms	
tvdh	Vertical Timing High Duration	9.5	-	ms	
thdl	Horizontal Timing Low Duration	250	-	µs	
thdh	Horizontal Timing Low Duration	250	-	µs	

AC characteristics of Tearing Effect Signal LPM Mode (Frame Rate = 1 Hz, Ta=25°C)

The signal's rise and fall times (tf, tr) are stipulated to be equal to or less than 15ns.

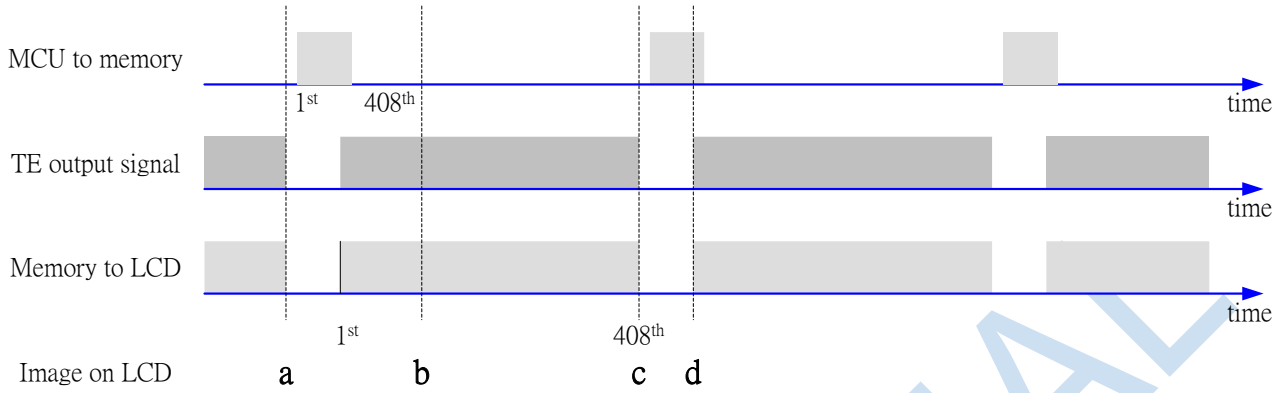


The Tearing Effect Output Line is fed back to the MPU and should be used as shown below to avoid Tearing Effect:

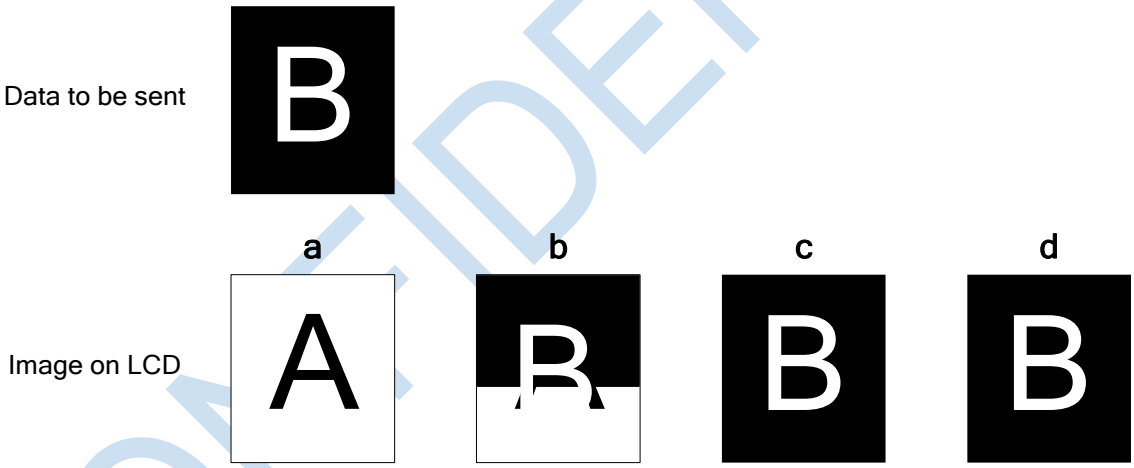
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7.5.3 Example 1: MPU Write is faster than panel read



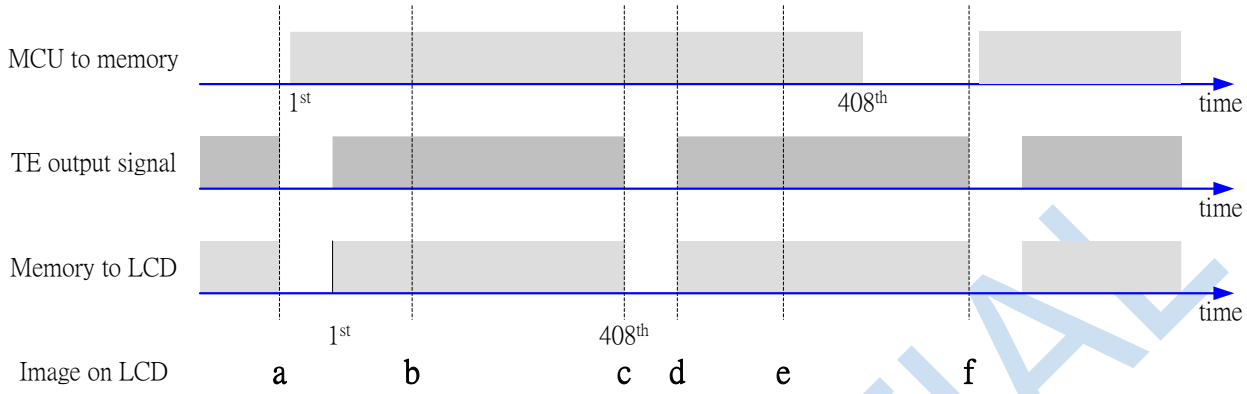
Data write to Frame Memory is now synchronized to the Panel Scan. It should be written during the vertical sync pulse of the Tearing Effect Output Line. This ensures that data is always written ahead of the panel scan and each Panel Frame refresh has a complete new image:



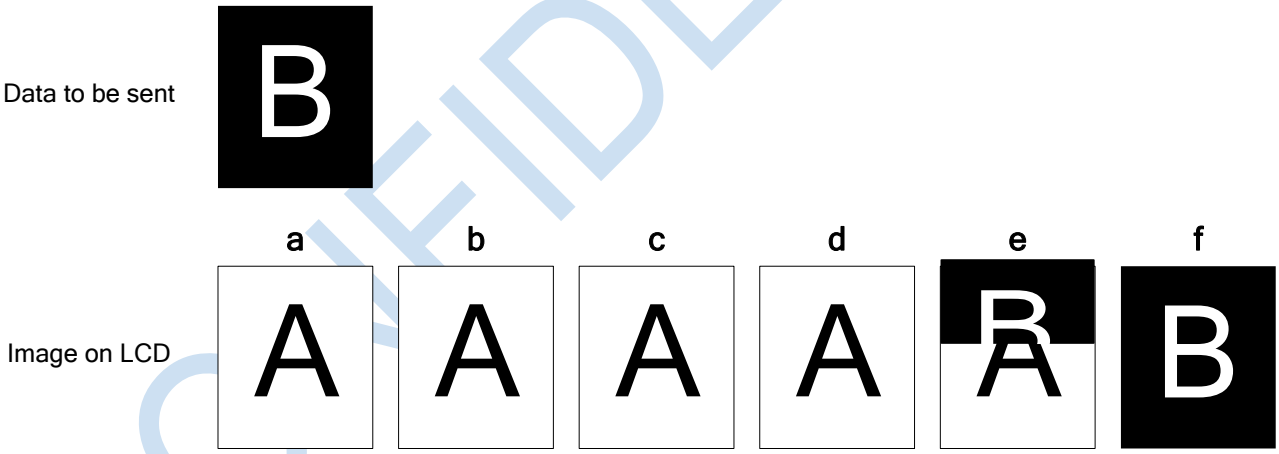
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7.5.4 Example 2: MPU write is slower than panel read



The MPU to Frame Memory write begins just after Panel Read has commenced i.e. after one horizontal sync pulse of the Tearing Effect Output Line. This allows time for the image to download behind the Panel Read pointer and finishing download during the subsequent Frame before the Read Pointer “catches” the MPU to Frame memory write position.



7.6 Oscillation Circuit

This is on-chip Oscillator without external resistor. When the internal oscillator is used, connect CLS to VDDI; when the external oscillator is used, connect CLS to VSS and input external clock to CL pin. This oscillator signal is used by the voltage booster and display timing generation circuit.

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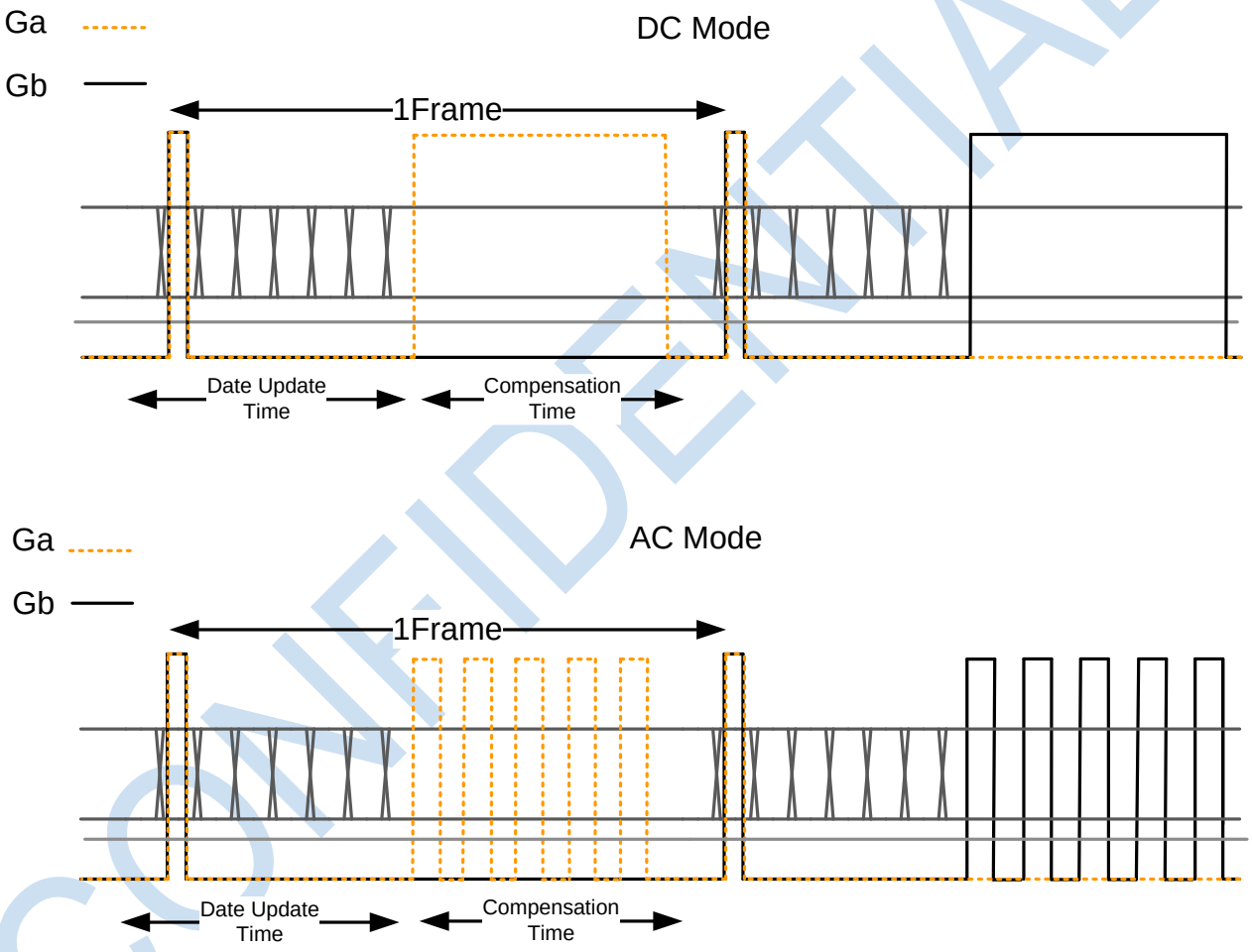
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7.7 Reset

Setting RSTB pin to “L” (hardware reset) can initialize internal function. Generally, VDDI is not stable at the time that the system power is just turned ON. The hardware reset or software reset is required to initialize internal registers after VDDI is stable. Initialization by RSTB pin or command SWRESET is essential before operating.

7.8 Dual TFT Waveform

The driving waveform of two transistors are as below, including data updating and compensation time.



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7.9 Power ON/OFF Sequence

VDDI and VDDA can be applied in any order.

VDDA and VDDI can be power down in any order.

During power off, if LCD is in the Sleep Out mode, VDDA and VDDI must be powered down minimum 120msec after RSTB has been released.

During power off, if LCD is in the Sleep In mode, VDDI or VDDA can be powered down minimum 0msec after RSTB has been released.

CSB can be applied at any timing or can be permanently grounded. RSTB has priority over CSB.

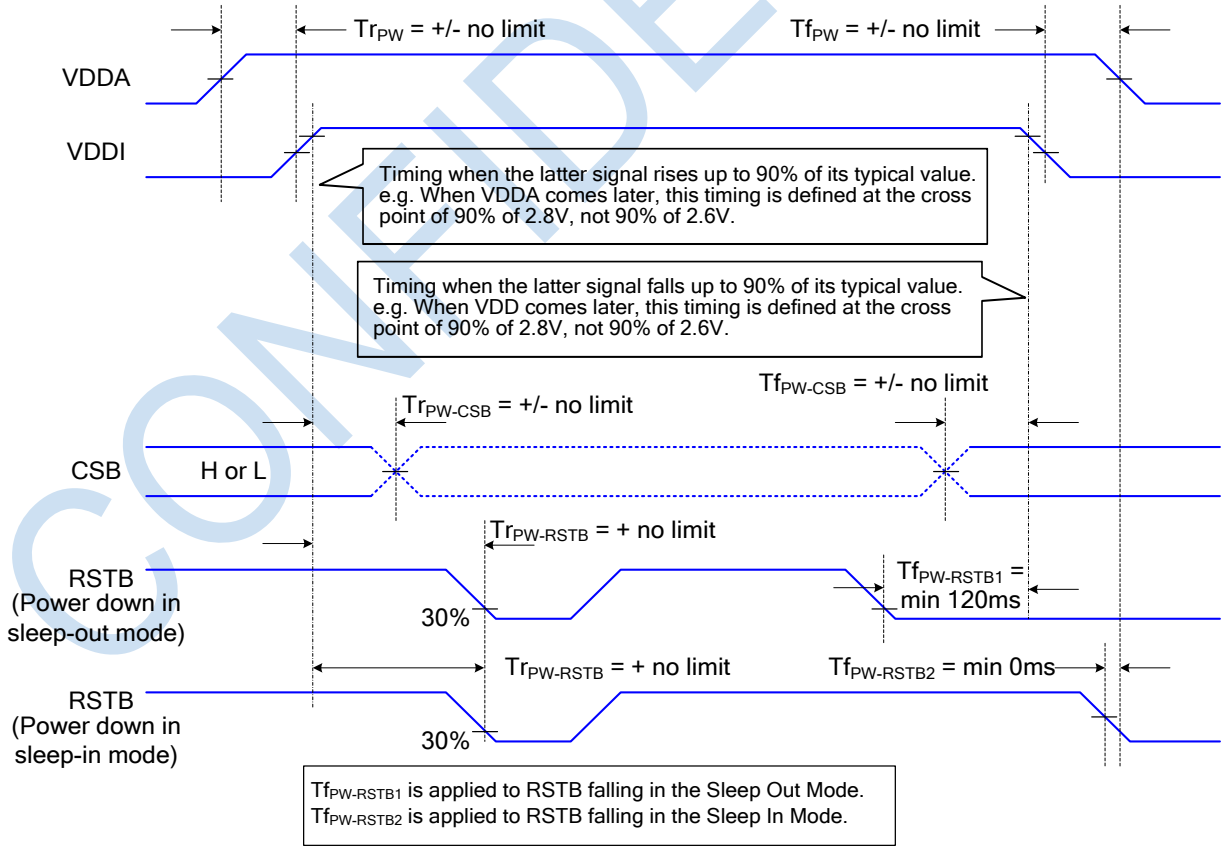
Note 1: There will be no damage to the display module if the power sequences are not met.

Note 2: There will be no abnormal visible effects on the display panel during the Power On/Off Sequences.

Note 3: There will be no abnormal visible effects on the display between end of Power On Sequence and before receiving Sleep Out command. Also between receiving Sleep In command and Power Off Sequence.

Note 4: If RSTB line is not held stable by host during Power On Sequence as defined in the sequence below, then it will be necessary to apply a Hardware Reset (RSTB) after Host Power On Sequence is complete to ensure correct operation. Otherwise function is not guaranteed.

The power on/off sequence is illustrated below

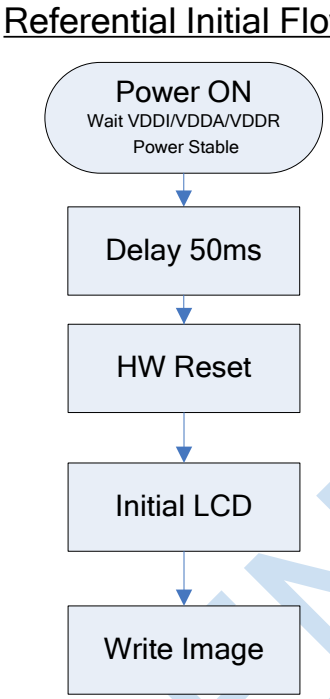


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7.10 Referential Initial Setup Flow

7.10.1 Initial Setting Flow



7.10.2 Referential Initial Code

```
Void Initialization_ ST7305 (void)
 {
        Write(Command, 0xD6);       // NVM Load Control
        Write(Data, 0x17);
        Write(Data, 0x02);
        Write(Command, 0xD1);       //Booster Enable
        Write(Data, 0x01);
        Write(Command, 0xC0);       //Gate Voltage Control
        Write(Data, 0x0E)            // VGH=15V
        Write(Data, 0x0A)            // VGL=-10V
        Write(Command, 0xC1);       //VSHP Setting
        Write(Data, 0x41);          //VSHP1=5V
        Write(Data, 0x41);          //VSHP2=5V
        Write(Data, 0x41);          //VSHP3=5V
        Write(Data, 0x41);          //VSHP4=5V
        Write(Command, 0xC2);       //VSLP Setting
        Write(Data, 0x32);          //VSLP1=1V
        Write(Data, 0x32);          //VSLP2=1V
```

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```
Write(Data, 0x32);          //VSLP3=1V
Write(Data, 0x32);          //VSLP4=1V
Write(Command, 0xC4);       //VSHN Setting
Write(Data, 0x46);          //VSHN1=-3.9V
Write(Data, 0x46);          //VSHN2=-3.9V
Write(Data, 0x46);          //VSHN3=-3.9V
Write(Data, 0x46);          //VSHN4=-3.9V
Write(Command, 0xC5);       //VSLN Setting
Write(Data, 0x46);          //VSLN1=-0.4V
Write(Data, 0x46);          //VSLN2=-0.4V
Write(Data, 0x46);          //VSLN3=-0.4V
Write(Data, 0x46);          //VSLN4=-0.4V
Write(Command, 0xB2);       //Frame Rate Control
Write(Data, 0x02);          //HPM=16Hz ; LPM=1Hz
Write(Command, 0xB3);       //Update Period Gate EQ Control in HPM
Write(Data, 0xE5);
Write(Data, 0xF6);
Write(Data, 0x05);          //HPM EQ Control
Write(Data, 0x46);
Write(Data, 0x77);
Write(Data, 0x77);
Write(Data, 0x77);
Write(Data, 0x77);
Write(Data, 0x76);
Write(Data, 0x45);
Write(Command, 0xB4);       //Update Period Gate EQ Control in LPM
Write(Data, 0x05);          //LPM EQ Control
Write(Data, 0x46);
Write(Data, 0x77);
Write(Data, 0x77);
Write(Data, 0x77);
Write(Data, 0x77);
Write(Data, 0x76);
Write(Data, 0x45);
Write(Command, 0x62);       //Gate Timing Control
Write(Data, 0x32);          //Enable/Disable Compensation
Write(Data, 0x03);
```

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```
        Write(Data, 0x1F);
        Write(Command, 0xB7);       //Source EQ Enable
        Write(Data, 0x13);
        Write(Command, 0xB0);       //Gate Line Setting
        Write(Data, 0x66);          //408 line
        Write(Command, 0x11);       //Sleep-out
        Delay_ms (120);

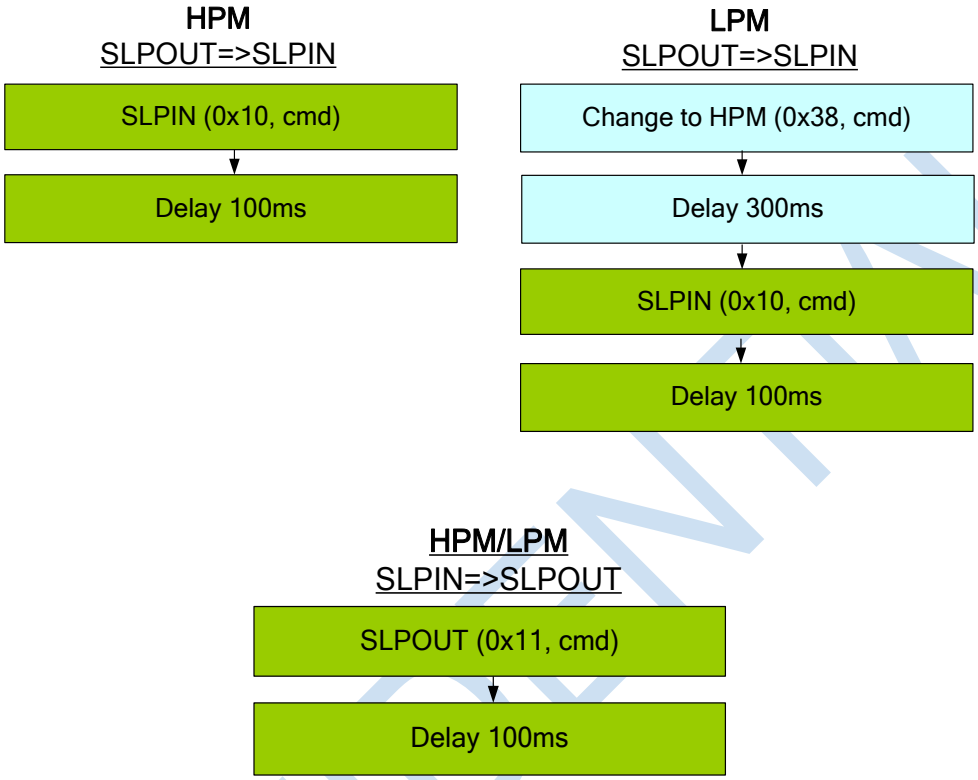
        Write(Command, 0xD8);       //OSC Setting
        Write(Data, 0xA6);
        Write(Data, 0xE9);
        Write(Command, 0xC9);       //Source Voltage Select
        Write(Data, 0x00);          //VSHP1; VSLP1 ; VSHN1 ; VSLN1
        Write(Command, 0x36);       //Memory Data Access Control
        Write(Data, 0x48);
        Write(Command, 0x3A);       //Data Format Select
        Write(Data, 0x10);
        Write(Command, 0xB8);       //Panel Setting
        Write(Data, 0x00);          // 2line interlace and 2line panel layout
        Write(Command, 0x2A);       //Column Address Setting
        Write(Data, 0x0D);
        Write(Data, 0x2E);
        Write(Command, 0x2B);       //Row Address Setting
        Write(Data, 0x00);
        Write(Data, 0xCB);
        Write(Command, 0x35);       //TE Setting
        Write(Data, 0x00);
        Write(Command, 0xD0);       //Enable Auto Power down
        Write(Data, 0xFF);
        Write(Command, 0x39);       //LPM ON
        Write(Command, 0x29);       //Display ON
}
```

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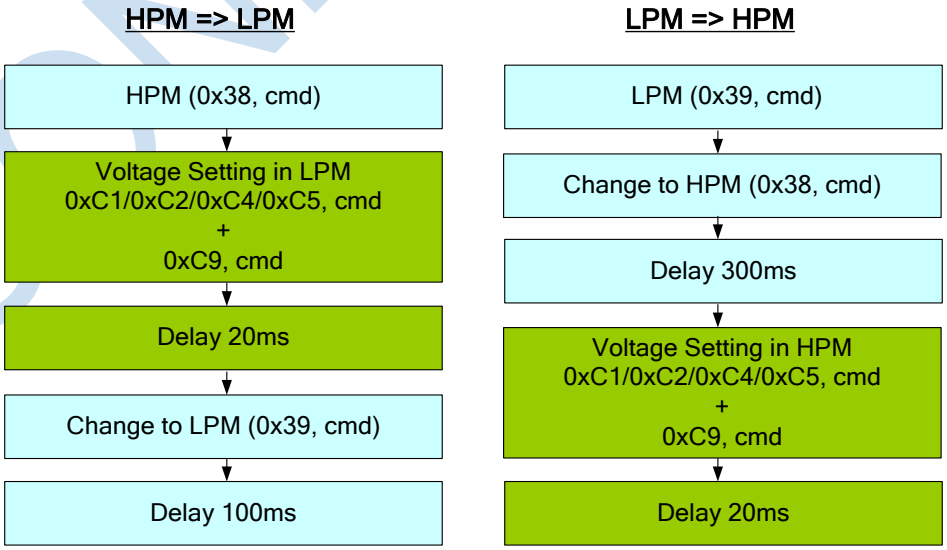
7.11 Sleep-In/Out Sequence

Sleep-in/out mode can be used to save power when display function is not required. To enter or exit from Sleep-in/out mode, refer to the following sequence.



7.12 Power Mode (HPM/LPM) Sequence

To switch between HPM and LPM, follow the sequences below.



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8 COMMAND

8.1 Command Table1

Instruction	A0	R/W	ERD	D7	D6	D5	D4	D3	D2	D1	D0	Hex	Function
1. NOP	0	↑	1	0	0	0	0	0	0	0	0	(00h)	No operation
2.SWRESET	0	↑	1	0	0	0	0	0	0	0	1	(01h)	Software reset
3.RDDID	0	↑	1	0	0	0	0	0	1	0	0	(04h)	Read display ID
	1	1	↑	ID17	ID16	ID15	ID14	ID13	ID12	ID11	ID10		ID1 read
	1	1	↑	ID27	ID26	ID25	ID24	ID23	ID22	ID21	ID20		ID2 read
	1	1	↑	ID37	ID36	ID35	ID34	ID33	ID32	ID31	ID30		ID3 read
4.RDDST	0	↑	1	0	0	0	0	1	0	0	1	(09h)	Read display status
	1	1	↑	BSTON	MY	MX	MV	0	GS	DO	0		Display status1
	1	1	↑	0	XDE	0	BPS	LPM	PTLON	SLOUT	NORON		Display status2
	1	1	↑	VSON	0	INVON	0	0	DISON	TEON	0		Display status3
5.RDDPM	0	↑	1	0	0	0	0	1	0	1	0	(0Ah)	Read display Power Mode
	1	1	↑	BSTON	LPM	PTLON	SLPOUT	NORON	DISON	0	0		Display Power Status
6.SLPIN	0	↑	1	0	0	0	1	0	0	0	0	(10h)	Sleep in
7.SLPOUT	0	↑	1	0	0	0	1	0	0	0	1	(11h)	Sleep out
8.PTLON	0	↑	1	0	0	0	1	0	0	1	0	(12h)	Partial on (240 duty set)
9.PTLOFF	0	↑	1	0	0	0	1	0	0	1	1	(13h)	Partial off (408 duty set)
10.INVOFF	0	↑	1	0	0	1	0	0	0	0	0	(20h)	Display inversion off
11.INVON	0	↑	1	0	0	1	0	0	0	0	1	(21h)	Display inversion on
12.DISPOFF	0	↑	1	0	0	1	0	1	0	0	0	(28h)	Display off
13.DISPON	0	↑	1	0	0	1	0	1	0	0	1	(29h)	Display on
14.CASET	0	↑	1	0	0	1	0	1	0	1	0	(2Ah)	Column address set
	1	↑	1	0	0	XS5	XS4	XS3	XS2	XS1	XS0		X address start: 13≦XS≦X
	1	↑	1	0	0	XE5	XE4	XE3	XE2	XE1	XE0		X address end S≦XE≦X
15.RASET	0	↑	1	0	0	1	0	1	0	1	1	(2Bh)	Row address set
	1	↑	1	YS7	YS6	YS5	YS4	YS3	YS2	YS1	YS0		Y address start: 0≦YS≦Y
	1	↑	1	YE7	YE6	YE5	YE4	YE3	YE2	YE1	YE0		Y address end: S≦YE≦Y
16.RAMWR	0	↑	1	0	0	1	0	1	1	0	0	(2Ch)	Memory write
	1	↑	1	D1[7]	D1[6]	D1[5]	D1[4]	D1[3]	D1[2]	D1[1]	D1[0]		Write data
17.TEOFF	0	↑	1	0	0	1	1	0	1	0	0	(34h)	Tearing effect line off
18.TEON	0	↑	1	0	0	1	1	0	1	0	1	(35h)	Tearing effect line on
	1	↑	1	0	0	0	0	0	0	0	TEM		
19.MADCTL	0	↑	1	0	0	1	1	0	1	1	0	(36h)	Memory data access control

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Instruction	A0	R/W	ERD	D7	D6	D5	D4	D3	D2	D1	D0	Hex	Function
	1	↑	1	MY	MX	MV	0	DO	GS	0	0		
20.VSCSAD	0	↑	1	0	0	1	1	0	1	1	1	(37h)	Vertical scrolling start address
	1	↑	1	0	0	0	0	0	0	0	VSP8		
	1	↑	1	VSP7	VSP6	VSP5	VSP4	VSP3	VSP2	VSP1	VSP0		
21.HPM	0	↑	1	0	0	1	1	1	0	0	0	(38h)	High power mode
22.LPM	0	↑	1	0	0	1	1	1	0	0	1	(39h)	Low power mode
23.DTFORM	0	↑	1	0	0	1	1	1	0	1	0	(3Ah)	Data format select
	1	↑	1	0	0	0	XDE	0	0	0	BPS		
24.RAMWRC	0	↑	1	0	0	1	1	1	1	0	0	(3Ch)	Write memory continue
	1	↑	1	D1[7]	D1[6]	D1[5]	D1[4]	D1[3]	D1[2]	D1[1]	D1[0]		Write data
	1	↑	1	Dx[7]	Dx[6]	Dx[5]	Dx[4]	Dx[3]	Dx[2]	Dx[1]	Dx[0]		
	1	↑	1	Dn[7]	Dn[6]	Dn[5]	Dn[4]	Dn[3]	Dn[2]	Dn[1]	Dn[0]		
	1	1	↑	Dn[7]	Dn[6]	Dn[5]	Dn[4]	Dn[3]	Dn[2]	Dn[1]	Dn[0]		
25.TESCAN	0	↑	1	0	1	0	0	0	1	0	0	(44h)	Set tear scanline
	1	↑	1	0	0	0	0	0	0	0	N8		
	1	↑	1	N7	N6	N5	N4	N3	N2	N1	N0		
26.RDID1	0	↑	1	1	1	0	1	1	0	1	0	(DAh)	Read ID1
	1	1	↑	ID17	ID16	ID15	ID14	ID13	ID12	ID11	ID10		Read parameter
27.RDID2	0	↑	1	1	1	0	1	1	0	1	1	(DBh)	Read ID2
	1	1	↑	ID27	ID26	ID25	ID24	ID23	ID22	ID21	ID20		Read parameter
28,RDID3	0	↑	1	1	1	0	1	1	1	0	0	(DCh)	Read ID3
	1	1	↑	ID37	ID36	ID35	ID34	ID33	ID32	ID31	ID30		Read parameter

"-": Don't care

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8.1.1 NOP (00h)

00H	NOP (No Operation)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
NOP	0	↑	1	0	0	0	0	0	0	0	0	(00h)
Parameter	No Parameter											
Description	This command is empty command.											

8.1.2 SWRESET (01h): Software Reset

01H	SWRESET (Software Reset)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
SWRESET	0	↑	1	0	0	0	0	0	0	0	1	(01h)
Parameter	No Parameter-											
Description	-The display module performs a software reset, registers are written with their SW reset default values. -Frame memory contents are unaffected by this command.											
Restriction	It will be necessary to wait 5msec before sending new command following software reset. The display module loads all display suppliers' factory default values to the registers during this 5msec. If software reset is sent during sleep in mode, it will be necessary to wait 120msec before sending sleep out command. Software reset command cannot be sent during sleep out sequence.											

8.1.3 RDDID (04h): Read Display ID

04H	RDDID (Read Display ID)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDDID	0	↑	1	0	0	0	0	0	1	0	0	(04h)
1st parameter	1	1	↑	ID17	ID16	ID15	ID14	ID13	ID12	ID11	ID10	
2nd parameter	1	1	↑	ID27	ID26	ID25	ID24	ID23	ID22	ID21	ID20	
3rd parameter	1	1	↑	ID37	ID36	ID35	ID34	ID33	ID32	ID31	ID30	
Description	-This read byte returns 24-bit display identification information. -The 1st parameter (ID17 to ID10): LCD module's manufacturer ID. -The 2nd parameter (ID26 to ID20): LCD module/driver version ID -The 3rd parameter (ID37 to UD30): LCD module/driver ID. -Commands RDID1/2/3(DAh, DBh, DCh) read data correspond to the parameters 1,2,3 of the command 04h, respectively.											
Default Value (D7 to D0)	1st parameter= 0x00h 2nd parameter =0x00h 3rd parameter = 0x00h											

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8.1.4 RDDST (09h): Read Display Status

09H	RDDST(Read Display Status)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDDST	0	↑	1	0	0	0	0	1	0	0	1	(09h)
1st parameter	1	1	↑	BSTON	MY	MX	MV	0	GS	DO	0	
2nd parameter	1	1	↑	0	XDE	0	BPS	LPM	PTLON	SLOUT	NORON	
3rd parameter	1	1	↑	VSON	0	INVON	0	0	DISON	TEON	0	

Description

- This command indicates the current status of the display as described in the table below

Bit	Description	Value
BSTON	Booster Voltage Status	'1' =Booster on, (when BSTEN (D1h) D0='1') '0' =Booster off, (when BSTEN (D1h) D0='0')
MY	Row Address Order	'1' =Decrement, (Bottom to Top, when MADCTL (36h) D7='1') '0' =Increment, (Top to Bottom, when MADCTL (36h) D7='0')
MX	Column Address Order	'1' =Decrement, (Right to Left, when MADCTL (36h) D6='1') '0' =Increment, (Left to Right, when MADCTL (36h) D6='0')
MV	Row/Column Exchange	'1' = Row/column exchange, (when MADCTL (36h) D5='1') '0' = Normal, (when MADCTL (36h) D5='0')
DO	Data Order	'0' =Left to right (When MADCTL (36h) D3='1') '1' = Right to left (When MADCTL (36h) D3='0', using in MX=1)
GS	Gate Scan Order	'0' = Data refresh Top to Bottom (when MADCTL (36h) D2='0') '1' = Data refresh Bottom to Top (when MADCTL (36h) D2='1')
XDE	Data Up Down Switch	'0' = Switch ON(when DTFORM (3Ah) D4='1', Using with MY=1) '1' = Switch OFF (when DTFORM (3Ah) D4='0')
BPS	Bytes Per Pixel Select	'0' = 4 writes operations for 24-bit data (when DTFORM (3Ah) D0='0') '1' = 3 writes operations for 24-bit data (when DTFORM (3Ah) D0='1')
LPM	Power Mode Select	'0' = HPM, '1' = LPM
PTLON	Partial Mode ON/OFF	'0' = OFF, '1' = ON
SLPOUT	Sleep In/Out	'0' = In, '1' = Out
NORON	Display Normal Mode On/Off	'0' = Scroll or Partial ON, '1' = Normal,
VSON	Scroll Mode On/Off	'0' = Off, '1' = On
INVON	Inversion Status	'0' = Normal, '1' = Inverse
DISON	Display On/Off	'1' = On, "0" = Off
TEON	Tearing effect line on/off	'1' = On, "0" = Off

Default Value (D7 to D0)	1st parameter= 0x00h 2nd parameter =0x40h

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	3rd parameter = 0x00h

8.1.5 RDDPM (0Ah): Read Display Power Mode

0AH	RDDPM (Read Display Power Mode)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDDPM	0	↑	1	0	0	0	0	1	0	1	0	(0Ah)
Parameter	1	1	↑	BSTON	LPM	PTLON	SLPOUT	NORON	DISON	0	0	

Description

- This command indicates the current status of the display as described in the table below:

Bit	Description	Value
BSTON	Booster Voltage Status	'1' =Booster on, (when BSTEN (D1h) D0='1') '0' =Booster off, (when BSTEN (D1h) D0='0')
LPM	Power Mode Select	'0' = High Power Mode, '1' = Low Power Mode
PTLON	Partial Mode ON/OFF	'0' = OFF, '1' = ON
SLPOUT	Sleep In/Out	'0' = In, '1' = Out
NORON	Display Normal Mode On/Off	'0' = Scroll or Partial ON, '1' = Normal,
DISON	Display On/Off	'1' = On, "0" = Off

Default Value (D7 to D0)	0x08h

8.1.6 SLPIN (10h): Sleep in

10H	SLPIN (Sleep In)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
SLPIN	0	↑	1	0	0	0	1	0	0	0	0	(10h)
Parameter	No Parameter											
Description	-This command causes the LCD module to enter the minimum power consumption mode. -In this mode the DC/DC converter is stopped, internal oscillator is stopped, and panel scanning is stopped. -MCU interface and memory are still working and the memory keeps its contents.											
Restriction	-This command has no effect when module is already in sleep in mode. Sleep in mode can only be left by the sleep out command (11h). -It will be necessary to wait 5msec before sending any new commands to a display module following this command to allow time for the supply voltages and clock circuits to stabilize. -It will be necessary to wait 100msec after sending sleep out command (when in sleep in mode) before sending an sleep in command.											

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8.1.7 SLPOUT (11h): Sleep Out

11H	SLPOUT (Sleep Out)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
SLPOUT	0	↑	1	0	0	0	1	0	0	0	1	(11h)
Parameter	No Parameter											
Description	-This command turn off sleep mode. -In this mode the DC/DC converter is enable, internal display oscillator is started, and panel scanning is started.											
Restriction	-This command has no effect when module is already in sleep in mode. Sleep in mode can only be left by the sleep out command (11h). -It will be necessary to wait 5msec before sending any new commands to a display module following this command to allow time for the supply voltages and clock circuits to stabilize. -It will be necessary to wait 100msec after sending sleep out command (when in sleep in mode) before sending an sleep in command.											

8.1.8 PTLON (12h): Partial On

12H	PTLON (Partial On)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
PTLON	0	↑	1	0	0	0	1	0	0	1	0	(12h)
Parameter	No Parameter											
Description	-This command turns on Partial mode. When partial on, display window become 240duty. -To leave Partial mode, the Partial off command (13h) should be written.											

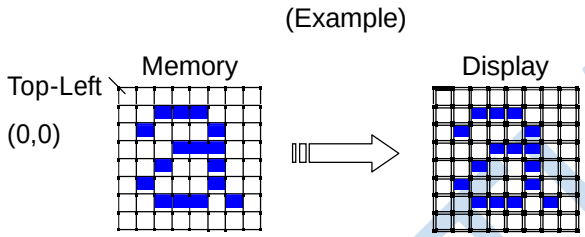
8.1.9 PTLOFF (13h): Partial Off

13H	PTLOFF (Partial Off)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
PTLOFF	0	↑	1	0	0	0	1	0	0	1	1	(13h)
Parameter	No Parameter											
Description	-This command turns the display to partial off mode. When partial off, display window become 408duty. -Exit from NORON by the partial mode on command.											

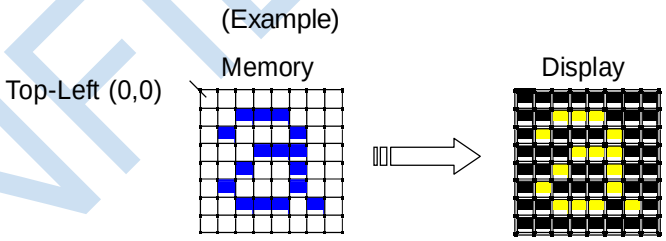
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8.1.10 INVOFF (20h): Display Inversion Off

20H	INVOFF (Display Inversion Off)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
INVOFF	0	↑	1	0	0	1	0	0	0	0	0	(20h)
Parameter	No Parameter											
Description	-This command is used to recover from display inversion mode. 											

8.1.11 INVON (21h): Display Inversion On

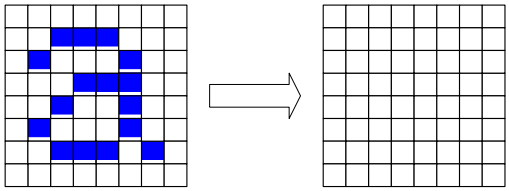
21H	INVON (Display Inversion On)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
INVON	0	↑	1	0	0	1	0	0	0	0	1	(21h)
Parameter	No Parameter											
Description	-This command is used to enter into display inversion mode. 											

8.1.12 DISPOFF (28h): Display Off

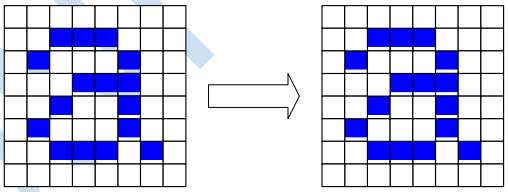
28H	DISPOFF (Display Off)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
DISPOFF	0	↑	1	0	0	1	0	1	0	0	0	(28h)
Parameter	No Parameter											
Description	- This command is used to enter into DISPLAY OFF mode. In this mode, the output from Frame Memory is disabled and blank page inserted. - This command makes no change of contents of frame memory. - This command does not change any other status.											

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	- There will be no abnormal visible effect on the display. - Exit from this command by Display On (29h) **(Example)** **Memory** **Display** 

8.1.13 DISPON (29h): Display On

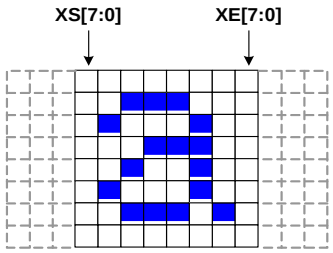
29H	DISPON (Display On)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
DISPON	0	↑	1	0	0	1	0	1	0	0	1	(29h)
Parameter	No Parameter											
Description	- This command is used to recover from DISPLAY OFF mode. - Output from the Frame Memory is enabled. - This command makes no change of contents of frame memory. - This command does not change any other status. **(Example)** **Memory** **Display** 											

8.1.14 CASET (2Ah): Column Address Set

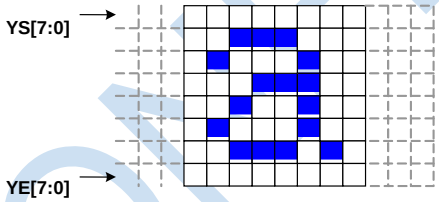
2AH	CASET (Column Address Set)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
CASET	0	↑	1	0	0	1	0	1	0	1	0	(2Ah)
1st parameter	1	↑	1	0	0	XS5	XS4	XS3	XS2	XS1	XS0	
2th parameter	1	↑	1	0	0	XE5	XE4	XE3	XE2	XE1	XE0	
Description	-The value of XS [5:0] and XE [5:0] are referred when RAMWR command comes. -Each value represents one column line in the Frame Memory.											

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Restriction	XS [5:0] always must be equal to or less than XE [5:0] When XS [5:0] or XE [5:0] is greater than maximum address like below, data of out of range will be ignored.(Parameter range: 13(0Dh)≤ XS [5:0] < XE [5:0] ≤ 46 (2Eh)
Default Value (D7 to D0)	1st parameter= 0x00h 2nd parameter =0x00h

8.1.15 RASET (2Bh): Row Address Set

2BH	RASET (Row Address Set)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RASET	0	↑	1	0	0	1	0	1	0	1	1	(2Bh)
1nd parameter	1	↑	1	YS7	YS6	YS5	YS4	YS3	YS2	YS1	YS0	
2th parameter	1	↑	1	YE7	YE6	YE5	YE4	YE3	YE2	YE1	YE0	
Description	-This command is used to defined area of frame memory where MCU can access. -The value of YS [7:0] and YE [7:0] are referred when RAMWR command comes. -Each value represents one page line in the Frame Memory. 											
Restriction	YS [7:0] always must be equal to or less than YE [15:0] When YS [7:0] or YE [7:0] is greater than maximum address like below, data of out of range will be ignored.(Parameter range: 0 ≤ YS [7:0] < YE [7:0] ≤ 203 (CBh)											
Default Value (D7 to D0)	1st parameter= 0x00h 2nd parameter =0x00h											

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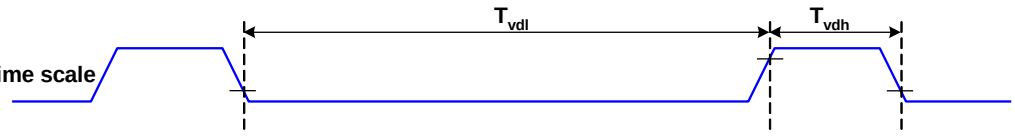
8.1.16 RAMWR (2Ch): Memory Write

2CH	RAMWR (Memory Write)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RAMWR	0	↑	1	0	0	1	0	1	1	0	0	(2Ch)
1st parameter	1	↑	1	D1[7]	D1[6]	D1[5]	D1[4]	D1[3]	D1[2]	D1[1]	D1[0]	
…	1	↑	1	Dx[7]	Dx[6]	Dx[5]	Dx[4]	Dx[3]	Dx[2]	Dx[1]	Dx[0]	
N parameter	1	↑	1	Dn[7]	Dn[6]	Dn[5]	Dn[4]	Dn[3]	Dn[2]	Dn[1]	Dn[0]	
Description	-This command is used to transfer data from MCU to frame memory. -When this command is accepted, the column register and the page register are reset to the start column/start page positions. -The start column/start page positions are different in accordance with MADCTL setting. -Sending any other command can stop frame write.											

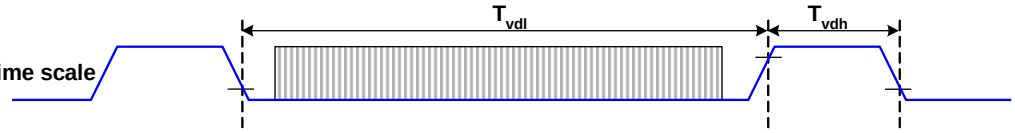
8.1.17 TEOFF (34h): Tearing Effect Line OFF

34H	TEOFF (Tearing Effect Line OFF)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
TEOFF	0	↑	1	0	0	1	1	0	1	0	0	(34h)
Parameter	No Parameter											
Description	-This command is used to turn OFF (Active Low) the Tearing Effect output signal from the TE signal line.											

8.1.18 TEON (35h): Tearing Effect Line On

35H	TEON (Tearing Effect Line On)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
TEON	0	↑	1	0	0	1	1	0	1	0	1	(35h)
Parameter	1	↑	1	0	0	0	0	0	0	0	TEM	
Description	-This command is used to turn ON the Tearing Effect output signal from the TE signal line. -The Tearing Effect Line On has one parameter, which describes the mode of the Tearing Effect Output Line: -When TEM =’0’: The Tearing Effect output line consists of V-Blanking information only **Vertical time scale**  -When TEM =’1’: The Tearing Effect output Line consists of both V-Blanking and H-Blanking information											

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	Vertical time scale  Note: During Sleep In Mode with Tearing Effect Line On, Tearing Effect Output pin will be active Low.
Default Value (D7 to D0)	0x00h

8.1.19 MADCTL (36h): Memory Data Access Control

36H	MADCTL (Memory Data Access Control)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
MADCTL	0	↑	1	0	0	1	1	0	1	1	0	(36h)
Parameter	1	↑	1	MY	MX	MV	0	DO	GS	0	0	

Description

-This command defines read/ write scanning direction of frame memory.

Bit	NAME	DESCRIPTION
D7	MY	Page Address Order
D6	MX	Column Address Order
D5	MV	Page/Column Order
D3	DO	Data Output Order
D2	GS	Gate Scan Order

-Bit Assignment

Bit D7- Page Address Order (MY)

"0" = Top to Bottom (When MADCTL D7="0").

"1" = Bottom to Top (When MADCTL D7="1").

Bit D6- Column Address Order (MX)

"0" = Left to Right (When MADCTL D6="0").

"1" = Right to Left (When MADCTL D6="1").

Bit D5- Page/Column Order (MV)

"0" = Column Direction Mode (When MADCTL D5="0").

"1" = Page Direction Mode (When MADCTL D5="1")

Bit D3- Data Order (DO)

"0" = Left to Right (When MADCTL D3="0" , using in MX=0)

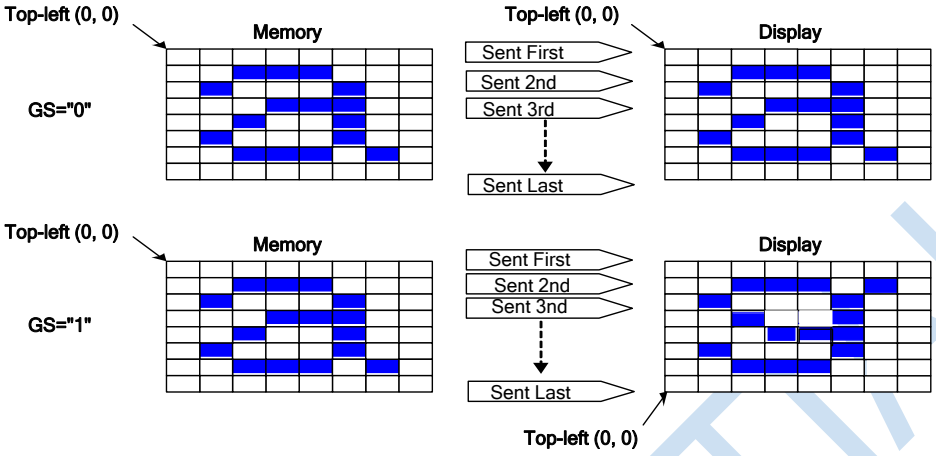
"1" = Right to Left (When MADCTL D3="1", using in MX=1)

Bit D2- Gate Scan Order (GS)

"0" =Data refresh Top to Bottom (When MADCTL D2="0")

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	"0" =Data refresh Bottom to Top (When MADCTL D2="1") **Gate Scan Order (GS)** 
Default Value (D7 to D0)	0x00h

8.1.20 VSCSAD (37h): Vertical Scroll Start Address of RAM

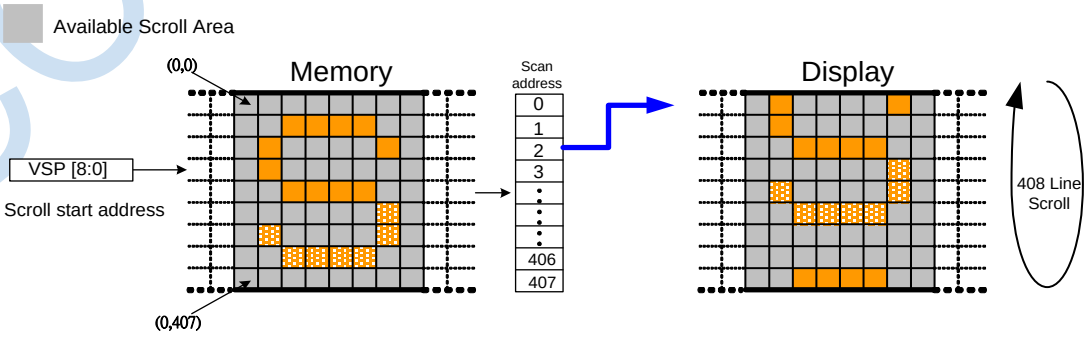
37H	VSCSAD (Vertical Scroll Address of RAM)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
VSCSAD	0	↑	1	0	0	1	1	0	1	1	1	(37h)
1st parameter	1	↑	1	0	0	0	0	0	0	0	VSP8	
2ND parameter	1	↑	1	VSP7	VSP6	VSP5	VSP4	VSP3	VSP2	VSP1	VSP0	

Description

- The Vertical Scrolling Start Address command has two parameters which describe which line in the Frame Memory will be written as the first line after the last line on the display as illustrated below: .

Example:

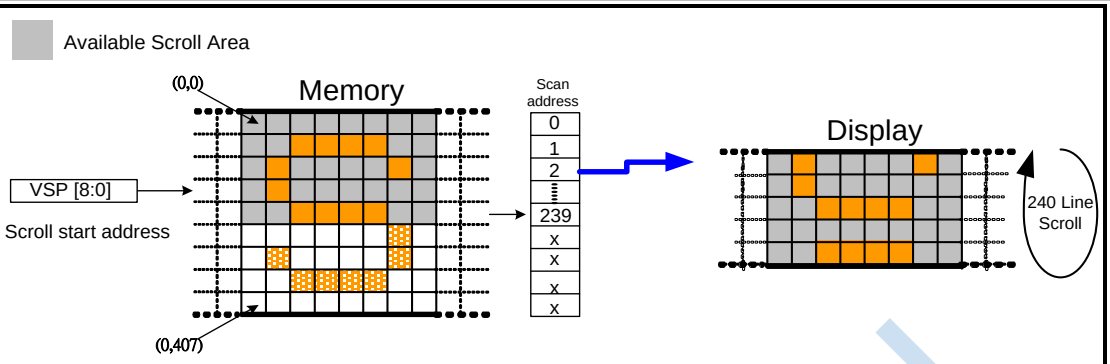
When partial off (408 Duty) and VSP = '2' (0<=VSP[8:0] <=407)



When partial on (240 Duty) and VSP = '2' (0<=VSP[8:0] <=239)

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Default Value (D7 to D0)	1st parameter = 0x00h 2nd parameter = 0x00h

8.1.21 HPM (38h): High Power Mode ON

38H	HPM (High Power Mode ON)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
HPM	0	↑	1	0	0	1	1	1	0	0	0	(38h)
Parameter	No Parameter											
Description	-This command is used to switch display drive mode to high power mode. In HPM, IC drivers LCD with frame rate (16 to 53Hz) that set by **FRCTRL (0xB2h)**. -Default setting of IC is HPM. -After reset, it is in high power mode.											

8.1.22 LPM (39h): Low Power Mode ON

39H	LPM (Low Power Mode ON)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
LPM	0	↑	1	0	0	1	1	1	0	0	1	(39h)
Parameter	No Parameter											
Description	-This command is used to switch display drive mode to low power mode. In LPM, IC drivers LCD with frame rate (1 to 8Hz) that set by **FRCTRL (0xB2h)** command.											

8.1.23 DTFORM (3Ah): Data Format Select

3AH	DTFORM (Data Format Select)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
DTFORM	0	↑	1	0	0	1	1	1	0	1	0	(3Ah)
Parameter	1	↑	1	0	0	0	XDE	0	0	0	BPS	
Description	-This command defines data format select.											

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	<table><tr><th>Bit</th><th>NAME</th><th>DESCRIPTION</th></tr><tr><td>D4</td><td>XDE</td><td>Data Up Down Switch</td></tr><tr><td>D0</td><td>BPS</td><td>Bytes Per Pixel Select</td></tr></table> **Bit D4- Data Up Down Switch (XDE)** "0" = Switch ON (Using with MY=1) "1" = Switch OFF **Bit D0- Bytes Per Pixel Select (BPS)** "0" = 4 write operations for 24-bit data "1" = 3 write operations for 24-bit data
Default Value (D7 to D0)	0x00h

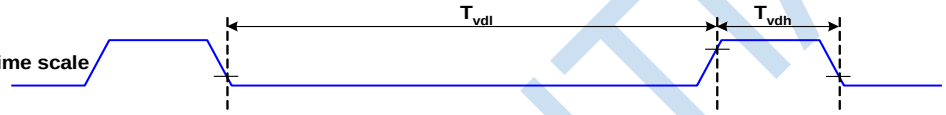
8.1.24 WRMEMC (3Ch): Write Memory Continue

3CH	WRMEMC (Write Memory Continue)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
WRMEMC	0	↑	1	0	0	1	1	1	1	0	0	(3Ch)
1ST parameter	1	↑	1	D1[7]	D1[6]	D1[5]	D1[4]	D1[3]	D1[2]	D1[1]	D1[0]	
⋮	1	↑	1	Dx[7]	Dx[6]	Dx[5]	Dx[4]	Dx[3]	Dx[2]	Dx[1]	Dx[0]	
Nth parameter	1	↑	1	Dn[7]	Dn[6]	Dn[5]	Dn[4]	Dn[3]	Dn[2]	Dn[1]	Dn[0]	
Description	-This command transfers image data from the host processor to the display module's frame memory continuing from the pixel location following the previous write memory continue or memory write command. -If MV=0: Data is written continuing from the pixel location after the write range of the previous memory write or write memory continue. The column register is then incremented and pixels are written to the frame memory until the column register equals the end column (XE) value. The column register is then reset to XS and the page register is incremented. Pixels are written to the frame memory until the page register equals the end page (YE) value and the column register equals the XE value, or the host processor sends another command. If the number of pixels exceeds (XE-XS+1)*(YE-YS+1) the extra pixels are ignored. If MV=1: Data is written continuing from the pixel location after the write range of the previous memory write or write memory continue. The page register is then incremented and pixels are written to the frame memory until the page register equals the end page (YE) value. The page register is then reset to YS and the column register is incremented. Pixels are written to the frame memory until the column register equals the end column (XE) value and the page register equals the YE value, or the host processor sends another command. If the number of pixels exceeds (XE-XS+1)*(YE-YS+1) the extra pixels are ignored.											

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8.1.25 TESCAN (44h): Set Tear Scanline

44H	TESCAN (Set Tear Scanline)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
TESCAN	0	↑	1	0	1	0	0	0	1	0	0	(44h)
1st Parameter	1	↑	1	0	0	0	0	0	0	0	N8	
2nd parameter	1	↑	1	N7	N6	N5	N4	N3	N2	N1	N0	
Description	-This command turns on the display module's Tearing Effect output signal on the TE signal line when the display module reaches line N. The TE signal is not affected by changing MV. -The tearing effect line on has one parameter that describes the tearing effect output line mode. -The tearing effect output line consist of V-blanking information only. Vertical time scale  Note that set tear scanline with N=0 is equivalent to tearing effect line on with TEM=0. The tearing effect output line shall be active low when the display module is in sleep mode											
Default Value (D7 to D0)	1st parameter = 0x00h 2nd parameter = 0x00h											

8.1.26 RDID1 (DAh): Read ID1

DAH	RDID1 (Read ID1)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDID1	0	↑	1	1	1	0	1	1	0	1	0	(DAh)
Parameter	1	1	↑	ID17	ID16	ID15	ID14	ID13	ID12	ID11	ID10	
Description	-This read byte is used to track the LCD module IC version.											

8.1.27 RDID2 (DBh): Read ID2

DBH	RDID2 (Read ID2)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDID2	0	↑	1	1	1	0	1	1	0	1	1	(DBh)
Parameter	1	1	↑	ID27	ID26	ID25	ID24	ID23	ID22	ID21	ID20	
Description	-This read byte is used to track the LCD module/driver IC version.											

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8.1.28 RDID3 (DCh): Read ID3

DCH	RDID3 (Read ID3)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDID3	0	↑	1	1	1	0	1	1	1	0	0	(DCh)
Parameter	1	1	↑	ID37	ID36	ID35	ID34	ID33	ID32	ID31	ID30	
Description	-This read byte identifies the LCD module/driver.											

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8.2 Command Table2

Instruction	A0	R/W	ERD	D7	D6	D5	D4	D3	D2	D1	D0	Hex	Function
1. GTCON	0	↑	1	0	1	1	0	0	0	1	0	(62h)	Gate Timing Control
	1	↑	1	0	0	CMON	CMODE	GAB[3]	GAB[2]	GAB[1]	GAB[0]		
	1	↑	1	0	0	CLKDIV[1]	CLKDIV[0]	FCMPA[3]	FCMPA[2]	FCMPA[1]	FCMPA[0]		
	1	↑	1	CMPWA[7]	CMPWA[6]	CMPWA[5]	CMPWA[4]	CMPWA[3]	CMPWA[2]	CMPWA[1]	CMPWA[0]		
2.GATESET	0	↑	1	1	0	1	1	0	0	0	0	(B0h)	Gate Setting
	1	↑	1	0	GL[6]	GL[5]	GL[4]	GL[3]	GL[2]	GL[1]	GL[0]		
3.FSTCOM	0	↑	1	1	0	1	1	0	0	0	1	(B1h)	First Gate Setting
	1	↑	1	0	0	0	0	0	0	0	FST[8]		
	1	↑	1	FST[7]	FST[6]	FST[5]	FST[4]	FST[3]	FST[2]	FST[1]	FST[0]		
4.FRCTRL	0	↑	1	1	0	1	1	0	0	1	0	(B2h)	FR Control
	1	↑	1	0	0	0	HFRA	0	LFRA[2]	LFRA[1]	LFRA[0]		
5.GTUPEQH	0	↑	1	1	0	1	1	0	0	1	1	(B3h)	Update Period Gate EQ Control In HPM
	1	↑	1	1	GEQEN	1	1	0	1	0	1		
	1	↑	1	1	1	1	1	0	1	1	0		
	1	↑	1	0	GPCHPU0[2:0]			0	GPCHPU1[2:0]				
	1	↑	1	0	GPCHPU2[2:0]			0	GPCHPU3[2:0]				
	1	↑	1	0	GPCHPU4[2:0]			0	GPCHPU5[2:0]				
	1	↑	1	0	GPCHPU6[2:0]			0	GPCHPU7[2:0]				
	1	↑	1	0	GPCHPU8[2:0]			0	GPCHPU9[2:0]				
	1	↑	1	0	GPCHPU10[2:0]			0	GPCHPU11[2:0]				
	1	↑	1	0	GPCHPU12[2:0]			0	GPCHPU13[2:0]				
	1	↑	1	0	GPCHPU14[2:0]			0	GPCHPU15[2:0]				
6.GTUPEQL	0	↑	1	1	0	1	1	0	1	0	0	(B4h)	Update Period Gate EQ Control In LPM
	1	↑	1	0	GPCLPU0[2:0]			0	GPCLPU1[2:0]				
	1	↑	1	0	GPCLPU2[2:0]			0	GPCLPU3[2:0]				
	1	↑	1	0	GPCLPU4[2:0]			0	GPCLPU5[2:0]				
	1	↑	1	0	GPCLPU6[2:0]			0	GPCLPU7[2:0]				
	1	↑	1	0	GPCLPU8[2:0]			0	GPCLPU9[2:0]				
	1	↑	1	0	GPCLPU10[2:0]			0	GPCLPU11[2:0]				
	1	↑	1	0	GPCLPU12[2:0]			0	GPCLPU13[2:0]				
	1	↑	1	0	GPCLPU14[2:0]			0	GPCLPU15[2:0]				

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Instruction	A0	R/W	ERD	D7	D6	D5	D4	D3	D2	D1	D0	Hex	Function
7.SOUEQ	0	↑	1	1	0	1	1	0	1	1	1	(B7h)	Source EQ Enable
	0	↑	1	0	0	0	SOUEQ	0	0	1	1		
8.PNLSET	0	↑	1	1	0	1	1	1	0	0	0	(B8h)	Panel Setting
	0	↑	1	0	DOTINV[1]	DOTINV[0]	0	DPSCN[1]	DPSCN[0]	LAY[1]	LAY[0]		
9.CLRAM	0	↑	1	1	0	1	1	1	0	1	1	(BBh)	CLR RAM
	1	↑	1	CLR	1	0	0	1	1	1	1		
10. GSMODE	0	↑	1	1	0	1	1	1	1	0	1	(BDh)	Gate Scan Mode Select
	1	↑	1	0	0	0	0	GSMODE[3]	GSMODE[2]	GSMODE[1]	GSMODE[0]		
11.GCTRL	0	↑	1	1	1	0	0	0	0	0	0	(C0h)	Gate Control
	1	↑	1	0	0	0	VGHA4	VGHA3	VGHA2	VGHA1	VGHA0		
	1	↑	1	0	0	0	VGLA4	VGLA3	VGLA2	VGLA1	VGLA0		
12.VSHPCTRL	0	↑	1	1	1	0	0	0	0	0	1	(C1h)	Source High Positive Voltage Control
	1	↑	1	VSHPA1[7]	VSHPA1[6]	VSHPA1[5]	VSHPA1[4]	VSHPA1[3]	VSHPA1[2]	VSHPA1[1]	VSHPA1[0]		
	1	↑	1	VSHPA2[7]	VSHPA2[6]	VSHPA2[5]	VSHPA2[4]	VSHPA2[3]	VSHPA2[2]	VSHPA2[1]	VSHPA2[0]		
	1	↑	1	VSHPA3[7]	VSHPA3[6]	VSHPA3[5]	VSHPA3[4]	VSHPA3[3]	VSHPA3[2]	VSHPA3[1]	VSHPA3[0]		
	1	↑	1	VSHPA4[7]	VSHPA4[6]	VSHPA4[5]	VSHPA4[4]	VSHPA4[3]	VSHPA4[2]	VSHPA4[1]	VSHPA4[0]		
13.VSLPCTRL	0	↑	1	1	1	0	0	0	0	1	0	(C2h)	Source Low Positive Voltage Control
	1	↑	1	VSLPA1[7]	VSLPA1[6]	VSLPA1[5]	VSLPA1[4]	VSLPA1[3]	VSLPA1[2]	VSLPA1[1]	VSLPA1[0]		
	1	↑	1	VSLPA2[7]	VSLPA2[6]	VSLPA2[5]	VSLPA2[4]	VSLPA2[3]	VSLPA2[2]	VSLPA2[1]	VSLPA2[0]		
	1	↑	1	VSLPA3[7]	VSLPA3[6]	VSLPA3[5]	VSLPA3[4]	VSLPA3[3]	VSLPA3[2]	VSLPA3[1]	VSLPA3[0]		
	1	↑	1	VSLPA4[7]	VSLPA4[6]	VSLPA4[5]	VSLPA4[4]	VSLPA4[3]	VSLPA4[2]	VSLPA4[1]	VSLPA4[0]		
14.VSHNCTRL	0	↑	1	1	1	0	0	0	1	0	0	(C4h)	Source High Negative Voltage Control
	1	↑	1	VSHNA1[7]	VSHNA1[6]	VSHNA1[5]	VSHNA1[4]	VSHNA1[3]	VSHNA1[2]	VSHNA1[1]	VSHNA1[0]		
	1	↑	1	VSHNA2[7]	VSHNA2[6]	VSHNA2[5]	VSHNA2[4]	VSHNA2[3]	VSHNA2[2]	VSHNA2[1]	VSHNA2[0]		
	1	↑	1	VSHNA3[7]	VSHNA3[6]	VSHNA3[5]	VSHNA3[4]	VSHNA3[3]	VSHNA3[2]	VSHNA3[1]	VSHNA3[0]		
	1	↑	1	VSHNA4[7]	VSHNA4[6]	VSHNA4[5]	VSHNA4[4]	VSHNA4[3]	VSHNA4[2]	VSHNA4[1]	VSHNA4[0]		
15.VSLNCTRL	0	↑	1	1	1	0	0	0	1	0	1	(C5h)	Source Low
	1	↑	1	VSLNA1[7]	VSLNA1[6]	VSLNA1[5]	VSLNA1[4]	VSLNA1[3]	VSLNA1[2]	VSLNA1[1]	VSLNA1[0]		

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Instruction	A0	R/W	ERD	D7	D6	D5	D4	D3	D2	D1	D0	Hex	Function
	1	↑	1	VSLNA2[7]	VSLNA2[6]	VSLNA2[5]	VSLNA2[4]	VSLNA2[3]	VSLNA2[2]	VSLNA2[1]	VSLNA2[0]		Negative
	1	↑	1	VSLNA3[7]	VSLNA3[6]	VSLNA3[5]	VSLNA3[4]	VSLNA3[3]	VSLNA3[2]	VSLNA3[1]	VSLNA3[0]		Voltage
	1	↑	1	VSLNA4[7]	VSLNA4[6]	VSLNA4[5]	VSLNA4[4]	VSLNA4[3]	VSLNA4[2]	VSLNA4[1]	VSLNA4[0]		Control
16.VSHLSEL	0	↑	1	1	1	0	0	1	0	0	1	(C9h)	Source Voltage Select
	1	↑	1	0	0	0	0	0	0	VSHLSEL[1:0]			
17. Power Mode CTRL	0	↑	1	1	1	0	0	0	1	1	1	(C7h)	Ultra Low Power Mode Setting
	1	↑	1	PSCTRL0	1	0	0	0	0	0	1		
	1	↑	1	0	1	0	0	0	0	0	1		
	1	↑	1	0	0	1	0	0	PSCTRL1	1	0		
18.ID1SET	0	↑	1	1	1	0	0	1	0	1	0	(CAh)	ID1 Set
	1	↑	1	ID17	ID16	ID15	ID14	ID13	ID12	ID11	ID10		
19.ID2SET	0	↑	1	1	1	0	0	1	0	1	1	(CBh)	ID2 Set
	1	↑	1	ID27	ID26	ID25	ID24	ID23	ID22	ID21	ID20		
20.ID3SET	0	↑	1	1	1	0	0	1	1	0	0	(CCh)	ID3 Set
	1	↑	1	ID37	ID36	ID35	ID34	ID33	ID32	ID31	ID30		
21 AUTOPWR	0	↑	1	1	1	0	1	0	0	0	0	(D0h)	Enable Auto Power Down
	1	↑	1	AUTOPWR	1	1	1	1	1	1	1		
22.BSTEN	0	↑	1	1	1	0	1	0	0	0	1	(D1h)	Booster Enable
	1	↑	1	0	0	0	0	0	0	0	BSTEN		
23. NVLOADCTRL	0	↑	1	1	1	0	1	0	1	1	0	(D6h)	NV Load Ctrl
	1	↑	1	0	0	0	1	0	VS_EN	ID_EN	1		
	1	↑	1	0	0	0	0	0	NRDTIME	NRDSLP	0		
24. OSCSET	0	↑	1	1	1	0	1	1	0	0	0	(D8h)	OSC Setting
	1	↑	1	OSCEN	0	OSCSW2	0	0	OSCSW1	OSCSW0	0		
	1	↑	1	1	1	1	0	1	0	0	1		
25.NVMRD	0	↑	1	1	1	1	0	1	0	0	1	(E9h)	OTP

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Instruction	A0	R/W	ERD	D7	D6	D5	D4	D3	D2	D1	D0	Hex	Function
	1	1	↑	DO[7]	DO[6]	DO[5]	DO[4]	DO[3]	DO[2]	DO[1]	DO[0]		Data Read
26.EXTBCTRL	0	↑	1	1	1	1	0	1	1	0	0	(ECh)	EXTB Control
	1	↑	1	EXTB_CTRL [7:0]									
27.NVMCTRL1	0	↑	1	1	1	1	1	1	0	0	0	(F8h)	NVM WR/RD Control
	1	↑	1	0	1	0	1	1	0	1	0	(5Ah)	
	1	↑	1	0	1	1	0	1	0	0	1	(69h)	
	1	↑	1	1	1	1	0	1	1	1	0	(EEh)	
	1	↑	1	0	0	0	0	0	PROG	RD	0		
28.NVMCTRL2	0	↑	1	1	1	1	1	1	0	1	0	(FAh)	NVM Program Setting
	1	↑	1	0	0	PA[5]	PA[4]	PA[3]	PA[2]	PA[1]	PA[0]		
	1	↑	1	PIN[7]	PIN[6]	PIN[5]	PIN[4]	PIN[3]	PIN[2]	PIN[1]	PIN[0]		
29.NVMRDEN	0	↑	1	1	1	1	1	1	0	1	1	(FBh)	Enable NVM Read
	1	↑	1	0	1	0	0	1	0	1	0	(4Ah)	
	1	↑	1	1	0	1	0	0	1	0	1	(A5h)	
30.NVPROM	0	↑	1	1	1	1	1	1	1	0	0	(FCh)	Enable NVM Program
	1	↑	1	0	0	1	0	1	0	0	1	(29h)	
	1	↑	1	1	0	1	0	0	1	0	1	(A5h)	

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8.2.1 GTCON (62h): Gate Timing Control

62H	GTCON (Gate Timing Control)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
GTCON	0	↑	1	0	1	1	0	0	0	1	0	(62h)
1st parameter	1	↑	1	0	0	CMON	CMODE	GAB[3]	GAB[2]	GAB[1]	GAB[0]	
2nd parameter	1	↑	1	0	0	CLKDIV[1]	CLKDIV[0]	FCMPA[3]	FCMPA[2]	FCMPA[1]	FCMPA[0]	
3rd parameter	1	↑	1	CMPWA[7]	CMPWA[6]	CMPWA[5]	CMPWA[4]	CMPWA[3]	CMPWA[2]	CMPWA[1]	CMPWA[0]	

Description

-This command is used to set gate timing during the compensation period in low power mode. The frame period is composed of display update period and compensation period.

-As the frequency decreases, turning on this function and matching the appropriate waveform is more beneficial for slowing down the aging rate and phenomenon of the TFT components.

CMON: Compensation ON/OFF. “0”=>OFF ; “1”=>ON

CMODE: Compensation Mode. “0”=>DC Mode ; “1”=>AC Mode.

GAB[3:0]: GA and GB Frame Interval. GAB is set within the rage of 0 to 15 frames.

CLKDIV[1:0]: Compensation CLK Divide

CLKDIV [1:0]	Compensation CLK Divide
00	Divide by 1
01	Divide by 2
10	Divide by 4
11	Divide by 8

FCMPA[3:0]: First Compensation Pulse to Data Upload End. Must be set within the rage of 0 to 15 (1 to 16 gate).

CMPWA[7:0]: Compensation Pulse Width. Must be set within the rage of 0 to 255.

Default Value (D7 to D0)

1st Parameter=0x32h

2nd parameter=0x03h

3rd parameter= 0x1Fh

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8.2.2 GATESET (B0h): Gate Line Setting

B0H	GATESET (Gate Line Setting)											
Inst / Para	D/CX	WRX	RDX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
GATESET	0	↑	1	1	0	1	1	0	0	0	0	(B0h)
1st parameter	1	↑	1	0	GL[6]	GL[5]	GL[4]	GL[3]	GL[2]	GL[1]	GL[0]	
Description	**GL[6:0]** specifies the gate line of the module in 4-line basis. This command must be selected before sleep-out. Do not change this command while the display is turned on. The relationship between the parameter GL[6:0] and the number of display lines is shown below.											
Default Value (D7 to D0)	0x00h											

GL[6]	GL[5]	GL[4]	GL[3]	GL[2]	GL[1]	GL[0]	Gate Line of Module
0	0	0	0	0	0	1	4
0	0	0	0	0	1	0	8
0	0	0	0	0	1	1	12
:	:	:	:	:	:	:	:
1	1	0	0	1	0	0	406
1	1	0	0	1	0	1	407
1	1	0	0	1	1	0	408

8.2.3 FSTCOM (B1h): First Gate Setting

B1H	FSTCOM (First Gate Setting)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
FSTCOM	0	↑	1	1	0	1	1	0	0	0	1	(B1h)
1st parameter	1	↑	1	0	0	0	0	0	0	0	FST[8]	
2nd parameter	1	↑	1	FST[7]	FST[6]	FST[5]	FST[4]	FST[3]	FST[2]	FST[1]	FST[0]	
Description	**FST[8:0]** specifies "the first output gate number – 1 " that mapping to the RAM page address 0. Continuously increasing or decreasing the first gate setting results in vertical-scrolling in 4-line basis. For detail setting value, please see the table as below.											

FST8	FST7	FST6	FST5	FST4	FST3	FST2	FST1	FST0	Line address
0	0	0	0	0	0	0	0	0	1
0	0	0	0	0	0	0	1	1	4
0	0	0	0	0	0	1	1	1	8
:	:	:	:	:	:	:	:	:	:
1	0	0	1	0	1	1	1	1	400
1	1	0	0	1	0	1	1	1	404
1	1	0	0	0	1	0	1	1	408

Example:

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	If FST[8:0]=7h, Gate 8 would output the data of RAM page address 0.
Default Value (D7 to D0)	1st Parameter=0x00h 2nd parameter=0x00h

8.2.4 FRCTRL (B2h): Frame Rate Control

B2H	FRCTRL (Frame Rate Control)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
FRCTRL	0	↑	1	1	0	1	1	0	0	1	0	(B2h)
1st parameter	1	↑	1	0	0	0	HFRA	0	LFRA[2]	LFRA[1]	LFRA[0]	

Description

HFRA: Frame rate control in high power mode (HPM).

Parameter Setting of 0xD8h Command	HFRA	Frame rate (Hz)
Parameter 1: 0xA6h (OSCSW[2:0]=111)	0	16
Parameter 2: 0XE9h	1	32
Parameter 1: 0x80h (OSCSW[2:0]=000)	0	26.5
Parameter 2: 0XE9h	1	53

LFRA[2:0]: Frame rate control in low power mode (LPM)

LFRA [2:0]	Frame rate (Hz)
0h	0.25
1h	0.5
2h	1
3h	2
4h	4
5h	8

Note: The value of frame rate setting should be determined by the module loading and display effect.

Default Value (D7 to D0)	0x12h

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8.2.5 GTUPEQH (B3h): Update Period Gate EQ Control in HPM

B3H	GTUPEQH (Update Period Gate EQ Control in HPM)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
GTUPEQH	0	↑	1	1	0	1	1	0	0	1	1	(B3h)
1st parameter	1	↑	1	1	GEQEN	1	0	0	1	0	1	
2nd parameter	1	↑	1	1	1	1	1	0	1	1	0	
3rd parameter	1	↑	1	0	GPCHPU0[2:0]			0	GPCHPU1[2:0]			
4th parameter	1	↑	1	0	GPCHPU2[2:0]			0	GPCHPU3[2:0]			
5th parameter	1	↑	1	0	GPCHPU4[2:0]			0	GPCHPU5[2:0]			
6th parameter	1	↑	1	0	GPCHPU6[2:0]			0	GPCHPU7[2:0]			
7rd parameter	1	↑	1	0	GPCHPU8[2:0]			0	GPCHPU9[2:0]			
8th parameter	1	↑	1	0	GPCHPU10[2:0]			0	GPCHPU11[2:0]			
9th parameter	1	↑	1	0	GPCHPU12[2:0]			0	GPCHPU13[2:0]			
10th parameter	1	↑	1	0	GPCHPU14[2:0]			0	GPCHPU15[2:0]			

Description

-This command is used to set gate EQ waveform in update period.

GEQEN: Gate EQ Enable (HPM/LPM both)

GEQEN	Gate EQ Enable
0	Disable
1	Enable

GPCHPU0~ GPCHPU15: Gate EQ Control in HPM at 0~15 PWM

Gate EQ of HPM update period includes 16 PWMs to achieve..

GPCHPU0~15[2:0]	Gate EQ Voltage
000	Floating
001	VGL
100	GND
101	NAVDD
110	AVDD
111	VGH

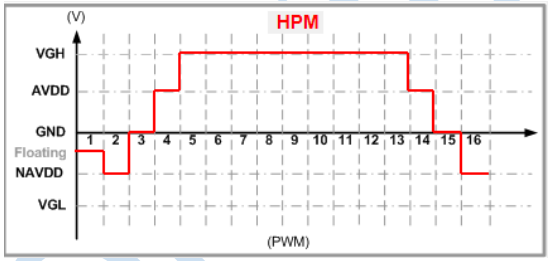
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Default HPM Update EQ Setting.

HPM PWM	Default Parameter	Voltage
GPCHPU0	000	Floating
GPCHPU1	101	NAVDD
GPCHPU2	100	GND
GPCHPU3	110	AVDD
GPCHPU4	111	VGH
GPCHPU5	111	VGH
GPCHPU6	111	VGH
GPCHPU7	111	VGH
GPCHPU8	111	VGH
GPCHPU9	111	VGH
GPCHPU10	111	VGH
GPCHPU11	111	VGH
GPCHPU12	111	VGH
GPCHPU13	110	AVDD
GPCHPU14	100	GND
GPCHPU15	101	NAVDD

Default HPM Update EQ Waveform as below



Default Value (D7 to D0)

1st Parameter=0xE5h

2nd parameter=0xF6h

3rd parameter=0x05h

4th parameter=0x46h

5th parameter=0x77h

6th parameter=0x77h

7rd parameter= 0x77h

8th parameter=0x77h

9th parameter=0x76h

10th parameter=0x45h

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8.2.6 GTUPEQL (B4h): Update Period Gate EQ Control in LPM

B4H	GTUPEQL (Update Period Gate EQ Control in LPM)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
GTUPEQL	0	↑	1	1	0	1	1	0	1	0	0	(B4h)
1st parameter	1	↑	1	0	GPCLPU0[2:0]			0	GPCLPU1[2:0]			
2nd parameter	1	↑	1	0	GPCLPU2[2:0]			0	GPCLPU3[2:0]			
3rd parameter	1	↑	1	0	GPCLPU4[2:0]			0	GPCLPU5[2:0]			
4th parameter	1	↑	1	0	GPCLPU6[2:0]			0	GPCLPU7[2:0]			
5th parameter	1	↑	1	0	GPCLPU8[2:0]			0	GPCLPU9[2:0]			
6th parameter	1	↑	1	0	GPCLPU10[2:0]			0	GPCLPU11[2:0]			
7rd parameter	1	↑	1	0	GPCLPU12[2:0]			0	GPCLPU13[2:0]			
8th parameter	1	↑	1	0	GPCLPU14[2:0]			0	GPCLPU15[2:0]			

Description

GPCLPU0~ GPCLPU15: Gate EQ Control in LPM at 0~15 PWM

Gate EQ of LPM update period includes 16 PWMs to achieve.

GPCLPU0~15[2:0]	Gate EQ Voltage
000	Floating
001	VGL
100	GND
101	NAVDD
110	AVDD
111	VGH

Default HPM Update EQ Setting.

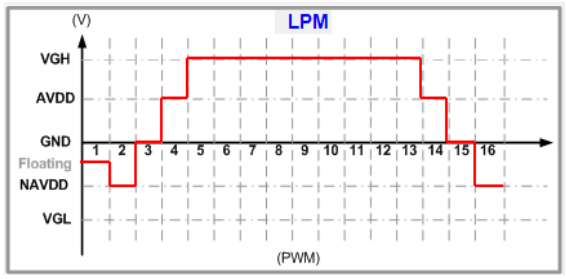
LPM PWM	Default Parameter	Voltage
GPCLPU0	000	Floating
GPCLPU1	101	NAVDD
GPCLPU2	100	GND
GPCLPU3	110	AVDD
GPCLPU4	111	VGH
GPCLPU5	111	VGH
GPCLPU6	111	VGH
GPCLPU7	111	VGH
GPCLPU8	111	VGH
GPCLPU9	111	VGH
GPCLPU10	111	VGH

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	GPCLPU11	111	VGH
	GPCLPU12	111	VGH
	GPCLPU13	110	AVDD
	GPCLPU14	100	GND
	GPCLPU15	101	NAVDD

Default LPM Update EQ Waveform as below



Default Value (D7 to D0)	1st Parameter=0x05h 2nd parameter=0x46h 3rd parameter=0x77h 4th parameter=0x77h 5th parameter=0x77h 6th parameter=0x77h 7rd parameter= 0x76h 8th parameter=0x45h

8.2.7 SOUEQ (B7h): Source EQ Enable

B7H	SOUEQ (Source EQ Enable)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
SOUEQ	0	↑	1	1	0	1	1	0	1	1	1	(B7h)
1st parameter	1	↑	1	0	0	0	SOUEQ	0	0	1	1	
Description	**SOUEQ** "0": Disable Source EQ. "1": Enable Source EQ.											
Default Value (D7 to D0)	0x13h											

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8.2.8 PNLSET (B8h): Panel Setting

B8H	PNLSET (Panel Setting)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
PNLSET	0	↑	1	1	0	1	1	1	0	0	0	(B8h)
Parameter	1	↑	1	0	DOTINV[1]	DOTINV[0]	0	DPSCN[1]	DPSCN[0]	LAY[1]	LAY[0]	

Description

-This command is used to set the gate scan and panel layout.

DOTINV[1:0]:

DOTINV [1:0]	Column/Dot Inversion
00	Column inversion
01	1-Dot inversion
10	2-Dot inversion

DPSCN[1:0]: Display Gate Scan Mode Select

DPSCN [1:0]	Gate Scan Mode
00	Two Line Interval
01	One Line Interval
10	Frame interval

LAY[1:0]: Panel Layout Select

LAY [1:0]	Panel Layout
00	Two Line Interlace
01	One Line Interlace
10	Non Interlace

Default Value (D7 to D0)	0x0Ah

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8.2.9 CLRAM (BBh): Enable Clear RAM

BBH	CLRAM (Enable Clear RAM)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
CLRAM	0	↑	1	1	0	1	1	1	0	1	1	(BBh)
Parameter	1	↑	1	CLR	1	0	0	1	1	1	1	
Description	-This command is used to clear RAM data to 0x00. Ensure to enable clear RAM in Display OFF mode. CLR=0 ; Disable Clear RAM CLR=1 ; Enable Clear RAM											

8.2.10 GSMODE (BDh): Gate Scan Mode (Panel Layout) Select

BDH	GSMODE (Gate Scan Mode (Panel Layout) Select)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
GSMODE	0	↑	1	1	0	1	1	1	1	0	1	(BDh)
Parameter	1	↑	1	0	0	0	0	GSMODE[3]	GSMODE[2]	GSMODE[1]	GSMODE[0]	
Description	-This command is used to select the gate scan mode which depends on panel layout. GSMODE[3:0]=0000 ; Cascade TFT Panel Layout GSMODE[3:0]=0010 ; Single TFT Panel Layout											
Default Value (D7 to D0)	0x00h											

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8.2.11 GCTRL (C0h): Gate Voltage Control

C0H	GCTRL (Gate Voltage Control)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
GCTRL	0	↑	1	1	1	0	0	0	0	0	0	(C0h)
1st parameter	1	↑	1	0	0	0	VGHA4	VGHA3	VGHA2	VGHA1	VGHA0	
2nd parameter	1	↑	1	0	0	0	VGLA4	VGLA3	VGLA2	VGLA1	VGLA0	

Description

-This command is used to adjust gate voltage.

VGHA[4:0]: VGH Setting.

VGHA[4:0]	VGH (V)
00h	8.0
01h	8.5
02h	9.0
03h	9.5
04h	10.0
05h	10.5
06h	11.0
07h	11.5
08h	12.0
09h	12.5
0Ah	13.0
0Bh	13.5
0Ch	14.0
0Dh	14.5
0Eh	15.0 (default)
0Fh	15.5
10h	16.0
11h	16.5

VGLA[4:0]: VGL Setting.

VGLA[4:0]	VGL (V)
02h	-6.0
03h	-6.5
04h	-7.0
05h	-7.5
06h	-8.0
07h	-8.5
08h	-9.0
09h	-9.5
0Ah	-10.0 (default)
0Bh	-10.5
0Ch	-11.0
0Dh	-11.5
0Eh	-12.0
0Fh	-12.5
10h	-13.0
11h	-13.5
12h	-14.0
13h	-14.5
14h	-15.0

Default Value (D7 to D0)

1st Parameter=0x0Eh

2nd parameter=0x0Ah

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8.2.12 VSHPCTRL (C1h): Source High Positive Voltage Control

C1H	VSHPCTRL (Source High Positive Voltage Control)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
VSHPCTRL	0	↑	1	1	1	0	0	0	0	0	1	(C1h)
1st parameter	1	↑	1	VSHPA1[7]	VSHPA1[6]	VSHPA1[5]	VSHPA1[4]	VSHPA1[3]	VSHPA1[2]	VSHPA1[1]	VSHPA1[0]	
2nd parameter	1	↑	1	VSHPA2[7]	VSHPA2[6]	VSHPA2[5]	VSHPA2[4]	VSHPA2[3]	VSHPA2[2]	VSHPA2[1]	VSHPA2[0]	
3rd parameter	1	↑	1	VSHPA3[7]	VSHPA3[6]	VSHPA3[5]	VSHPA3[4]	VSHPA3[3]	VSHPA3[2]	VSHPA3[1]	VSHPA3[0]	
4th parameter	1	↑	1	VSHPA4[7]	VSHPA4[6]	VSHPA4[5]	VSHPA4[4]	VSHPA4[3]	VSHPA4[2]	VSHPA4[1]	VSHPA4[0]	

Description

-This command is used to adjust source high positive voltage in different mode.

The calculation of VSHPAx is as shown below:

VSHPAx= 3.7+0.02x VSHPAx[7:0]

VSHPA1[7:0] : Source High Positive Voltage Mode1

VSHPA2[7:0] : Source High Positive Voltage Mode2

VSHPA3[7:0] : Source High Positive Voltage Mode3

VSHPA4[7:0] : Source High Positive Voltage Mode4

The suggestion of usable VSHPAx voltage is shown below

VSHPAx[7]	VSHPAx[6]	VSHPAx[5]	VSHPAx[4]	VSHPAx[3]	VSHPAx[2]	VSHPAx[1]	VSHPAx[0]	VSHPAx(V)
0	0	0	0	0	0	0	0	3.70
0	0	0	0	0	0	0	1	3.72
0	0	0	0	0	0	1	0	3.74
:	:	:	:	:	:	:	:	:
0	1	0	0	0	0	0	1	5.00 (default)
:	:	:	:	:	:	:	:	:
0	1	1	1	0	0	0	1	5.96
0	1	1	1	0	0	1	0	5.98
0	1	1	1	0	0	1	1	6.00

Default Value (D7 to D0)

1st Parameter=0x41h

2nd parameter=0x41h

3rd parameter=0x41h

4th parameter=0x41h

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8.2.13 VSLPCTRL (C2h): Source Low Positive Voltage Control

C2H	VSLPCTRL (Source Low Positive Voltage Control)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
VSLPCTRL	0	↑	1	1	1	0	0	0	0	1	0	(C2h)
1st parameter	1	↑	1	VSLPA1[7]	VSLPA1[6]	VSLPA1[5]	VSLPA1[4]	VSLPA1[3]	VSLPA1[2]	VSLPA1[1]	VSLPA1[0]	
2nd parameter	1	↑	1	VSLPA2[7]	VSLPA2[6]	VSLPA2[5]	VSLPA2[4]	VSLPA2[3]	VSLPA2[2]	VSLPA2[1]	VSLPA2[0]	
3rd parameter	1	↑	1	VSLPA3[7]	VSLPA3[6]	VSLPA3[5]	VSLPA3[4]	VSLPA3[3]	VSLPA3[2]	VSLPA3[1]	VSLPA3[0]	
4th parameter	1	↑	1	VSLPA4[7]	VSLPA4[6]	VSLPA4[5]	VSLPA4[4]	VSLPA4[3]	VSLPA4[2]	VSLPA4[1]	VSLPA4[0]	

Description

-This command is used to adjust source low positive voltage in different mode.

The calculation of VSLPAx is as shown below:

VSLPAx= 0.02x VSLPAx [7:0]

VSLPA1 [7:0] : Source Low Positive Voltage Mode1

VSLPA2 [7:0] : Source Low Positive Voltage Mode2

VSLPA3 [7:0] : Source Low Positive Voltage Mode3

VSLPA4 [7:0] : Source Low Positive Voltage Mode4

The suggestion of usable VSLPAx voltage is shown below

VSLPAx[7]	VSLPAx[6]	VSLPAx[5]	VSLPAx[4]	VSLPAx[3]	VSLPAx[2]	VSLPAx[1]	VSLPAx[0]	VSLPAx(V)
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	1	0.02
0	0	0	0	0	0	1	0	0.04
:	:	:	:	:	:	:	:	:
0	1	1	0	0	0	1	0	1.00 (default)
:	:	:	:	:	:	:	:	:
0	1	1	0	0	0	0	1	1.96
0	1	1	0	0	0	1	0	1.98
0	1	1	0	0	0	1	1	2.00

Default Value (D7 to D0)

1st Parameter=0x32h

2nd parameter=0x32h

3rd parameter=0x32h

4th parameter=0x32h

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8.2.14 VSHNCTRL (C4h): Source High Negative Voltage Control

C4H	VSHNCTRL (Source High Negative Voltage Control)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
VSHNCTRL	0	↑	1	1	1	0	0	0	1	0	0	(C4h)
1st parameter	1	↑	1	VSHNA1[7]	VSHNA1[6]	VSHNA1[5]	VSHNA1[4]	VSHNA1[3]	VSHNA1[2]	VSHNA1[1]	VSHNA1[0]	
2nd parameter	1	↑	1	VSHNA2[7]	VSHNA2[6]	VSHNA2[5]	VSHNA2[4]	VSHNA2[3]	VSHNA2[2]	VSHNA2[1]	VSHNA2[0]	
3rd parameter	1	↑	1	VSHNA3[7]	VSHNA3[6]	VSHNA3[5]	VSHNA3[4]	VSHNA3[3]	VSHNA3[2]	VSHNA3[1]	VSHNA3[0]	
4th parameter	1	↑	1	VSHNA4[7]	VSHNA4[6]	VSHNA4[5]	VSHNA4[4]	VSHNA4[3]	VSHNA4[2]	VSHNA4[1]	VSHNA4[0]	

Description

-This command is used to adjust source high negative voltage in different mode.

The calculation of VSHNAx is as shown below:

VSHNAx= -2.5-0.02x VSHNAx[7:0]

VSHNA1[7:0] : Source High Negative Voltage Mode1

VSHNA2[7:0] : Source High Negative Voltage Mode2

VSHNA3[7:0] : Source High Negative Voltage Mode3

VSHNA4[7:0] : Source High Negative Voltage Mode4

The suggestion of usable VSHNAx voltage is shown below

VSHNAx[7]	VSHNAx[6]	VSHNAx[5]	VSHNAx[4]	VSHNAx[3]	VSHNAx[2]	VSHNAx[1]	VSHNAx[0]	VSHNAx(V)
0	0	0	0	0	0	0	0	-2.50
0	0	0	0	0	0	0	1	-2.52
0	0	0	0	0	0	1	0	-2.54
:	:	:	:	:	:	:	:	:
0	1	0	0	0	1	1	0	-3.90 (default)
:	:	:	:	:	:	:	:	:
0	1	1	1	1	0	1	1	-4.96
0	1	1	1	1	1	0	0	-4.98
0	1	1	1	1	1	0	1	-5.00

Default Value (D7 to D0)

1st Parameter=0x46h

2nd parameter=0x46h

3rd parameter=0x46h

4th parameter=0x46h

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8.2.15 VSLNCTRL (C5h): Source Low Negative Voltage Control

C5H	VSLNCTRL (Source Low Positive Voltage Control)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
VSLNCTRL	0	↑	1	1	1	0	0	0	1	0	1	(C5h)
1st parameter	1	↑	1	VSLNA1[7]	VSLNA1[6]	VSLNA1[5]	VSLNA1[4]	VSLNA1[3]	VSLNA1[2]	VSLNA1[1]	VSLNA1[0]	
2nd parameter	1	↑	1	VSLNA2[7]	VSLNA2[6]	VSLNA2[5]	VSLNA2[4]	VSLNA2[3]	VSLNA2[2]	VSLNA2[1]	VSLNA2[0]	
3rd parameter	1	↑	1	VSLNA3[7]	VSLNA3[6]	VSLNA3[5]	VSLNA3[4]	VSLNA3[3]	VSLNA3[2]	VSLNA3[1]	VSLNA3[0]	
4th parameter	1	↑	1	VSLNA4[7]	VSLNA4[6]	VSLNA4[5]	VSLNA4[4]	VSLNA4[3]	VSLNA4[2]	VSLNA4[1]	VSLNA4[0]	

Description

-This command is used to adjust source low positive voltage in different mode.

The calculation of VSLNAx is as shown below:

VSLNAx=1- 0.02x VSLNAx [7:0]

VSLNA1 [7:0] : Source Low Negative Voltage Mode1

VSLNA2 [7:0] : Source Low Negative Voltage Mode2

VSLNA3 [7:0] : Source Low Negative Voltage Mode3

VSLNA4 [7:0] : Source Low Negative Voltage Mode4

The suggestion of usable VSLNAx voltage is shown below

VSLNAx[7]	VSLNAx[6]	VSLNAx[5]	VSLNAx[4]	VSLNAx[3]	VSLNAx[2]	VSLNAx[1]	VSLNAx[0]	VSLNAx(V)
0	0	0	0	0	0	0	0	1.00
0	0	0	0	0	0	0	1	0.98
0	0	0	0	0	0	1	0	0.96
:	:	:	:	:	:	:	:	:
0	1	0	0	0	1	1	0	-0.40 (default)
:	:	:	:	:	:	:	:	:
1	0	0	0	1	0	1	0	-1.76
1	0	0	0	1	0	1	1	-1.78
1	0	0	0	1	1	0	0	-1.80

Default Value (D7 to D0)

1st Parameter=0x46h

2nd parameter=0x46h

3rd parameter=0x46h

4th parameter=0x46h

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8.2.16 Power Mode CTRL (C7h): Ultra Low Power Mode Setting

62H	Power Mode CTRL (Ultra Low Power Mode Setting)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Power CTRL	0	↑	1	1	1	0	0	0	1	1	1	(C7h)
1st parameter	1	↑	1	PSCTRL0	1	0	0	0	0	0	1	
2nd parameter	1	↑	1	0	1	0	0	0	0	0	1	
3rd parameter	1	↑	1	0	0	1	0	0	PSCTRL1	1	0	
Description	-This command is used to save more power consumption. -The panel loading and visual quality should be included in the evaluation.											

PSCTRL [1:0]	Ultra Low Power Mode Setting
00	Disable Ultra Low Power Mode
11	Enable Ultra Low Power Mode

Default Value (D7 to D0)	1st Parameter=0x41h 2nd parameter=0x41h 3rd parameter= 0x22h

8.2.17 VSHLSEL (C9h): Source Voltage Select

C9H	VSHLSEL (Source Voltage Select)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
VSHLSEL	0	↑	1	1	1	0	0	1	0	0	1	(C9h)
Parameter	1	↑	1	0	0	0	0	0	0	VSHLSEL[1:0]		
Description	-This command is used to switch source voltage.											

VSHLSEL [1:0]	Source Voltage
00	VSHPA1/ VSLPA1/ VSHNA1/ VSLNA1
01	VSHPA2/ VSLPA2/ VSHNA2/ VSLNA2
10	VSHPA3/ VSLPA3/ VSHNA3/ VSLNA3
11	VSHPA4/ VSLPA4/ VSHNA4/ VSLNA4

Default Value (D7 to D0)	Parameter=0x00h

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8.2.18 ID1SET (CAh): ID1 Setting

CAH	ID1SET (ID1 Code Setting)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
ID1SET	0	↑	1	1	1	0	0	1	0	1	0	(CAh)
Parameter	1	↑	1	ID17	ID16	ID15	ID14	ID13	ID12	ID11	ID10	
Description	**ID1[7:0]:** ID1 Setting.											
Default Value (D7 to D0)	Parameter=0x00h											

8.2.19 ID2SET (CBh): ID2 Setting

CBH	ID2SET (ID2 Code Setting)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
ID2SET	0	↑	1	1	1	0	0	1	0	1	1	(CBh)
Parameter	1	↑	1	ID27	ID26	ID25	ID24	ID23	ID22	ID21	ID20	
Description	**ID2[7:0]:** ID2 Setting.											
Default Value (D7 to D0)	Parameter=0x00h											

8.2.20 ID3SET (CCh): ID3 Setting

CCH	ID3SET (ID3 Code Setting)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
ID3SET	0	↑	1	1	1	0	0	1	1	0	0	(CCh)
Parameter	1	↑	1	ID37	ID36	ID35	ID34	ID33	ID32	ID31	ID30	
Description	**ID3[7:0]:** ID3 Setting.											
Default Value (D7 to D0)	Parameter=0x00h											

8.2.21 AUTOPWRCTRL (D0h): Enable Auto Power Down

D0H	AUTOPWRCTRL (Enable Auto Power Down)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
AUTOPWRCTRL	0	↑	1	1	1	0	1	0	0	0	0	(D0h)
Parameter	1	↑	1	AUTOPWR	1	1	1	1	1	1	1	
Description	-This command is used to enable/disable auto power down **AUTOPWR:** "0": Disable Auto Power down. "1": Enable Auto Power down											
Default Value (D7 to D0)	Parameter=0xFFh											

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8.2.22 BSTEN (D1h): Booster Enable

D1H	BSTEN (Booster Enable)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
BSTEN	0	↑	1	1	1	0	1	0	0	0	1	(D1h)
Parameter	1	↑	1	0	0	0	0	0	0	0	BSTEN	
Description	**BSTEN:** "0": Disable Booster "1": Enable Booster											
Default Value (D7 to D0)	Parameter=0x00h											

8.2.23 NVMLOADCTRL (D6h): NVM Load Control

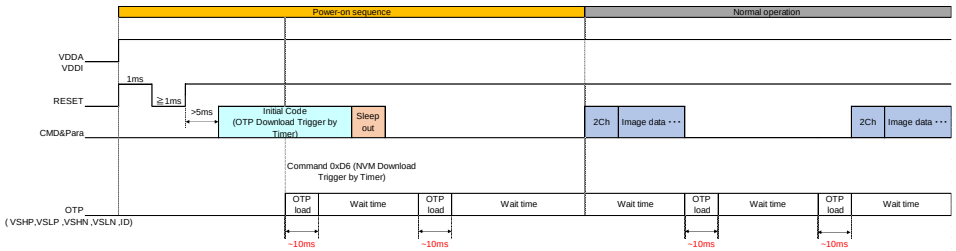
D6H	NVMLOADCTRL (NVM Load Control)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
NVMLOADCTRL	0	↑	1	1	1	0	1	0	1	1	0	(D6h)
1st parameter	1	↑	1	0	0	0	1	0	VS_EN	ID_EN	1	
2nd parameter	1	↑	1	0	0	0	0	0	NRDTIME	NRDSLP	0	

Description

-This command is used to control the items load by NVM and NVM Load control.

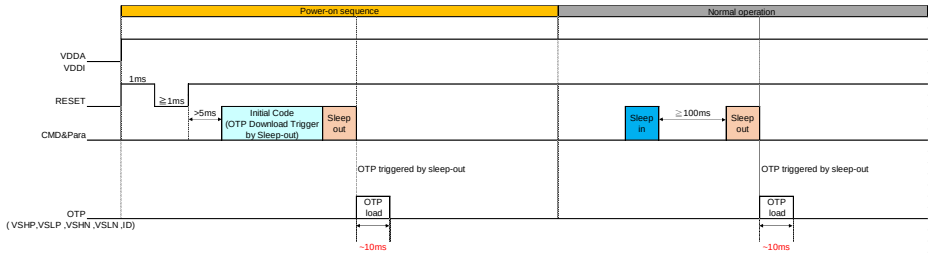
Bit	NVM Load Item	Value
VS_EN	Source High/Low Voltage	'0' = Disable Source High/Low Voltage Load '1' = Enable Source High/Low Voltage Load
ID_EN	ID1/ID2/ID3	'0' = Disable ID1/ID/ID3 Load '1' = Enable ID1/ID/ID3 Load
NRDTIME	NVM Load Control by Timer	"0": NVM load will not be trigger by Timer. "1": NVM load will be trigger by Timer
NRDSLP	NVM Load Control by Sleep-out	"0": NVM load will not be trigger by sleep Out. "1": NVM load will be trigger by sleep Out.

NRDTIME=1 : NVM download will be triggered by timer.



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	NRDSLP=1 : NVM download will be triggered by sleep out. 
Default Value (D7 to D0)	1st Parameter=0x17h 2nd parameter=0x00h

8.2.24 OSCSET (D8h): OSC Setting

D8H	OSCSET (OSC Setting)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
OSCEN	0	↑	1	1	1	0	1	1	0	0	0	(D8h)
1st parameter	1	↑	1	OSCEN	0	OSCSW2	0	0	OSCSW1	OSCSW0	0	
2nd parameter	1	↑	1	1	1	1	0	1	0	0	1	
Description	**OSCEN (OSC Enable/Disable)** **OSCEN** / **OSC Enable** 0 / Disable OSC 1 / Enable OSC **OSCSW[2:0] (OSC Switch)** **OSCSW[2:0]** / **HPM Frame Rate Max** 000 / 53Hz 111 / 32Hz											
Default Value (D7 to D0)	1st Parameter=0x26h 2nd parameter=0xE9h											

8.2.25 NVMRD (E9h): NVM Data Read

E9H	NVMRD (NVM Data Read)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
NVMRD	0	↑	1	1	1	1	0	1	0	0	1	(E9h)
Parameter	1	↑	1	DO[7]	DO[6]	DO[5]	DO[4]	DO[3]	DO[2]	DO[1]	DO[0]	
Description	-This command is used to read register that load from NVM.											

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8.2.26 EXTBCTRL (ECh): EXTB Control

ECH	EXTBCTRL (EXTB Control)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
EXTBCTRL	0	↑	1	1	1	1	0	1	1	0	0	(ECh)
Parameter	1	↑	1	EXTB_CTRL[7:0]								
Description	**EXTB_CTRL[7:0]:** "5Ah": EXTB=H (VDD) "A5h": EXTB=L (VSS)											
Default Value (D7 to D0)	Parameter=0x5Ah											

8.2.27 NVMCTRL1 (F8h): NVM WR/RD Control

F8H	NVMCTRL1 (NVM WR/RD Control)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
NVMCTRL1	0	↑	1	1	1	1	1	1	0	0	0	(F8h)
1st parameter	1	↑	1	0	1	0	1	1	0	1	0	(5Ah)
2nd parameter	1	↑	1	0	1	1	0	1	0	0	1	(69h)
3rd parameter	1	↑	1	1	1	1	0	1	1	1	0	(EEh)
4th parameter	1	↑	1	0	0	0	0	0	PROG	RD	0	
Description	-NVM WR/RD mode select											
Default Value (D7 to D0)	1st Parameter=0x5Ah 2nd parameter=0x69h 3rd parameter=0xEEh 4th parameter=0x00h											

Bit	Value
PROG	'1' =Enable Program Mode '0' =Disable Program Mode
RD	'1' = Enable Read Mode '0' = Disable Read Mode

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8.2.28 NVMCTRL2 (FAh): NVM Program Setting

FAH	NVMCTRL2 (NVM Program Setting)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
NVMCTRL2	0	↑	1	1	1	1	1	1	0	1	0	(FAh)
1st parameter	1	↑	1	0	0	PA[5]	PA[4]	PA[3]	PA[2]	PA[1]	PA[0]	
2nd parameter	1	↑	1	PIN[7]	PIN[6]	PIN[5]	PIN[4]	PIN[3]	PIN[2]	PIN[1]	PIN[0]	
Description	-NVM program register address and data setting.											
Default Value (D7 to D0)	1st Parameter=0x00h 2nd parameter=0x00h											

8.2.29 NVMRDEN (FBh): NVM Read Enable

FBH	NVMRDEN (NVM Read Enable)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
NVMRDEN	0	↑	1	1	1	1	1	1	0	1	1	(FBh)
1st parameter	1	↑	1	0	1	0	0	1	0	1	0	(4Ah)
2nd parameter	1	↑	1	1	0	1	0	0	1	0	1	(A5h)
Description	-Enable NVM read mode.											

8.2.30 NVMPROM (FCh): NVM Program Enable

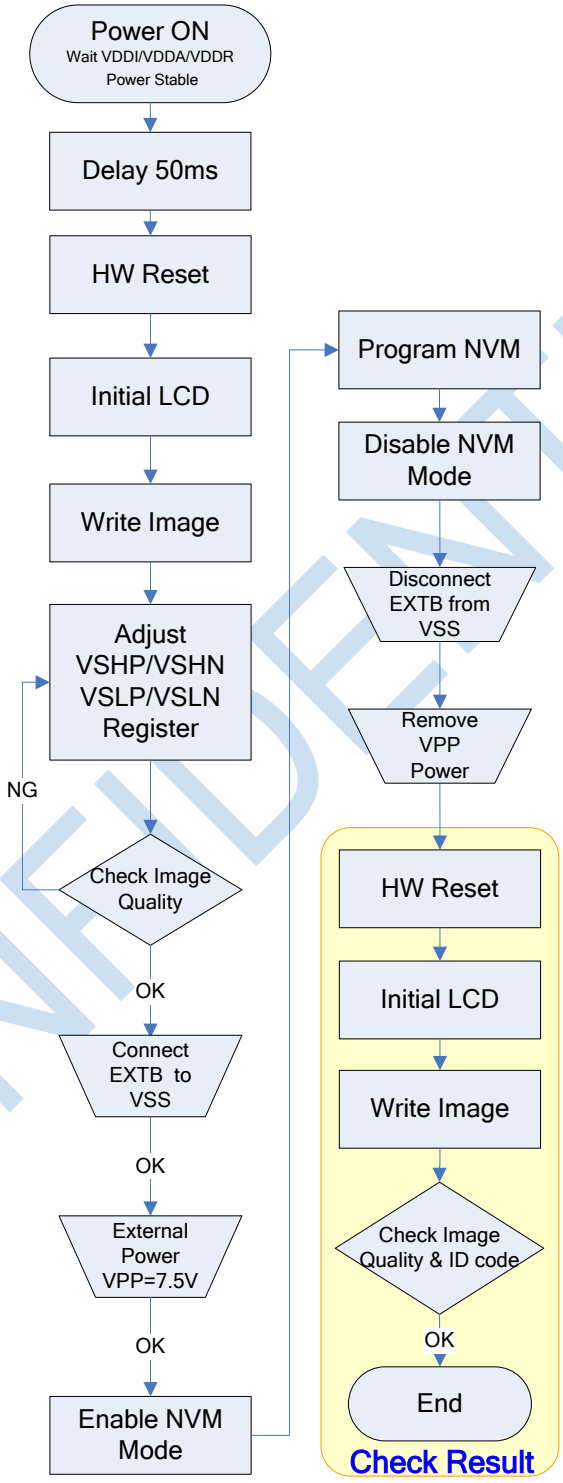
FCH	NVMPROM (NVM Program Enable)											
Inst / Para	A0	RWR	ERD	D7	D6	D5	D4	D3	D2	D1	D0	HEX
NVMPROM	0	↑	1	1	1	1	1	1	1	0	0	(FCh)
1st parameter	1	↑	1	0	0	1	0	1	0	0	1	(29h)
2nd parameter	1	↑	1	1	0	1	0	0	1	0	1	(A5h)
Description	-Enable NVM program mode.											

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9 NVM PROGRAMMING FLOW

NVM Programming Flow



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10 ABSOLUTE MAXIMUM RATING

Item	Symbol	Rating	Unit
I/O Power Supply Voltage	VDDI	- 0.3 ~ +4.0	V
Analog Power Supply Voltage	VDDA	- 0.3 ~ +4.0	V
Reference Power Supply Voltage	VDDR	- 0.3 ~ +4.0	V
LCD Driver Supply Voltage	AVDD	- 0.3 ~ +6.4	V
	VSH-VSL	-0.3 ~ +6.2	V
	VGH-VGL	-0.3 ~ +33.0	V
Logic Input Voltage Range	VIN	-0.3 ~ VDDI + 0.5	V
Logic Output Voltage Range	VO	-0.3 ~ VDDI + 0.5	V
Operating Temperature Range	TOPR	-30 ~ +85	℃
Storage Temperature Range	TSTG	-40 ~ +125	℃

Table 2 Maximum Ratings

Note: If one of the above items is exceeded its maximum limitation momentarily, the quality of the product may be degraded. Absolute maximum limitation, therefore, specify the values exceeding which the product may be physically damaged. Be sure to use the product within the recommend range.

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11 DC CHARACTERISTICS

Parameter	Symbol	Condition	Specification			Unit	Related Pins
			MIN.	TYP.	MAX.		
Power & Operation Voltage							
Analog Power Supply (Normal Mode)	VDDA/VDDR	Analog Power	2.55		3.6	V	
Digital Power Supply (Normal Mode)	VDDI	I/O Supply Voltage	1.65		3.6	V	
Power Supply (1.8V Mode)	VDDA/VDDR VDDI	Single Power	1.7	1.8	1.9	V	
Gate Driver High Voltage	VGH		8.0		16.5	V	Note 1
Gate Driver Low Voltage	VGL		-15.0		-6.0	V	
Gate Driver Supply Voltage		| VGH-VGL |	14.0		31.5	V	Note 2
Input / Output							
Logic-High Input Voltage	VIH		0.7VDDI		VDDI	V	
Logic-Low Input Voltage	VIL		VSS		0.3VDDI	V	
Logic-High Output Voltage	VOH	IOH = -1.0mA	0.8VDDI		VDDI	V	
Logic-Low Output Voltage	VOL	IOL = +1.0mA	VSS		0.2VDDI	V	
Input Leakage Current	IIL	IOH = -1.0mA	-0.1		+0.1	uA	
VCOM Voltage							
VCOM amplitude	VCOM			VSS		V	
Source Driver							
Source High Voltage (Positive)	VSHP		3.7		6.0	V	
Source Low Voltage (Positive)	VSLP		0		2.0	V	
Source High Voltage (Negative)	VSHN		-5.0		-2.5	V	
Source Low Voltage (Negative)	VSLN		-1.8		1.0	V	

Table 3 Basic DC Characteristics

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Notes:

1. When evaluating the maximum and minimum of VGH, VDD=2.8V.

2. The maximum value of |VGH-VGL| can not over 32.0V.

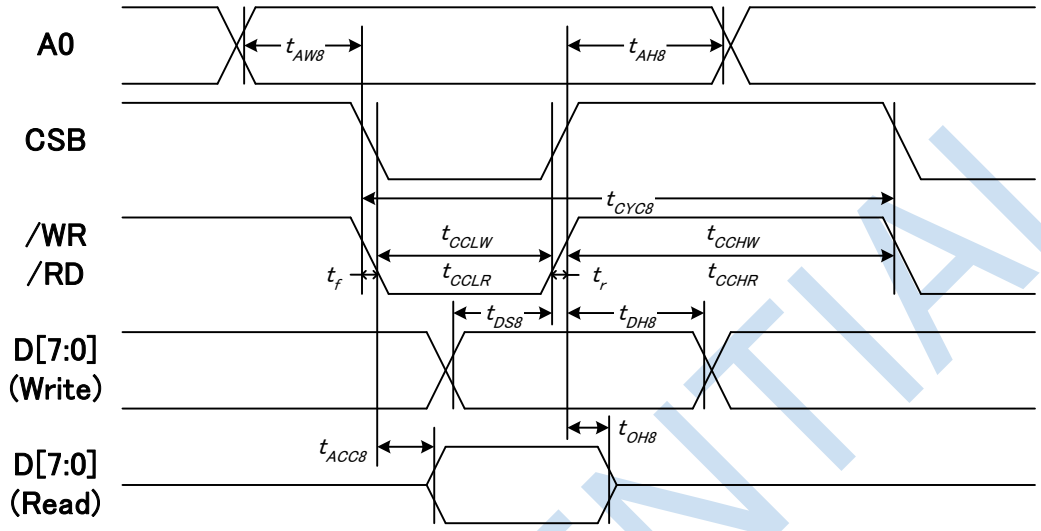
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12 AC CHARACTERISTICS

12.1 Interface Timing

12.1.1 System Bus Timing for Parallel 8080 MCU Interface



VDDI = 1.8~3.3V, Ta = 25°C

Item	Signal	Symbol	Condition	Min.	Max.	Unit
Address setup time	A0	tAW8		20	—	ns
Address hold time		tAH8		0	—	
System cycle time (WRITE)	/WR	tCYC8		160	—	
/WR L pulse width (WRITE)		tCCLW		70	—	
/WR H pulse width (WRITE)		tCCHW		70	—	
System cycle time (READ)	RD	tCYC8		400	—	
/RD L pulse width (READ)		tCCLR		180		
/RD H pulse width (READ)		tCCHR		180		
WRITE Data setup time	D[7:0]	tDS8		15	—	
WRITE Data hold time		tDH8		15	—	
READ access time		tACC8	CL = 30 pF	—	100	
READ Output disable time		tOH8	CL = 30 pF	10	110	

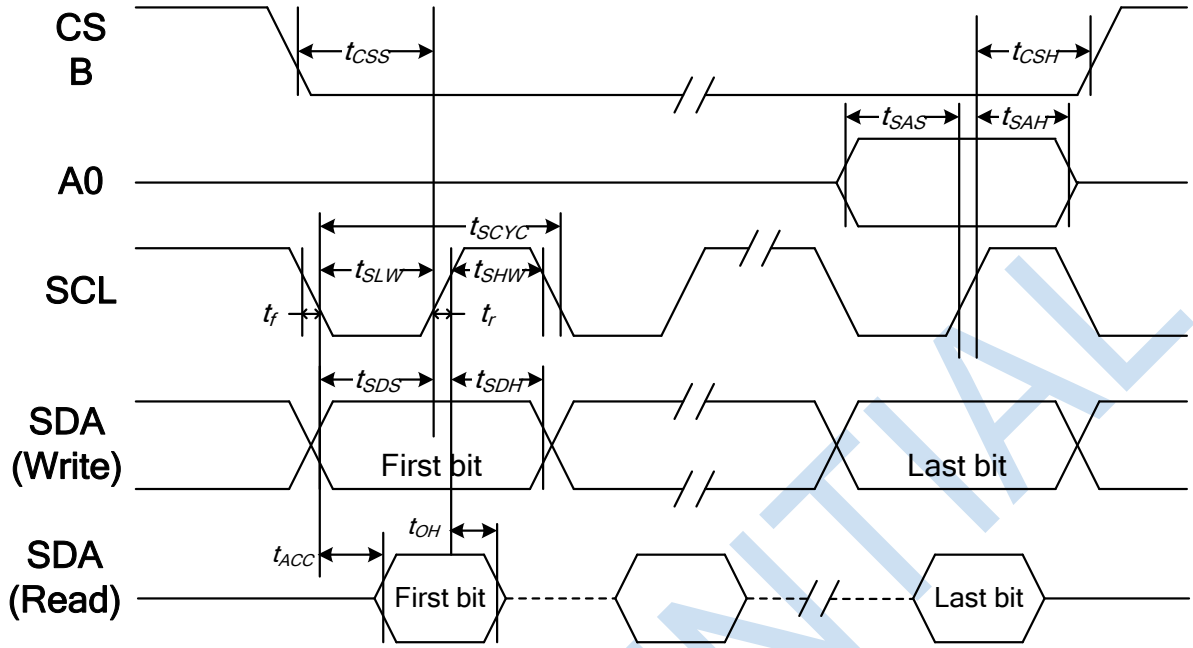
Note:

1. The input signal rise time and fall time (tr, tf) is specified at 15 ns or less. When the system cycle time is extremely fast, (tr + tf) ≦ (tCYC8 – tCCLW – tCCHW) for (tr + tf) ≦ (tCYC8 – tCCLR – tCCHR) are specified.
2. All timing is specified using 20% and 80% of VDDI as the reference.

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12.1.2 System Bus Timing for 4SPI MCU Interface



VDDI = 1.8~3.3V, Ta = 25°C

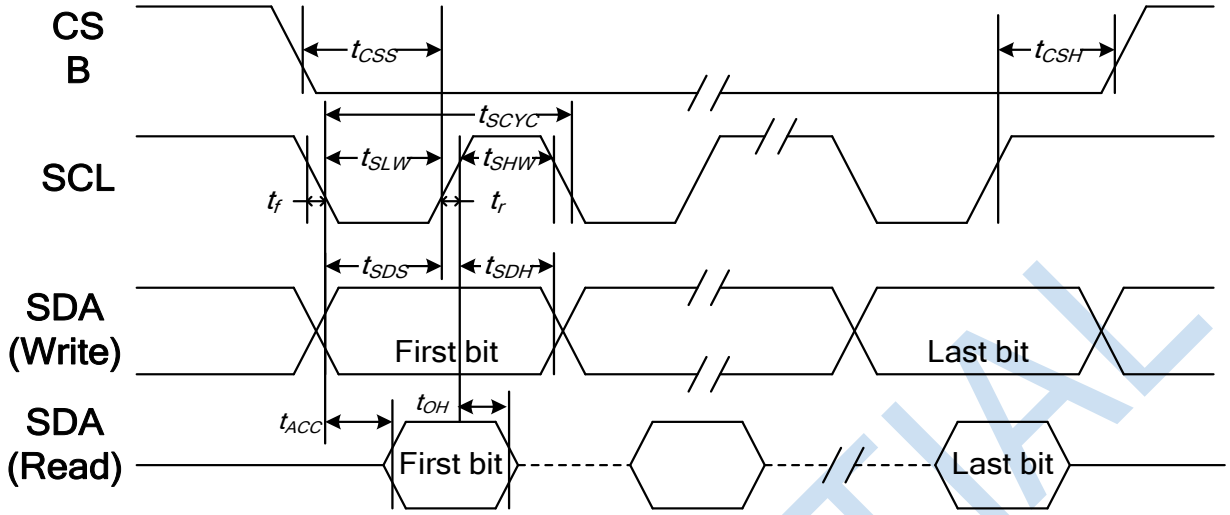
Item	Signal	Symbol	Condition	Min.	Max.	Unit
Serial clock period (Write)	SCL	tSCYC		30	—	ns
Serial clock period (Read)				150		
SCLK “H” pulse width (Write)		tSHW		15	—	
SCLK “H” pulse width (Read)				60		
SCLK “L” pulse width (Write)		tSLW		15	—	
SCLK “L” pulse width (Read)				60		
Address setup time	A0	tSAS		10	—	
Address hold time		tSAH		10	—	
Data setup time	SDA (Write)	tSDS		10	—	
Data hold time		tSDH		10	—	
Read data access time	SDA (Read)	tACC	For maximum CL=30p	10	50	
Read data output disable time		tOH	For minimum CL=8p	15	50	
CSB-SCLK time	CSB	tCSS		10	—	
CSB-SCLK time		tCSH		10	—	

Note:

1. The input signal rise and fall time (tr, tf) are specified at 15 ns or less.
2. All timing is specified using 20% and 80% of VDDI as the standard.

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12.1.3System Bus Timing for 3SPI MCU Interface



VDDI = 1.8~3.3V, Ta = 25°C

Item	Signal	Symbol	Condition	Min.	Max.	Unit
Serial clock period (Write)	SCL	tSCYC		30	—	ns
Serial clock period (Read)				150		
SCLK “H” pulse width (Write)		tSHW		15	—	
SCLK “H” pulse width (Read)				60		
SCLK “L” pulse width (Write)		tSLW		15	—	
SCLK “L” pulse width (Read)				60		
Data setup time	SDA (Write)	tSDS		10	—	
Data hold time		tSDH		10	—	
Data setup time	SDA (Read)	tACC	For maximum CL=30p	10	50	
Data hold time		tOH	For minimum CL=8p	15	50	
CSB-SCLK time	CSB	tCSS		10	—	
CSB-SCLK time		tCSH		10	—	

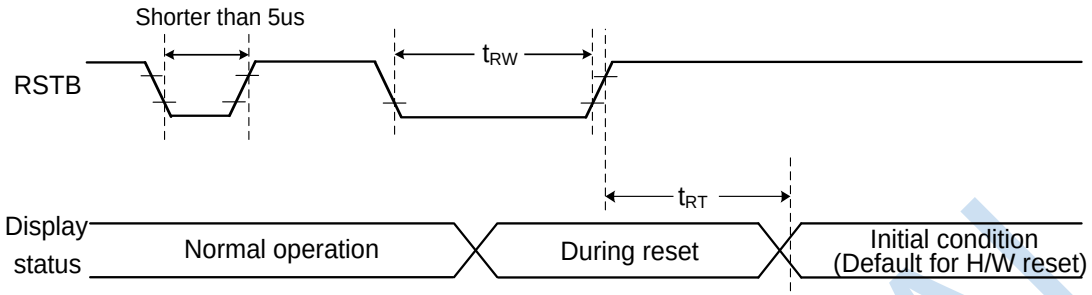
Note:

1. The input signal rise and fall time (tr, tf) are specified at 15 ns or less.
2. All timing is specified using 20% and 80% of VDDI as the standard.

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12.1.4 Reset Timing



VDDI = 1.8~3.3V, Ta = 25°C

Item	Symbol	Condition	Min.	Max.	Unit
Reset "L" pulse width	tRW		1	—	ms
Reset cancel	tRT	Note1, 5 (sleep-in mode)	—	5	ms
		Note1, 6, 7 (sleep-out mode)	—	120	ms

Notes:

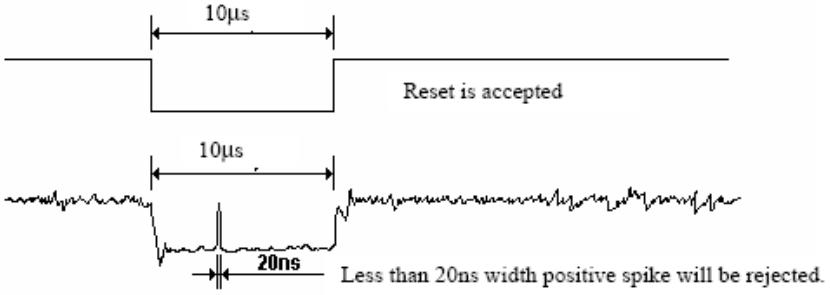
1. The reset cancel includes also required time for loading ID bytes, VSource setting and other settings from NVM (or similar device) to registers. This loading is done every time when there is HW reset cancel time (tRT) within 5 ms after a rising edge of RSTB.

2. Spike due to an electrostatic discharge on RSTB line does not cause irregular system reset according to the table below:

RSTB Pulse	Action
Shorter than 5us	Reset Rejected
Longer than 9us	Reset
Between 5us and 9us	Reset starts

3. During the Resetting period, the display will be blanked (The display is entering blanking sequence, which maximum time is 120 ms, when Reset Starts in Sleep Out –mode. The display remains the blank state in Sleep In –mode.) and then return to Default condition for Hardware Reset.

4. Spike Rejection also applies during a valid reset pulse as shown below:



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5. When Reset applied during Sleep In Mode.

6. When Reset applied during Sleep Out Mode.

7. It is necessary to wait 5msec after releasing RSTB before sending commands. Also Sleep Out command cannot be sent for 120msec.

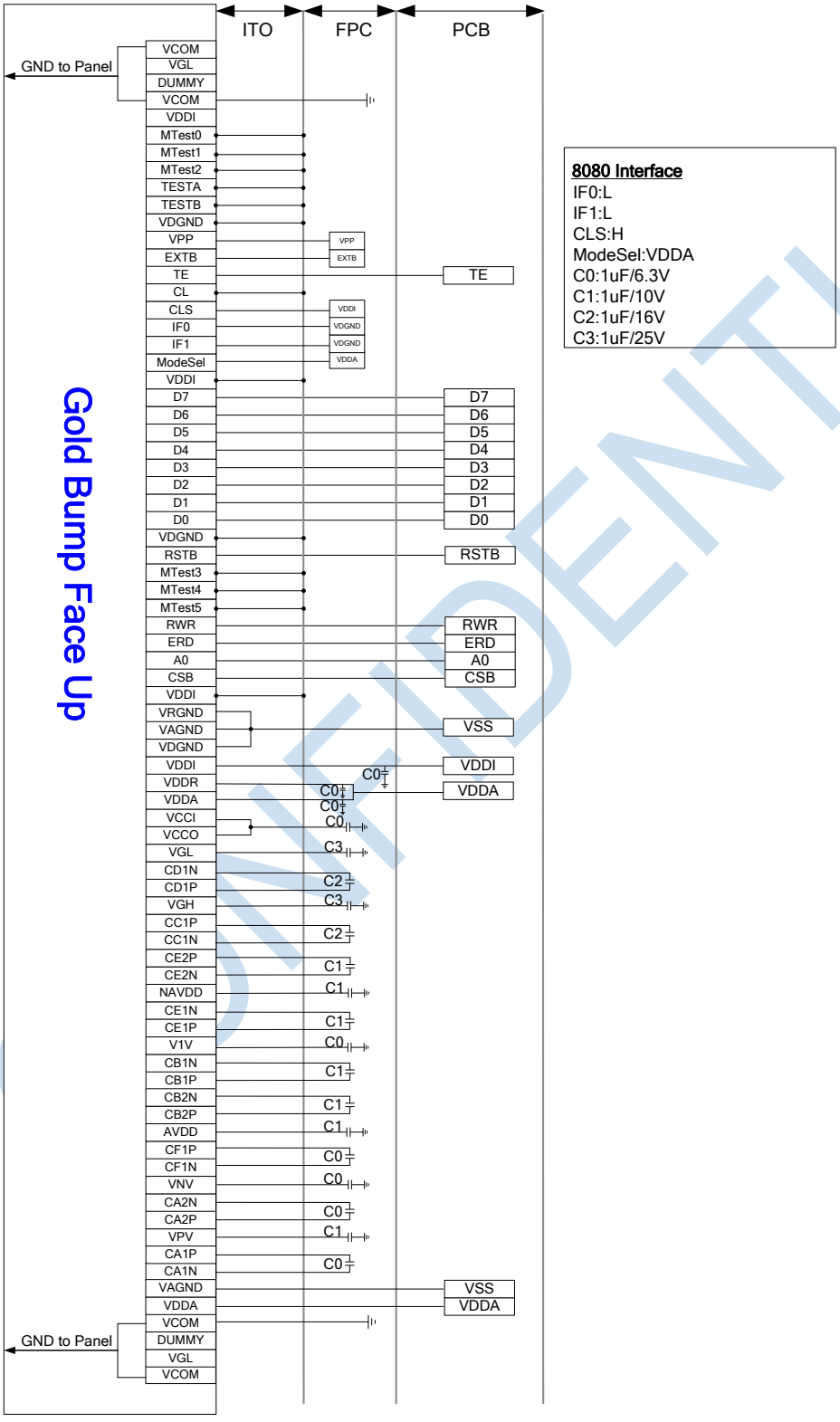
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13 APPLICATION NOTE

13.1 Parallel 8080 Interface

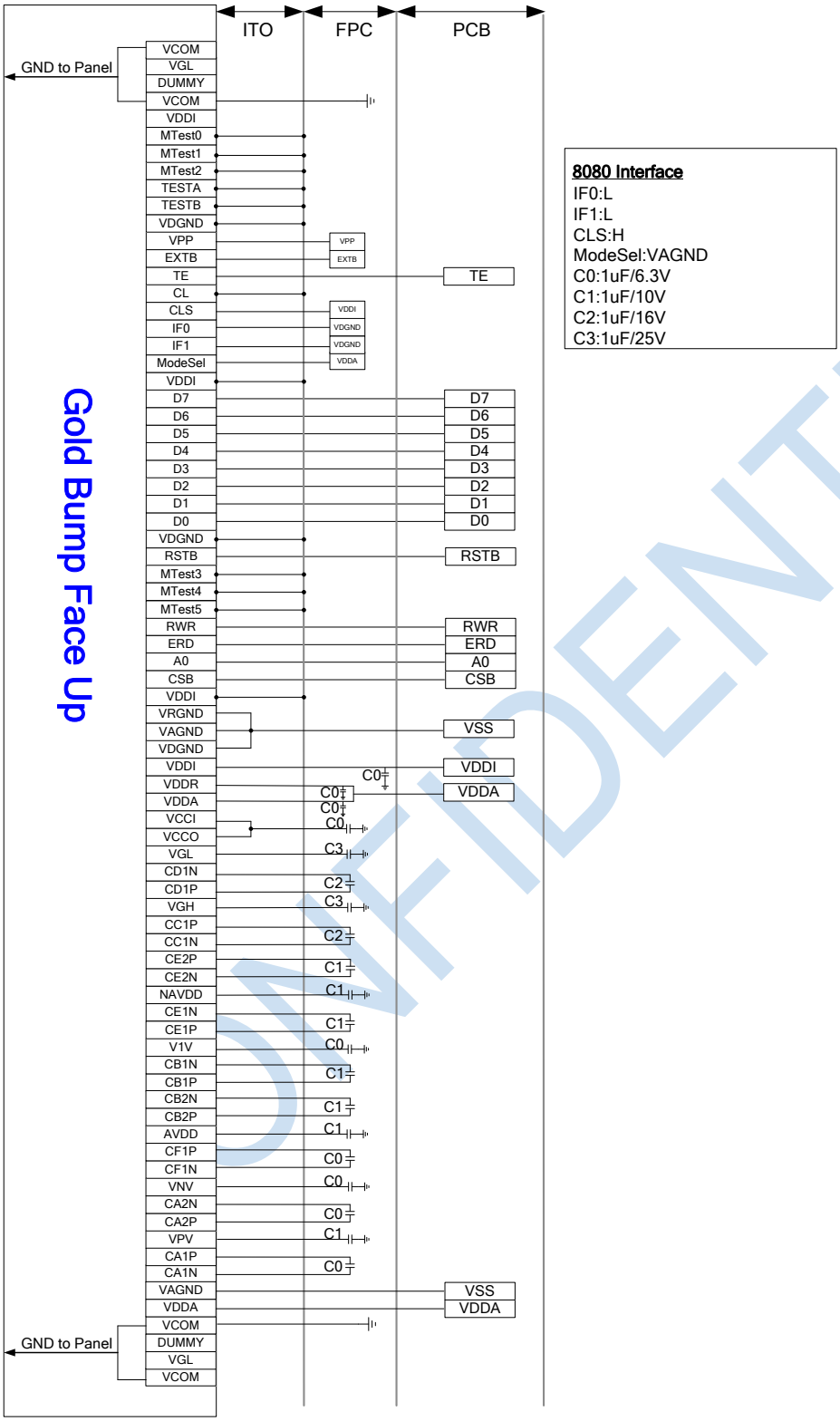
13.1.1 VDDA=3.3V



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13.1.2 VDDI=VDDA=1.8V

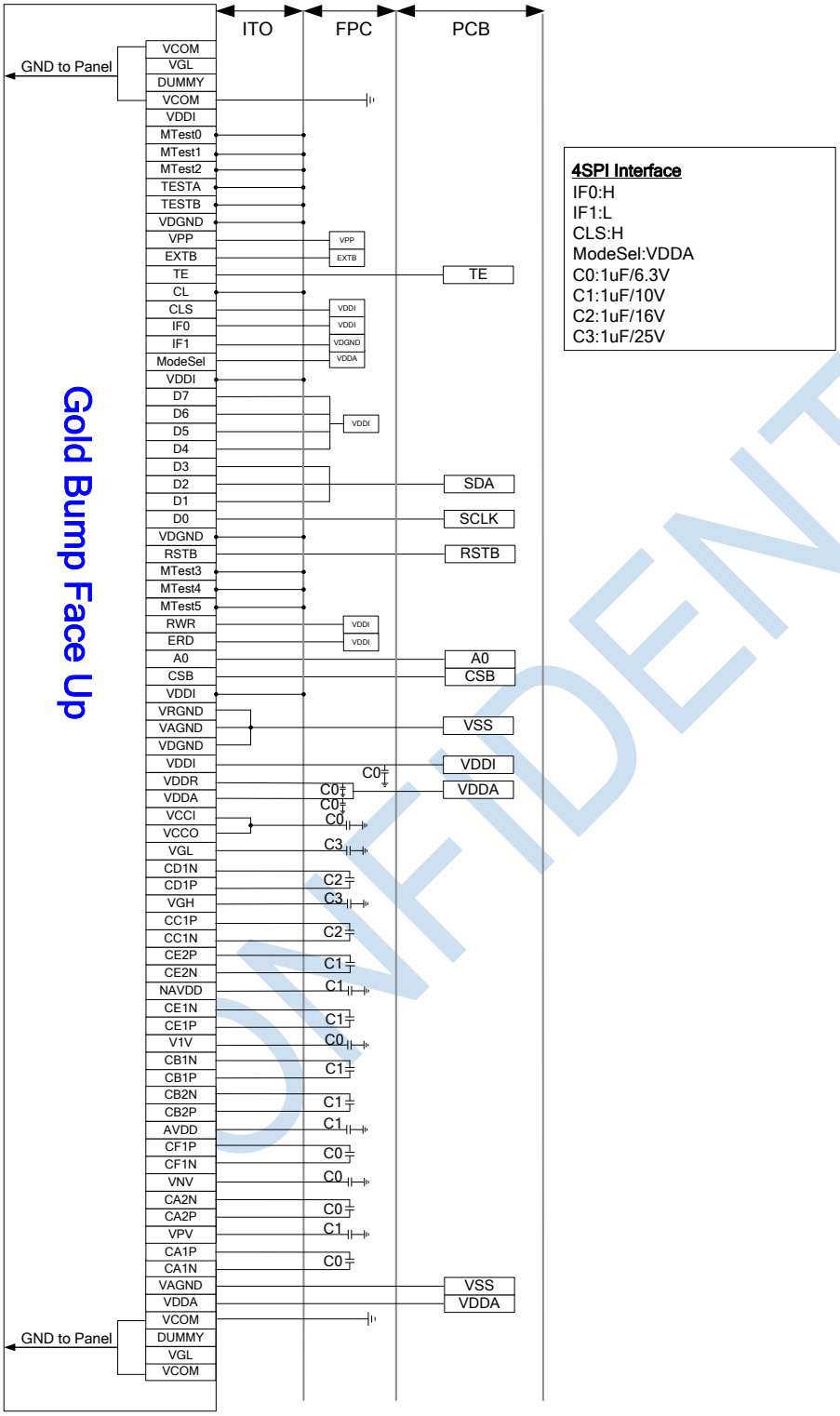


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13.2 Single 4-line SPI Interface

13.2.1 VDDA=3.3V

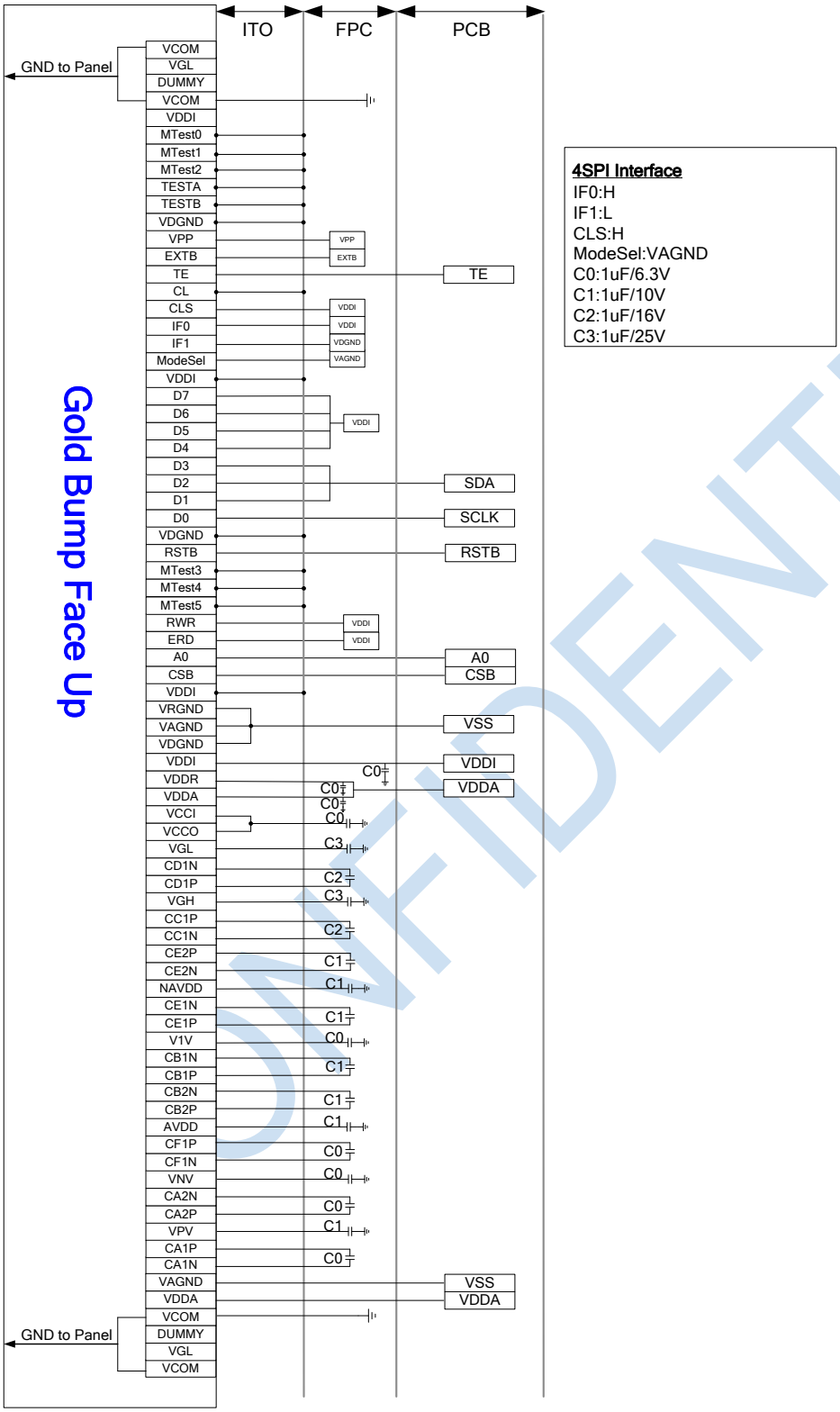


4SPI Interface
IF0:H
IF1:L
CLS:H
ModeSel:VDDA
C0:1uF/6.3V
C1:1uF/10V
C2:1uF/16V
C3:1uF/25V

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13.2.2 VDDI=VDDA=1.8V

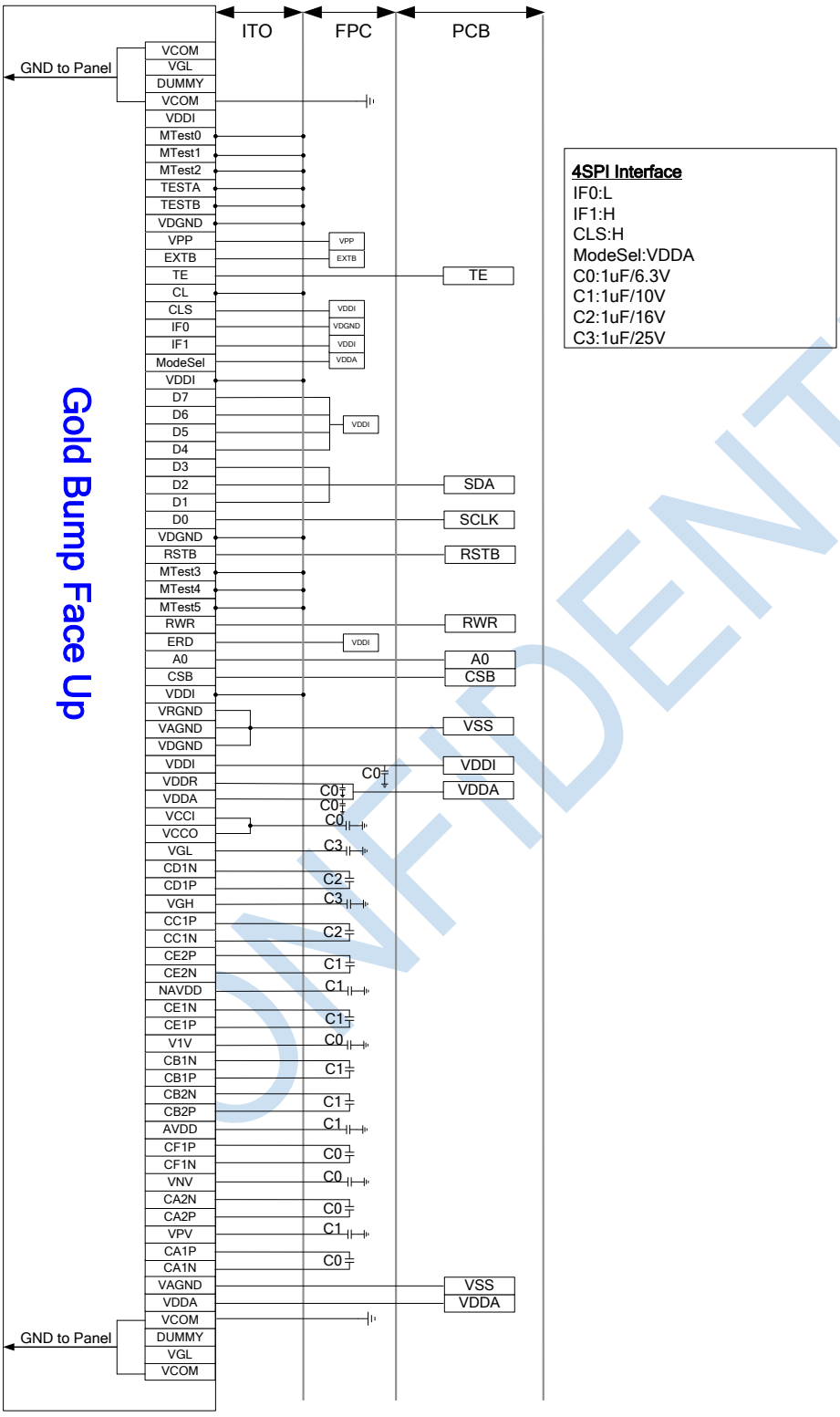


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13.3 Dual 4-line SPI Interface

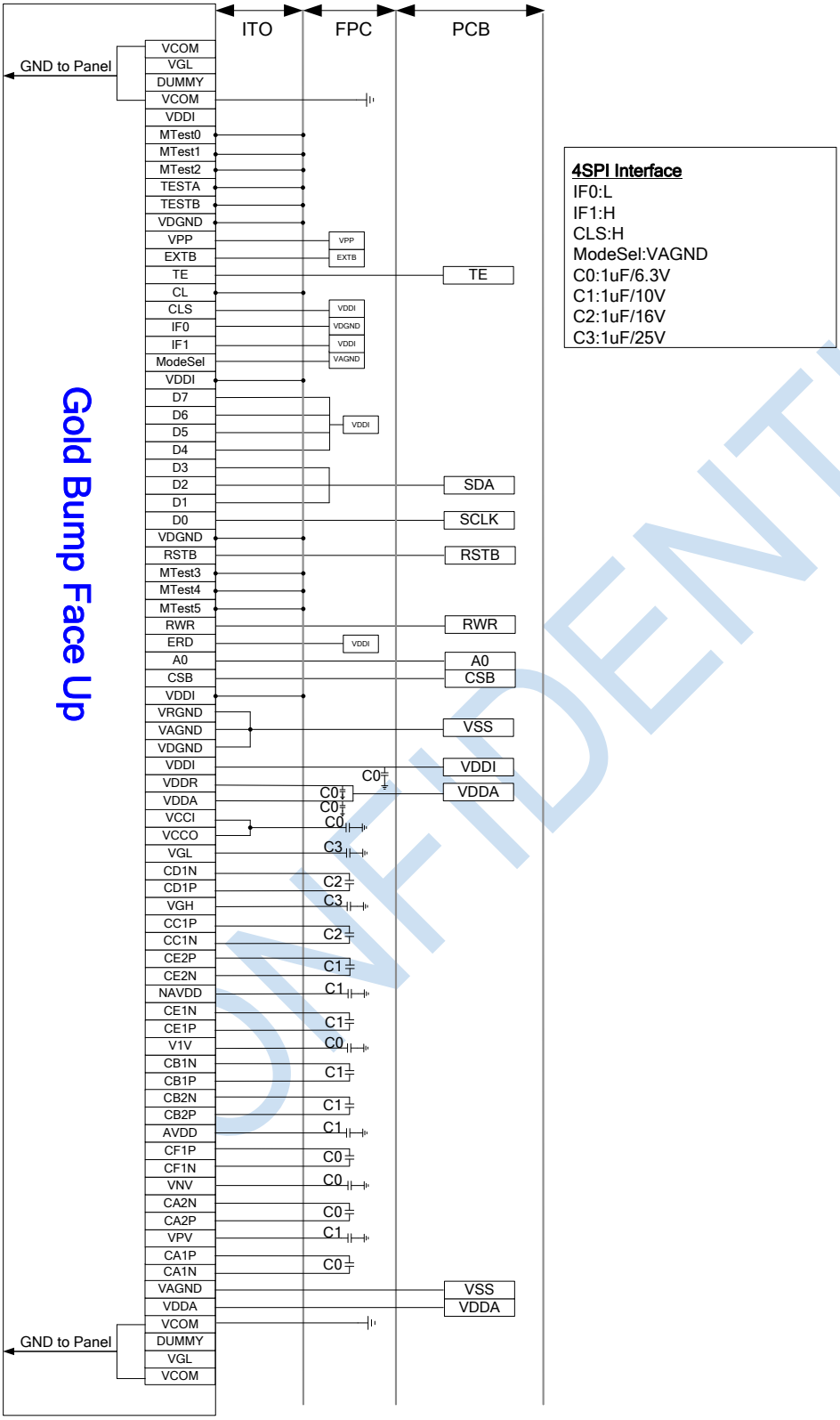
13.3.1 VDDA=3.3V



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13.3.2 VDDI=VDDA=1.8V

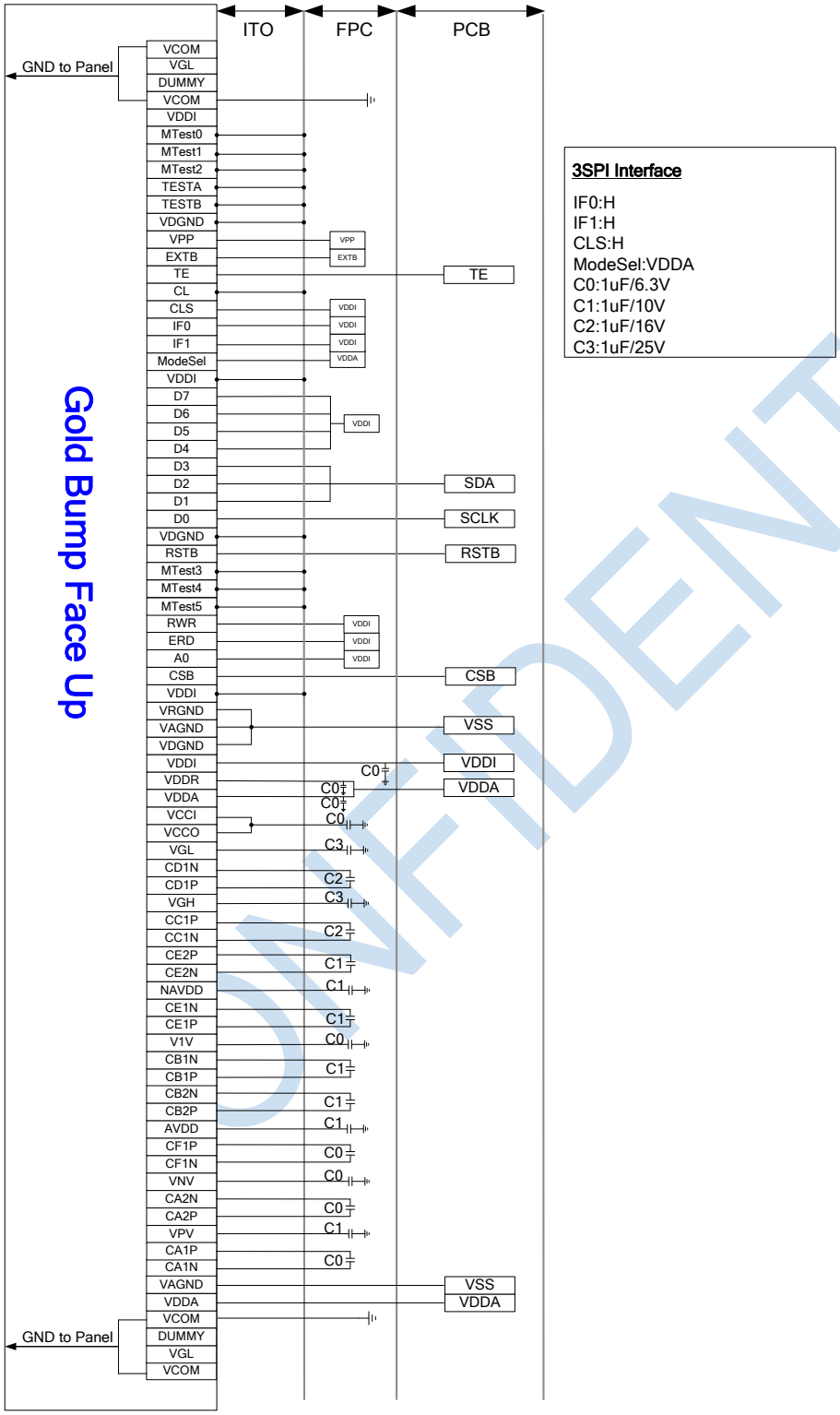


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13.4 3-line SPI Interface

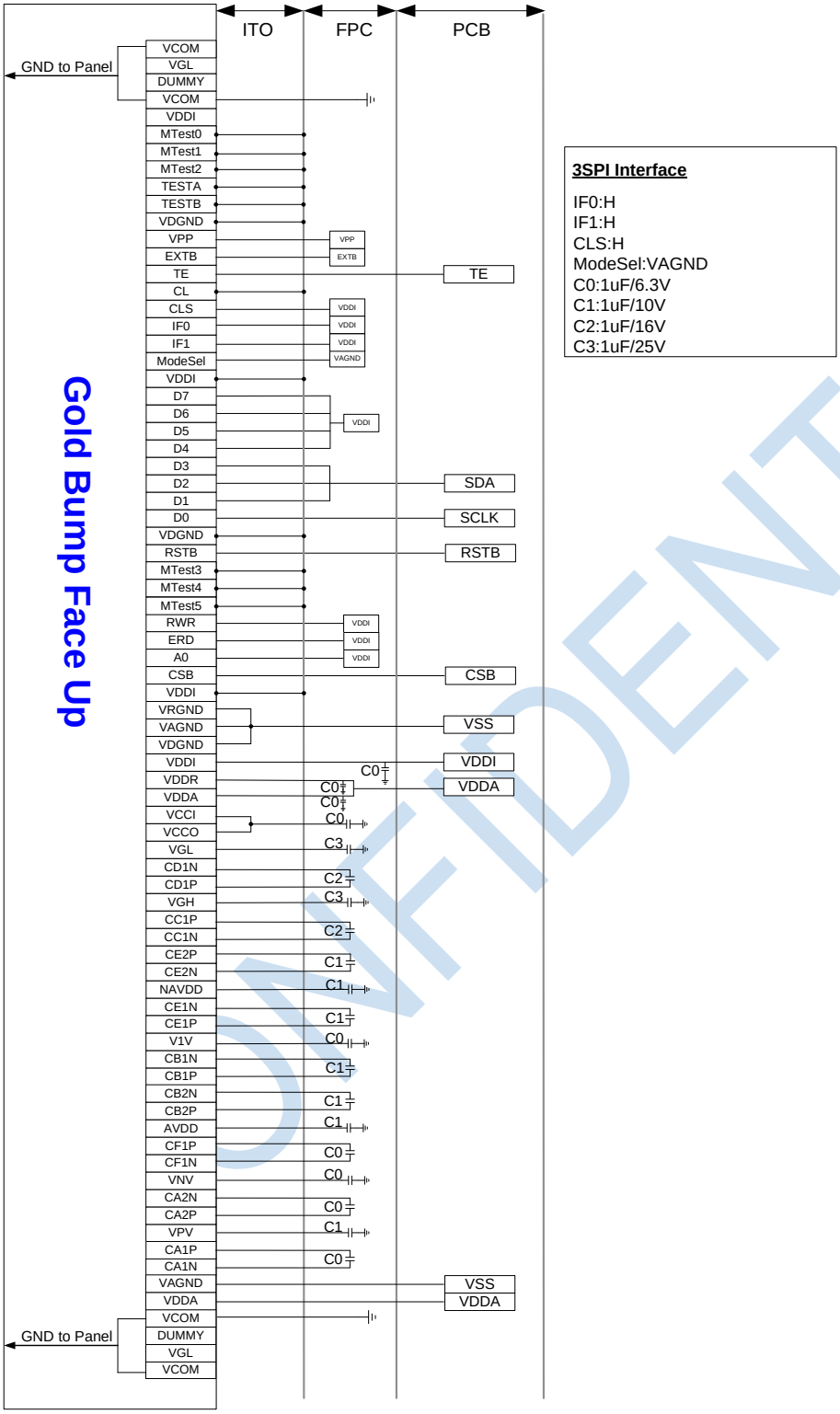
13.4.1 VDDA=3.3V



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13.4.2 VDDI=VDDA=1.8V



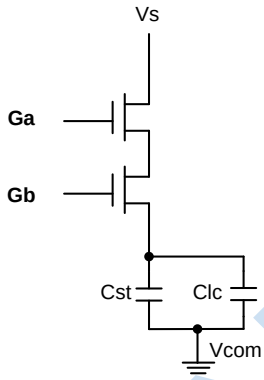
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14 PANEL LAYOUT

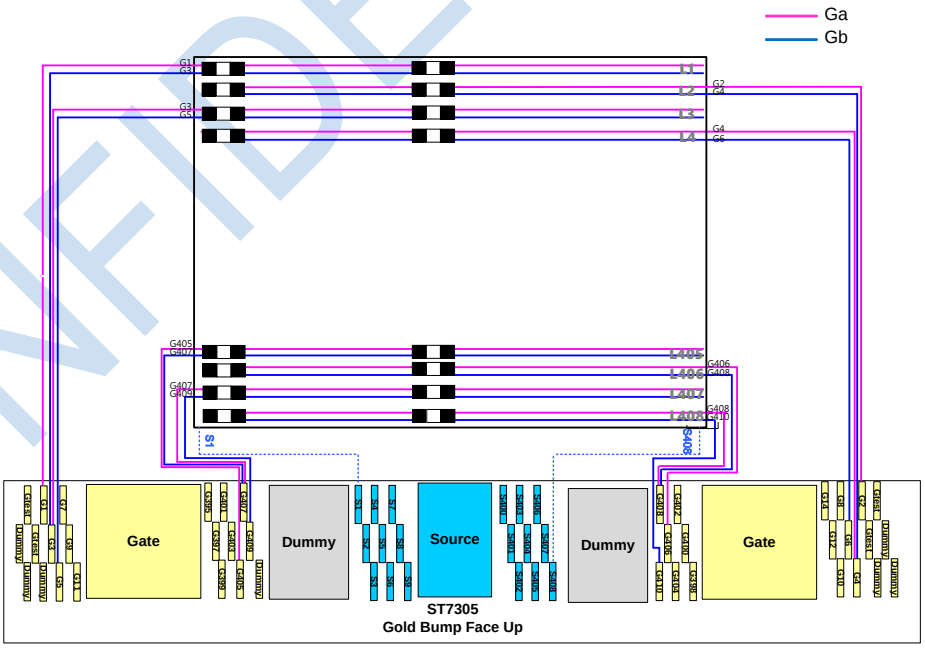
14.1 TFT Structure

The display module will operate at very low frame frequency to conserve power, so the recommended panel structure is two transistors for one line to control gate scan sequence. The benefits of this panel layout are to reduce leakage and enhance panel reliability.



14.2 Gate Layout

14.2.1 1-line interlace



Even Gate
Odd Gate

1Line-Interlace

Line	Ga	Gb
L1	G1	G3
L2	G2	G4
L3	G3	G5
L4	G4	G6
⋮	⋮	
L405	G405	G407
L406	G406	G408
L407	G407	G409
L408	G408	G410

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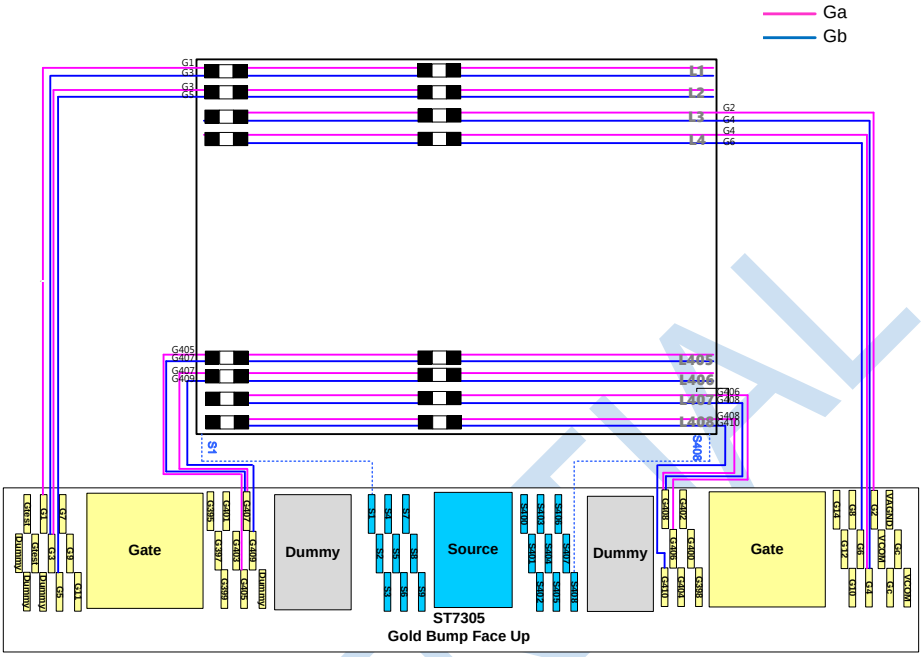
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14.2.2 2-line interlace

Even Gate
Odd Gate

2Line-Interlace

Line	Ga	Gb
L1	G1	G3
L2	G3	G5
L3	G2	G4
L4	G4	G6
⋮	⋮	
L405	G405	G407
L406	G407	G409
L407	G406	G408
L408	G408	G410

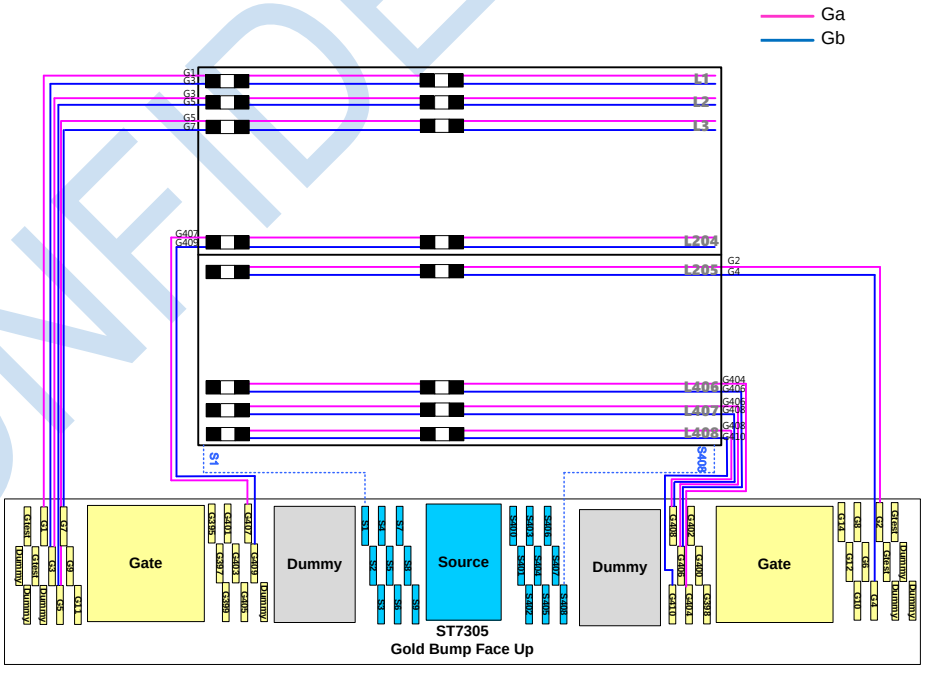


14.2.3 Non interlace

Even Gate
Odd Gate

Non-Interlace

Line	Ga	Gb
L1	G1	G3
L2	G3	G5
L3	G5	G7
⋮	⋮	
L204	G407	G409
L205	G2	G4
⋮	⋮	
L406	G404	G406
L407	G406	G408
L408	G408	G410



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15 REVISION HISTORY

Version	Date	Description
V1.0	2023/10	● Formal Version
V1.1	2023/11	● Add Command BDh ● Add D6h NVM Download Related Description
V1.2	2024/03	● Add B2h HPM Frame Rate 26.5Hz/53Hz and Related Description ● Add D8h OSC Switch Function and Related Description

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